

Using the Chapter Problems

Every chapter ends with a two-part problem section. Part I contains Core benchmarks to work without software: show the derivation, check feasibility or the defining identity, and state the economic point. Part II uses Proof and Applied problems as bridges to one named chapter case, which always appears last. Problem identifiers are stable across the book, course outline, and released handouts.

Before using Codex, Python, or Lean, write four lines: Seed identifies the hand result being extended; Fixed records the assumptions that do not change; Scaled names the single enlarged dimension or strengthened claim; Risk states the new failure mode. A verified artifact must reproduce the seed, include a boundary or deliberately failing test, report the validation command, and separate the checked mathematical statement from the economic interpretation.

Weekly handouts are release views of this architecture. The chapter problem section is the question-only source of record; solutions and instructor notes are maintained separately.

Microeconomic Theory I

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Preface to the Fall 2026 Edition

These notes provide a rigorous first course in graduate microeconomic theory. The Fall 2026 edition preserves the substantive coverage of the Fall 2024 course while improving the exposition, notation, organization, and portability of the source files. The central discipline throughout is to distinguish primitives, behavioral assumptions, optimization problems, equilibrium conditions, and empirical implications.

The material is cumulative. Part I begins with utility maximization and consumer demand, then develops duality, comparative statics, welfare, revealed preference, and abstract choice. Part II studies expected utility, stochastic choice, limited attention, and rational inattention. Part III turns to aggregation and representative consumers. Part IV develops production, competitive markets, and general equilibrium. Part V treats equilibrium restrictions, sequential trade, matching, and computable general equilibrium. Appendix A collects order theory, Appendix B records the exact versions of starred theorems, and the Glossary of Symbols provides linked notation.

Notation and Conventions

The following conventions apply unless a chapter explicitly states otherwise. They are collected here so that the same symbol does not silently change meaning across topics. In the PDF, a symbol shown in dark blue at its first substantive use links to its entry in the Glossary of Symbols; the symbol in the glossary links back to that first use. The LyX manuscript is the authoritative source for both the text and these links.

- There are L commodities, indexed by $\ell = 1, \dots, L$. Commodity bundles are column vectors $x \in X \subseteq \mathbb{R}_+^L$.
- Consumers are indexed by $i = 1, \dots, I$; firms are indexed by $j = 1, \dots, J$. Consumption is denoted by x , production by y , and endowments by ω .
- Prices are column vectors $p \in \mathbb{R}_{++}^L$, unless a result explicitly permits zero boundary prices. The scalar $w > 0$ denotes wealth. The budget set is $B(p, w) = \{x \in X : p \cdot x \leq w\}$.
- Set difference is denoted by $A \setminus B$. We write $\max, \min, \arg \max, \arg \min$ for the value and optimizer-set operators; each is typeset as a mathematical operator.
- Direct utility is u . Marshallian demand is $x(p, w)$, Hicksian demand is $h(p, \bar{u})$, indirect utility is $v(p, w)$, and the expenditure function is $e(p, \bar{u})$.
- The expression $a \cdot b$ denotes the Euclidean inner product. The Euclidean norm is $\|a\|_2$, and $\mathbf{1}_n$ denotes the all-ones column vector in $\mathbb{R}^n$. The notation $a^\top$ denotes transpose. A prime marks an alternative object of the same type, as in a' ; it is not used for transpose.
- Weak preference, strict preference, and indifference are denoted by $\succeq, \succ, \sim$, respectively. A choice correspondence is written $c : \mathcal{A} \rightrightarrows X$; the surrounding section defines the menu domain and states when the rule is single-valued.
- For a differentiable vector-valued function, $D_z f$ is the Jacobian of a vector-valued map

with respect to z , while $\nabla_z f$ is the column gradient of a scalar-valued function. Hessians are written $D_z^2 f$.

- For vectors, $a \leq b$ means coordinatewise inequality and $a \gg 0$ means every coordinate is strictly positive. The power set is 2^X .
- Probability and expectation are written $\Pr, \mathbb{P}, \mathbb{E}$ after the relevant probability space has been declared. The indicator is $\mathbb{I}\{\cdot\}$, and $\log$ denotes the natural logarithm.

Part I

The Classical Rational Consumer

Chapter 1

Utility Maximization and Consumer Demand

Question: What is the benchmark model, and what behavior does it generate?

The classical rational consumer chooses a feasible bundle to maximize a utility function. This chapter states that model, proves existence of demand, and develops its first-order characterization. Duality and welfare are treated in the next chapter. Throughout, $X = \mathbb{R}_+^L$, $L \geq 1$ is the fixed number of commodities, $\mathbf{P} = \mathbb{R}_{++}^L$, $\mathbf{W} = \mathbb{R}_{++}$.

1.1 The Benchmark Model

The benchmark model has three primitives: a consumption set X , a utility function $u : X \rightarrow \mathbb{R}$, and a feasible-set correspondence $(p, w) \mapsto B(p, w)$. At prices and wealth (p, w) , the consumer solves

$$x^* \in x(p, w) := \arg \max_{y \in B(p, w)} u(y).$$

Everything in the chapter follows from this optimization problem plus assumptions that are displayed where they are used.

Utility is ordinal in the deterministic benchmark: if ϕ is strictly increasing on $u(X)$, then u and $\phi \circ u$ generate exactly the same demand. We therefore take a utility representation as the primitive here without attaching cardinal or interpersonal meaning to its levels. Chapter 4 returns to the logically prior question of when an observed preference relation admits such a representation.

1.2 Commodity Space and Budget Sets

An economic commodity is a good or service whose quantity matters to the consumer. There are $L \geq 1$ commodities. A consumption bundle is a column vector $x = (x_1, \dots, x_L)^\top \in \mathbb{R}^L$, and x_ℓ denotes the quantity of commodity $\ell \in \{1, \dots, L\}$.

The consumption set $X \subseteq \mathbb{R}^L$ records the bundles that are feasible for the consumer. Unless a section states otherwise, we take $X = \mathbb{R}_+^L$, so negative consumption is ruled out. Other applications may impose upper bounds or interpret coordinates as net trades.

The consumer cannot generally choose every bundle in X . In the benchmark model, prices and wealth are the only constraints. Let $w > 0$ denote wealth and let $p \in \mathbb{R}_{++}^L$ denote the price vector. The budget set is $B(p, w) = \{x \in X : p \cdot x \leq w\}$, a subset of the consumption set.

1.3 Utility Maximization

We assume that u is continuous and $X = \mathbb{R}_+^L$.

The utility maximization problem (UMP) is:

$$\begin{aligned} \max_{x \geq 0} u(x) \\ \text{subject to } p \cdot x \leq w. \end{aligned}$$

Equivalently, define the budget set as $x \in B(p, w) = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}$.

Proposition 1.1. *If $p \gg 0$ and u is continuous, then the UMP has a solution.*

Proof. Because $B(p, w)$ contains the zero bundle, it is nonempty. For each $x \in B(p, w)$ and each commodity ℓ , strict positivity of prices gives $0 \leq x_\ell \leq w/p_\ell$; hence the budget set is bounded.

The map $x \mapsto p \cdot x$ is continuous, so $B(p, w) = \mathbb{R}_+^L \cap \{x : p \cdot x \leq w\}$ is closed. The Heine--Borel theorem therefore implies that $B(p, w)$ is compact.

The Weierstrass maximum theorem therefore gives a solution to the UMP.

□

We now prove further properties of the demand correspondence.

Proposition 1.2. *The Marshallian demand correspondence is*

$$x(p, w) := \arg \max_{y \in B(p, w)} u(y).$$

It is non-empty, compact-valued and upper hemicontinuous. The indirect utility function is

$$v(p, w) = \max_{y \in B(p, w)} u(y),$$

and it is continuous.

Proof. The result follows from Berge's maximum theorem once we verify that the budget correspondence $B(\cdot)$ is continuous. We already know that u is continuous and that $B(p, w)$ is compact, so $B(\cdot)$ is compact-valued.

Let $(p^n, w^n) \rightarrow (p, w)$. If $x^n \in B(p^n, w^n)$ and $x^n \rightarrow x$, then $p^n \cdot x^n \leq w^n$; taking limits gives $p \cdot x \leq w$, so the graph is closed. Moreover, near any (p, w) , prices are bounded away from zero and wealth is bounded above; hence the nearby budget sets are uniformly bounded. Closed graph plus this local boundedness gives upper hemicontinuity.

For the reverse inclusion, let $x \in B(p, w)$. If $p \cdot x < w$, then $x \in B(p^n, w^n)$ for all sufficiently large n . If $p \cdot x = w$, set

$$x^n = \frac{x}{\max\{1, (p^n \cdot x)/w^n\}}.$$

Then $x^n \in B(p^n, w^n)$ and $x^n \rightarrow x$. Hence the budget correspondence is continuous on $\mathbb{R}_{++}^L \times \mathbb{R}_{++}$, and Berge's theorem gives the stated properties. □

The next definitions separate the assumptions used in the comparative-statics results.

Definition 1.3. Local nonsatiation. A preference relation $\succeq$ is locally nonsatiated if, for every $x \in X$ and every $\varepsilon > 0$, there is $y \in X$ with $\|y - x\|_2 < \varepsilon$ and $y \succ x$.

Proposition 1.4. *If a utility function representing a locally nonsatiated preference relation generates the Marshallian demand correspondence, then every demanded bundle exhausts the budget: $p \cdot x = w$ for every $x \in x(p, w)$.*

Proof. If $p \cdot x < w$, local nonsatiation supplies a strictly preferred bundle arbitrarily close to x . For a sufficiently small neighborhood the bundle remains affordable, contradicting utility maximization. Therefore the budget constraint binds.

Proposition 1.5. *The same demand correspondence is homogeneous of degree zero. If preferences are convex, demand is convex-valued; if preferences are strictly convex, demand is single-valued.*

Proof. Homogeneity follows because $B(\lambda p, \lambda w) = B(p, w)$ for every $\lambda > 0$. If $x, y \in x(p, w)$ and preferences are convex, every mixture $\alpha x + (1 - \alpha)y$ is feasible and weakly preferred to either

maximizer, so it is also demanded. Under strict convexity, two distinct maximizers would have a feasible mixture strictly preferred to both, a contradiction. $\square$

1.4 Demand Correspondences

Definition 1.6. A Walrasian demand correspondence is a map

$$x : P \times W \rightrightarrows X$$

with nonempty values such that every $q \in x(p, w)$ satisfies $p \cdot q \leq w$. If demand is generated by a choice correspondence c , then $x = c \circ B$, where

$$B(p, w) = \{q \in X : p \cdot q \leq w\}.$$

Example 1.7. With two perfect-substitute goods and utility $u(q) = q_1 + q_2$,

$$x(p, w) = \begin{cases} \{(w/p_1, 0)\}, & p_1 < p_2, \\ \{(0, w/p_2)\}, & p_2 < p_1, \\ \{q \in \mathbb{R}_+^2 : p_1 q_1 + p_2 q_2 = w\}, & p_1 = p_2. \end{cases}$$

The equal-price value is the entire budget line, not a pair of independent intervals.

For later use, distinguish the following properties.

- A correspondence has a closed graph if $(p^n, w^n) \rightarrow (p, w)$, $q^n \in x(p^n, w^n)$, and $q^n \rightarrow q$ imply $q \in x(p, w)$. Closed graph and upper hemicontinuity are not equivalent without additional compactness or local-boundedness conditions.
- It is compact-valued if $x(p, w)$ is compact for every (p, w) . In Euclidean space, compactness is equivalent to being closed and bounded.
- It is convex valued if $\alpha q + (1 - \alpha)r \in x(p, w)$ whenever $q, r \in x(p, w)$ and $\alpha \in [0, 1]$.

1.5 First-Order and KKT Conditions

Assume that u is continuously differentiable and that a constraint qualification holds at the candidate optimum. The KKT conditions are then necessary. If u is concave, they are also sufficient for a global optimum; differentiability alone does not provide sufficiency.

The KKT conditions are as follows.

$$\mathcal{L} = u(x) + \lambda(w - p \cdot x)$$

Primal and dual feasibility require $x^* \geq 0$, $w - p \cdot x^* \geq 0$, and $\lambda^* \geq 0$.

$\partial_{x_\ell} u(x^*) \leq \lambda^* p_\ell$ and $x_\ell^* [\partial_{x_\ell} u(x^*) - \lambda^* p_\ell] = 0$ for all $\ell \in \{1, \dots, L\}$.

Complementary slackness is $\lambda^*(w - p \cdot x^*) = 0$.

This is a system with $L + 1$ unknowns.

Local nonsatiation implies that the consumer is on the budget hyperplane (the frontier). This means that

$$w - p \cdot x^* = 0.$$

For an interior solution with a positive multiplier and nonzero marginal utilities, the KKT conditions reduce to

$$\frac{\partial_{x_\ell} u(x^*)}{\partial_{x_k} u(x^*)} = \frac{p_\ell}{p_k} \quad \forall \ell, k \in \{1, \dots, L\}.$$

Note that for $L = 2$, $-\frac{p_1}{p_2}$ is the slope of the budget line and $-\frac{\partial_{x_1} u(x^*)}{\partial_{x_2} u(x^*)}$ is the slope of the indifference curve. Their equality is the familiar tangency condition expressed in terms of the marginal rate of substitution (MRS).

1.6 Workhorse Utility Specifications

A useful specification gives more than a closed-form answer: it identifies the assumptions behind expenditure shares, substitution, and corner choices. We derive several families from the same budget problem. Throughout, prices are strictly positive and wealth is positive; additional consumption-domain restrictions are stated for each model.

1.6.1 Homotheticity and Cobb-Douglas Demand

Definition 1.8. Homothetic preferences. A preference relation is homothetic if $x \succeq y$ implies $\alpha x \succeq \alpha y$ for every $\alpha > 0$ whenever the scaled bundles are feasible. Under the usual continuity and monotonicity assumptions, it admits a positively homogeneous utility representation; an arbitrary representation need not itself be homogeneous.

The Cobb-Douglas representation $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ is a standard example.

For L goods, fix positive taste weights a_ℓ with $\sum_{\ell=1}^L a_\ell = 1$. Cobb-Douglas utility is $u(x) = \prod_{\ell=1}^L x_\ell^{a_\ell}$. Every boundary bundle has utility zero and a strictly positive affordable bundle has positive utility, so the optimum is interior. Maximizing its logarithm gives $a_\ell/x_\ell = \lambda p_\ell$. Multiplying by x_ℓ and summing gives $\lambda = 1/w$, hence

$$x_\ell(p, w) = a_\ell \frac{w}{p_\ell}, \quad b_\ell(p, w) := \frac{p_\ell x_\ell(p, w)}{w} = a_\ell.$$

The expenditure share b_ℓ is the fraction of the budget allocated to good ℓ ; the share vector is $\mathbf{b} = (b_1, \dots, b_L)^\top$. Cobb-Douglas shares are constant even though quantities respond to prices. Each wealth elasticity is one, each own-price elasticity is minus one, and each uncompensated cross-price elasticity is zero.

Example 1.9. A fixed budget share. Let $a = (1/4, 3/4)$, $p = (2, 1)$, and $w = 80$. Demand is $(10, 60)$ and expenditure is $(20, 60)$. Doubling only the first price halves x_1 while leaving x_2 unchanged. Doubling wealth doubles both quantities. A constant share is a restriction on behavior, not a property of all homothetic preferences.

1.6.2 CES: A Family of Substitution Patterns

The constant-elasticity-of-substitution (CES) family varies substitution while retaining explicit demand. Keep the positive normalized weights above. Let $\rho < 1$ and define the substitution elasticity $\sigma = 1/(1 - \rho) > 0$. For $\rho \neq 0$, use

$$u_\rho(x) = \left(\sum_{\ell=1}^L a_\ell x_\ell^\rho \right)^{1/\rho}.$$

For $0 < \rho < 1$ this formula is defined on $\mathbb{R}_+^L$. For $\rho < 0$, define it first on $\mathbb{R}_{++}^L$ and extend it continuously by assigning zero utility to bundles with any zero coordinate. At $\rho = 0$, take the continuous limit $u_0(x) = \prod_{\ell} x_\ell^{a_\ell}$. Each representation is homogeneous of degree one. CES preferences are therefore homothetic, including cases in which expenditure shares vary with relative prices.

The optimum is interior. For $\rho \leq 0$, a positive bundle dominates every boundary bundle in utility. For $0 < \rho < 1$, a small transfer of expenditure toward a missing good is beneficial because its marginal utility becomes arbitrarily large near zero. For $\rho \neq 0$, an equivalent objective on the positive orthant, including the sign needed when $\rho < 0$, is

$$\tilde{u}_\rho(x) = \frac{\sum_\ell a_\ell x_\ell^\rho - 1}{\rho}, \quad \partial_{x_\ell x_\ell}^2 \tilde{u}_\rho = a_\ell(\rho - 1)x_\ell^{\rho-2} < 0.$$

At $\rho = 0$, instead use $\tilde{u}_0(x) = \sum_\ell a_\ell \log x_\ell$, with own second derivatives $-a_\ell/x_\ell^2 < 0$. In both cases all cross derivatives vanish, so the objective is strictly concave. Its first-order conditions $a_\ell x_\ell^{\rho-1} = \lambda p_\ell$, together with budget exhaustion, identify the unique maximizer. For negative ρ , maximizing the untransformed sum of powers would reverse the intended ranking. Taking ratios and then imposing the budget gives

$$\frac{x_\ell}{x_k} = \left(\frac{a_\ell/p_\ell}{a_k/p_k} \right)^\sigma, \quad x_\ell(p, w) = \frac{w a_\ell^\sigma p_\ell^{-\sigma}}{\sum_{k=1}^L a_k^\sigma p_k^{1-\sigma}}.$$

The same demand formula applies at $\sigma = 1$ and gives Cobb-Douglas. It satisfies Walras' law, has degree zero in (p, w) , and scales linearly with wealth. Its shares are

$$b_\ell(p) = \frac{a_\ell^\sigma p_\ell^{1-\sigma}}{\sum_k a_k^\sigma p_k^{1-\sigma}}.$$

Except in the Cobb-Douglas case, a taste weight is generally not an expenditure share. Raising a good's price lowers its share if $\sigma > 1$ and raises its share if $0 < \sigma < 1$, even though its purchased quantity falls in both cases. The name CES refers to the response of a quantity ratio to a relative price:

$$-\frac{d \log(x_\ell/x_k)}{d \log(p_\ell/p_k)} = \sigma.$$

This relative-demand elasticity differs from an individual good's own-price elasticity. Differentiating the demand formula at fixed wealth gives, for $k \neq \ell$,

$$\begin{aligned} \frac{\partial \log x_\ell}{\partial \log w} &= 1, \\ \frac{\partial \log x_\ell}{\partial \log p_\ell} &= -\sigma(1 - b_\ell) - b_\ell, \\ \frac{\partial \log x_\ell}{\partial \log p_k} &= (\sigma - 1)b_k. \end{aligned}$$

Thus every good is normal. Distinct goods are gross substitutes when $\sigma > 1$ and gross complements when $0 < \sigma < 1$: these terms describe the cross-price response at fixed wealth. Chapter 2 separates this response into substitution and purchasing-power effects. In particular, a negative uncompensated cross effect need not imply a negative compensated cross effect.

Example 1.10. Substitution and relative expenditure. Let $a_1 = a_2 = 1/2$, $\rho = 1/2$, $p = (1, 2)$,

and $w = 120$. Then $\sigma = 2$ and the denominator is $1/4 + 1/8 = 3/8$, giving $x = (80, 20)$. The quantity ratio is four but the expenditure ratio is two. The budget check is $80 + 2(20) = 120$. The own-price elasticity of good 1 is $-4/3$, rather than -2 .

Example 1.11. Gross complementarity. Let $a_1 = a_2 = 1/2$, $\rho = -1$, $p = (1, 4)$, and $w = 90$. Here $\sigma = 1/2$, demand is $(30, 15)$, and shares are $(1/3, 2/3)$. Raising only p_2 to 9 gives $(22.5, 7.5)$. Both quantities fall: the greater purchasing-power burden outweighs the relative-price incentive to consume more of good 1.

Example 1.12. Weights versus shares in three goods. Let $a = (0.5, 0.3, 0.2)$, $\sigma = 2$, $p = (1, 1, 1)$, and $w = 38$. The denominator is $0.25 + 0.09 + 0.04 = 0.38$, so $x = (25, 9, 4)$. Shares are $(25/38, 9/38, 4/38)$, not the taste weights. The same weights under Cobb-Douglas give $(19, 11.4, 7.6)$. A change in curvature with weights fixed generally changes the reference allocation as well as price responses.

Calibration makes the distinction operational. Given positive reference shares $b_\ell^0 = p_\ell^0 x_\ell^0 / w^0$ and a specified σ , weights reproducing that observation are

$$a_\ell = \frac{(b_\ell^0)^{1/\sigma} (p_\ell^0)^{(\sigma-1)/\sigma}}{\sum_k (b_k^0)^{1/\sigma} (p_k^0)^{(\sigma-1)/\sigma}}.$$

One positive observation therefore cannot identify the substitution elasticity: each specified σ admits calibrated weights. Different elasticities can fit the same initial allocation and predict different responses to a price change.

1.6.3 Limiting Cases: Substitutes and Complements

Holding weights fixed, $\rho \rightarrow 0$ gives Cobb-Douglas, $\rho \uparrow 1$ gives $u(x) = \sum_\ell a_\ell x_\ell$, and $\rho \rightarrow -\infty$ gives $u(x) = \min_\ell x_\ell$. Fixed weights disappear in the last limit. Unequal required proportions instead arise by applying the CES aggregator to x_ℓ/d_ℓ , with $d_\ell > 0$, before taking that limit. The endpoint models must be solved directly; their solutions need not satisfy the interior CES conditions.

Example 1.13. Perfect substitutes and a demand correspondence. For $u(x) = 3x_1 + x_2$, $p = (2, 1)$, and $w = 12$, utility per dollar is $3/2$ for good 1 and 1 for good 2, so the unique choice is $(6, 0)$. At $p = (3, 1)$ the two ratios tie: every nonnegative bundle satisfying $3x_1 + x_2 = 12$ maximizes utility. Reporting a single demand formula would omit valid choices.

Example 1.14. Perfect complements and a kink. For $u(x) = \min\{x_1/2, x_2\}$, the consumer buys $x_1 = 2x_2$. At $p = (1, 3)$ and $w = 20$, the chosen bundle is $(8, 4)$ because $2(4) + 3(4) = 20$. In

general $x_2 = w/(2p_1 + p_2)$ and $x_1 = 2w/(2p_1 + p_2)$. Tangency of differentiable indifference curves is unavailable at the optimum.

1.6.4 Subsistence and Stone-Geary Demand

To allow expenditure shares to vary with wealth, introduce a nonnegative subsistence bundle $\gamma \in \mathbb{R}_+^L$. On $X = \gamma + \mathbb{R}_+^L$, let $u(x) = \prod_{\ell} (x_{\ell} - \gamma_{\ell})^{a_{\ell}}$ with positive normalized weights. Suppose $w > p \cdot \gamma$. Setting $z = x - \gamma$ leaves discretionary wealth $w - p \cdot \gamma$, so Cobb-Douglas demand in z gives

$$x_{\ell}(p, w) = \gamma_{\ell} + a_{\ell} \frac{w - p \cdot \gamma}{p_{\ell}}.$$

At $w = p \cdot \gamma$, the only feasible bundle is γ ; below that level this subsistence-constrained model has no feasible bundle. It is not a model of behavior below subsistence. Except in special cases, these preferences are not homothetic about the origin and their budget shares depend on wealth.

Example 1.15. A necessity and discretionary spending. Let $\gamma = (2, 1)$, $a = (1/2, 1/2)$, $p = (2, 1)$, and $w = 15$. Subsistence costs 5, leaving 10 to allocate; demand is $(4.5, 6)$ and shares are $(0.6, 0.4)$. At wealth 25, demand is $(7, 11)$ and shares are $(0.56, 0.44)$. The first good's share falls even though its quantity increases.

1.6.5 Quasilinearity and Wealth-Dependent Corners

Definition 1.16. Quasilinear preferences. A numeraire good L is quasilinear when preferences admit a representation

$$u(x) = g(x_1, \dots, x_{L-1}) + x_L.$$

Such preferences are invariant to a common translation of both bundles in the numeraire whenever the translated bundles remain feasible. The converse is not implied by translation invariance alone. On a product domain $Z \times \mathbb{R}$, one sufficient formulation adds strict monotonicity in the numeraire and numeraire solvability: for a fixed $z_0 \in Z$ and every $z \in Z$, there is a unique $g(z)$ such that $(z, 0) \sim (z_0, g(z))$. Translation invariance then gives $(z, m) \sim (z_0, g(z) + m)$, which yields the displayed representation.

A nonnegative numeraire creates a boundary that matters for demand. Let $u(x_1, x_2) = 2\sqrt{x_1} + x_2$ on $\mathbb{R}_+^2$, normalize $p_2 = 1$, and write $r = p_1 > 0$. Substituting the budget gives the strictly concave problem $\max_{0 \leq x_1 \leq w/r} \{2\sqrt{x_1} - rx_1 + w\}$. Its unique solution is

$$x_1(r, w) = \min\{r^{-2}, w/r\}, \quad x_2(r, w) = \max\{w - r^{-1}, 0\}.$$

Example 1.17. When zero wealth effects stop applying. At $r = 2$ and $w = 3$, demand is $(1/4, 5/2)$: extra wealth is spent entirely on good 2. At $w = 1/4$, demand is $(1/8, 0)$ and all wealth is spent on good 1. The claim that quasilinear demand has no wealth effect for the non-numeraire good requires an interior numeraire choice, here $w > 1/r$.

1.6.6 From CES to Flexible Translog Demand

CES imposes one common, constant substitution elasticity. Applied demand analysis often needs substitution patterns that vary with relative prices. A translog specification uses a quadratic polynomial in logarithms, allowing such variation. Direct-utility, indirect-utility, and expenditure translog specifications are different parametric restrictions: the dual of a direct translog utility need not be translog. Chapter 2 develops a homothetic translog expenditure model, derives its demand shares, and checks the price region on which it represents a classical rational consumer. This order lets us explain the model through expenditure minimization rather than present an unexplained demand formula.

For further practice, Problems 1.2-1.5 compare these specifications. The artists case study, Problem 10.4, uses CES fan demand as the first stage of a model of streaming revenue, concert production, and ticket pricing; its consumer part can be completed now.

1.7 Problems

Part I: By hand

Problem 1.1 [Core]. Budget scaling and Cobb-Douglas demand.

Let $X = \mathbb{R}_+^2$ and $B(p, w) = \{x \in X : p \cdot x \leq w\}$, where $p \gg 0$ and $w > 0$.

(a) Prove that $B(\lambda p, \lambda w) = B(p, w)$ for every $\lambda > 0$. Explain why every demand correspondence generated by preference maximization is homogeneous of degree zero in (p, w) .

(b) Suppose preferences are locally nonsatiated. Prove that every chosen bundle satisfies Walras' law, $p \cdot x = w$.

(c) For $u(x) = x_1^\alpha x_2^{1-\alpha}$ with $0 < \alpha < 1$, derive Marshallian demand and verify both properties directly.

Problem 1.2 [Core]. CES curvature, demand, and expenditure shares.

For $\rho < 1$ and $\rho \neq 0$, let $u_\rho(x) = (\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho)^{1/\rho}$, with the Cobb-Douglas limit used at $\rho = 0$ and the boundary extension stated in the text. Write $\sigma = 1/(1 - \rho)$.

(a) Derive demand from the first-order conditions. Explain why the optimum is interior for $\rho < 1$ and why maximizing the sum of powers is incorrect when $\rho < 0$.

(b) At $p = (1, 4)$ and $w = 120$, calculate quantities and expenditure shares for $\sigma \in \{1/2, 1, 2\}$. Check budget exhaustion.

(c) Raise only the second price to 9. Recalculate the three demands and explain why a smaller quantity can have a larger expenditure share.

(d) Derive the own-price, cross-price, and wealth elasticities. Distinguish the elasticity of a quantity ratio from an individual demand elasticity.

Problem 1.4 [Core]. Substitutes, complements, and boundary choices.

Consider separately $u^A(x) = 2x_1 + x_2$ and $u^B(x) = \min\{x_1/3, x_2\}$ on $\mathbb{R}_+^2$, at strictly positive prices and wealth.

(a) Derive the full demand correspondence for u^A , including the price ratio at which it is set-valued.

(b) Derive demand for u^B . At $p = (2, 3)$ and $w = 36$, compute both models' choices.

(c) Explain why interior differentiable tangency conditions do not characterize these optima.

(d) Obtain the ratio $x_1 = 3x_2$ as a CES limit. Explain why holding the usual taste weights fixed does not preserve unequal Leontief proportions.

Part II: Household demand across subsistence and discretionary regimes

— chapter use case and verified scale-up

Problem 1.3 [Applied]. CES calibration and observational equivalence.

At prices $p^0 = (1, 2, 4)$, a household with wealth $w^0 = 120$ chooses $x^0 = (30, 25, 10)$. Consider Chapter 1's CES representation with positive normalized weights.

(a) For a specified $\sigma > 0$, derive the weights reproducing this observation.

(b) Calibrate the weights for $\sigma = 1/2$ and $\sigma = 2$. For each calibration predict demand at $p^1 = (2, 2, 4)$ and wealth 120.

(c) Explain why fitting one observation does not identify substitution. Describe additional observations that could distinguish the models.

Chapter use case

Seed: Problems 1.1, 1.4. Fixed: all unlisted primitives and assumptions. Scaled: wealth from two hand calculations to a grid, with preferences and prices fixed. Tool: Codex + Python. Verification: Reproduce both hand wealth levels; classify infeasible, boundary, and interior regimes separately for each model; check feasibility, budget exhaustion, and continuity at every valid threshold.

Problem 1.5 [Applied]. Subsistence and quasilinear corners.

The first of two consumers has $u(x) = (x_1 - 1)^{1/3}(x_2 - 2)^{2/3}$ on $x \geq (1, 2)$. The second has $u(x) = 4\sqrt{x_1} + x_2$ on $\mathbb{R}_+^2$.

(a) Derive the first consumer's demand above subsistence expenditure. State what happens at and below that threshold.

(b) At $p = (2, 1)$, compute the first consumer's demands and shares at wealth 10 and 22. Distinguish homotheticity in discretionary consumption from homotheticity in total consumption.

(c) Normalize the second consumer's numeraire price to one and write $r = p_1$. Derive demand for all $r, w > 0$, including the zero-numeraire region.

(d) Identify where a zero wealth effect for the non-numeraire good is valid. Compare this restriction with the Stone--Geary model.

Chapter 2

Duality, Comparative Statics, and Welfare

Question: How do we compute the model's price responses and welfare implications?

Chapter 1 tells us which bundle the classical rational consumer chooses. This chapter studies the same problem from its value functions. Indirect utility and expenditure minimization yield demand identities, comparative statics, and money-metric welfare measures. Set-valued duality is stated first; differentiability and uniqueness are added only for envelope formulas such as Roy's identity and Shephard's lemma.

2.1 Indirect Utility

The indirect utility function records the maximum utility attainable from a budget:

$$v(p, w) = \max\{u(x) : x \in X, p \cdot x \leq w\}.$$

Thus $v : P \times W \rightarrow \mathbb{R}$ ranks price--wealth situations through the utility they permit.

Every Marshallian maximizer yields this same value: $v(p, w) = u(x^*)$ for every $x^* \in x(p, w)$.

Proposition 2.1. *Suppose that $u(\cdot)$ is a continuous utility function representing a locally nonsatiated preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}_+^L$. The indirect utility function $v(p, w)$ is:*

1. *Homogeneity of degree zero.*
2. *$v(p, w)$ is strictly increasing in w and nonincreasing in p_ℓ .*

3. $v(p, w)$ is quasiconvex in (p, w) . The set $\{(p, w) | v(p, w) \leq \bar{v}\}$ is convex.
4. $v(p, w)$ is continuous in p and w .

Proof. Homogeneity follows from $B(\lambda p, \lambda w) = B(p, w)$ for every $\lambda > 0$. If $p' \geq p$ componentwise, then $B(p', w) \subseteq B(p, w)$, so the value is nonincreasing in every price. Continuity follows from Berge's maximum theorem and the continuity and compact-valuedness of the budget correspondence. To prove strict monotonicity in wealth, let $w' > w$ and choose $x^* \in x(p, w)$. The bundle x^* lies strictly inside $B(p, w')$. Local nonsatiation supplies a bundle arbitrarily close to x^* that is strictly preferred; choosing it close enough keeps it in $B(p, w')$. Hence $v(p, w') > v(p, w)$.

For quasiconvexity, fix $\bar{v}$ and let $L_{\bar{v}} = \{(p, w) : v(p, w) \leq \bar{v}\}$. If $u(x) > \bar{v}$, then every $(p, w) \in L_{\bar{v}}$ must satisfy $p \cdot x > w$, because otherwise x would be affordable and the value would exceed $\bar{v}$. Conversely, this strict inequality for every x with $u(x) > \bar{v}$ rules out an affordable bundle with utility above $\bar{v}$. Hence

$$L_{\bar{v}} = \bigcap_{u(x) > \bar{v}} \{(p, w) : p \cdot x > w\}.$$

Each set in the intersection is a linear half-space in (p, w) , so the lower contour set is convex and v is quasiconvex.

2.2 Roy's Identity

Assume the maximizer is unique, denote it by $x(p, w)$, and assume the value function is differentiable. Then $v(p, w) = u(x(p, w))$. The envelope theorem applied to

$$v(p, w) = \max_{x \geq 0, p \cdot x \leq w} u(x)$$

gives

$$\partial_w v(p, w) = \lambda(p, w), \quad \nabla_p v(p, w) = -\lambda(p, w)x(p, w).$$

If demand is unique, the envelope derivatives exist, and $\partial_w v(p, w) = \lambda(p, w) > 0$, Roy's identity is

$$x(p, w) = -\frac{\nabla_p v(p, w)}{\partial_w v(p, w)}.$$

Example 2.2. Quasilinear demand and Roy's identity. Let good L be the numeraire, set

$p_L = 1$, and let utility be $u(q, q_L) = g(q) + q_L$ for $q \in \mathbb{R}_+^{L-1}$. Write $p_{-L} = (p_1, \dots, p_{L-1})$ for the nonnumeraire price vector. Feasibility of the nonnegative numeraire requires $p_{-L} \cdot q \leq w$. The indirect utility is

$$v(p, w) = w + \max_{q \geq 0, p_{-L} \cdot q \leq w} \{g(q) - p_{-L} \cdot q\}.$$

When the unconstrained maximizer of $g(q) - p_{-L} \cdot q$ is affordable and leaves a nonnegative numeraire, the wealth constraint inside the displayed maximum is slack and non-numeraire demands have no income effects. When the maximizer is unique and the envelope theorem applies, Roy's identity gives $x_k(p, w) = -\partial_{p_k} v(p, w)$ for $k < L$, and the numeraire demand is the residual $x_L = w - p_{-L} \cdot x_{-L}(p)$, where $x_{-L}(p)$ collects the demands for the nonnumeraire goods and p_{-L} is their price vector.

2.3 Expenditure Minimization

At prices p , the expenditure minimization problem (EMP) asks for the least expenditure required to attain utility $\bar{u}$:

$$e(p, \bar{u}) = \min\{p \cdot x : x \in X, u(x) \geq \bar{u}\}.$$

Proposition 2.3. *Suppose u is a continuous utility function representing a locally nonsatiated preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}_+^L$ and the price vector is $p \gg 0$. Then:*

(i) *If x^* is optimal in the UMP when $w > 0$ then x^* is optimal in the EMP when the required level of utility is $u(x^*)$. Moreover, the minimized expenditure level $p \cdot x^* = w$ is exactly the wealth w .*

(ii) *If x^* is optimal in the EMP when the required utility level is $\bar{u} > u(0)$ then x^* is optimal in the UMP when wealth is $p \cdot x^*$. Moreover, the maximized utility level $u(x^*) = \bar{u}$.*

The proof uses local nonsatiation to show that both constraints bind and then applies the definitions of the two optimization problems.

Expenditure function. The function $e(p, \bar{u})$ is the value of the EMP. Whenever a solution exists, $e(p, \bar{u}) = p \cdot x^*$ for every expenditure-minimizing bundle x^* .

Proposition 2.4. *Suppose that u is a continuous utility function representing a locally nonsatiated preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}_+^L$, with $p \in \mathbb{R}_{++}^L$ and the target utility in the attainable range. The expenditure $e(p, \bar{u})$ is*

(i) Homogeneous of degree one in p .

(ii) Nondecreasing in $\bar{u}$ and nondecreasing in p_ℓ for all ℓ . Strict increase in the utility target requires restricting attention to targets above the baseline utility and conditions ensuring that the minimum reaches the target exactly.

(iii) Concave in p

(iv) Continuous in $(p, \bar{u})$.

Proof. (i) Multiplying every price by $\alpha > 0$ does not change the feasible set and multiplies the objective by α ; hence $e(\alpha p, \bar{u}) = \alpha e(p, \bar{u})$. For (ii), increasing the utility target shrinks the feasible set, while increasing a price cannot reduce the expenditure on any fixed nonnegative bundle.

(iii) Let $p^\alpha = \alpha p + (1 - \alpha)p'$ with $\alpha \in [0, 1]$ and choose any $x^\alpha \in h(p^\alpha, \bar{u})$. Feasibility of x^α at each price vector gives

$$\begin{aligned} e(p^\alpha, \bar{u}) &= \alpha p \cdot x^\alpha + (1 - \alpha)p' \cdot x^\alpha \\ &\geq \alpha e(p, \bar{u}) + (1 - \alpha)e(p', \bar{u}). \end{aligned}$$

Hence the expenditure function is concave in prices.

(iv) Continuity in $(p, \bar{u})$.

Continuity follows from the standard expenditure-function theorem. The unbounded feasible set cannot be inserted directly into Berge's theorem. Instead, around any $(p, \bar{u})$ with strictly positive prices, one first obtains a common expenditure upper bound from one feasible bundle. Strict positivity of nearby prices then confines every optimizer to a common compact box. Berge's theorem applies to this locally compactified problem. Continuity of preferences and local nonsatiation ensure feasibility stability and that an optimum reaches the target utility.

□

2.4 Hicksian Compensated Demand

The Hicksian (compensated) demand correspondence is the solution set of the EMP: $h(p, \bar{u}) = \arg \min\{p \cdot x : x \in X, u(x) \geq \bar{u}\}$.

Proposition 2.5. *Suppose $u(\cdot)$ is continuous and represents a locally nonsatiated preference relation $\succeq$, prices satisfy $p \gg 0$, and the attainable target satisfies $\bar{u} > u(0)$. Then $h(p, \bar{u})$ is nonempty and compact-valued and has the following properties:*

(i) It is homogeneous of degree zero in p

(ii) It has no excess utility: For $x^* \in h(p, \bar{u})$, $u(x^*) = \bar{u}$

(iii) *Convexity and uniqueness: If $\succeq$ is convex, then $h(p, \bar{u})$ is convex; if $\succeq$ is strictly convex, then Hicksian demand is single-valued.*

2.5 KKT Conditions for Expenditure Minimization

Assume u is continuously differentiable and a constraint qualification holds at the candidate optimum. The KKT conditions below are necessary. If u is concave, they are also sufficient.

The Lagrangian for this problem is:

$$\mathcal{L}_{\text{EMP}} = p \cdot x - \lambda(u(x) - \bar{u}).$$

Stationarity and complementary slackness for the nonnegativity constraints give

$$p \geq \lambda \nabla_x u(x^*)$$

$$x_\ell^* [p_\ell - \lambda \nabla u(x^*)_\ell] = 0,$$

for some $\lambda \geq 0$, together with primal feasibility $x^* \geq 0$ and $u(x^*) \geq \bar{u}$, and complementary slackness $\lambda[u(x^*) - \bar{u}] = 0$.

2.6 Shephard's Lemma

Proposition 2.6. *Every Hicksian bundle is a supergradient of the concave expenditure function in prices. If $e(\cdot, \bar{u})$ is differentiable at p , Hicksian demand is unique there and Shephard's identity holds:*

$$\nabla_p e(p, \bar{u}) = h(p, \bar{u}).$$

Proof. Fix $x^h \in h(p, \bar{u})$. For every other price vector p' , the same bundle remains feasible, so

$$e(p', \bar{u}) \leq p' \cdot x^h = e(p, \bar{u}) + (p' - p) \cdot x^h.$$

Hence x^h is a supergradient of the concave function $e(\cdot, \bar{u})$. At a differentiability point the superdifferential is the singleton $\{\nabla_p e(p, \bar{u})\}$, so every Hicksian bundle equals that gradient and demand is unique.

□

2.7 Curvature of Hicksian Demand

Assume throughout this subsection that $e(\cdot, \bar{u})$ is twice continuously differentiable, so Hicksian demand is single-valued and differentiable. The Jacobian $D_p h(p, \bar{u})$ has entries $[D_p h(p, \bar{u})]_{k\ell} = \partial_{p_\ell} h_k(p, \bar{u})$.

Proposition 2.7. *The Hicksian price-effect matrix satisfies*

$$D_p h(p, \bar{u}) = D_p^2 e(p, \bar{u}).$$

It is symmetric and negative semidefinite, and $D_p h(p, \bar{u})p = 0$. The first claims follow from Shephard's identity, twice continuous differentiability, and concavity of the expenditure function. The last follows by differentiating the degree-zero homogeneity of Hicksian demand.

Definition 2.8. Negative semidefiniteness. A symmetric $L \times L$ matrix A is negative semidefinite if $z^\top A z \leq 0$ for every vector $z \in \mathbb{R}^L$.

Example 2.9. The following matrix is NSD:

$$A = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$z^\top A z = -2z_1^2 \leq 0$$

for any $z \in \mathbb{R}^2$.

2.8 Consumer Duality

The expenditure function is the value of the EMP: for every $x^h \in h(p, \bar{u})$, $e(p, \bar{u}) = p \cdot x^h$.

Proposition 2.10. *Consumer duality. Suppose that u is continuous and represents a locally nonsatiated preference relation on $X = \mathbb{R}_+^L$. For $p \gg 0$, $w > 0$, and every attainable target $\bar{u} \in u(X)$ with $\bar{u} > u(0)$, the demand and value functions satisfy*

$$h(p, v(p, w)) = x(p, w),$$

$$x(p, e(p, \bar{u})) = h(p, \bar{u}),$$

$$v(p, e(p, \bar{u})) = \bar{u},$$

$$e(p, v(p, w)) = w.$$

The first two equalities are equalities of correspondences; when demand is single-valued, the same notation denotes functions. The restriction $\bar{u} > u(0)$ ensures $e(p, \bar{u}) > 0$, so the Marshallian demand appearing in the second and third identities lies in the declared wealth domain. At the boundary target $u(0)$, the identities require an explicit extension of Marshallian demand to zero wealth.

Proof. Choose $x^m \in x(p, w)$. Local nonsatiation gives $p \cdot x^m = w$. If a bundle delivered utility at least $v(p, w)$ at expenditure strictly below w , local nonsatiation would provide a strictly preferred nearby bundle still inside $B(p, w)$, contradicting the definition of the value function. Thus the expenditure needed to reach $v(p, w)$ is exactly w , and the minimizers are exactly the Marshallian maximizers. For an attainable $\bar{u} > u(0)$, continuity and positive prices ensure that the EMP has a minimizer x^h and that its expenditure is positive. Its utility target binds: if $u(x^h) > \bar{u}$, continuity along the segment from 0 to x^h would produce the target utility at strictly lower expenditure. No bundle affordable at $e(p, \bar{u})$ can yield utility strictly above $\bar{u}$, because the same segment argument would again contradict expenditure minimality. Hence $v(p, e(p, \bar{u})) = \bar{u}$, and budget exhaustion identifies the two demand correspondences. $\square$

2.9 Slutsky Equation

Assume in this section that Marshallian and Hicksian demands are single-valued and continuously differentiable and that the expenditure function is differentiable in prices. The Slutsky matrix is $S : \mathbf{P} \times \mathbf{W} \rightarrow \mathbb{R}^{L \times L}$, $S(p, w)_{\ell, k} = \partial_{p_k} x_\ell(p, w) + \partial_w x_\ell(p, w) x_k(p, w)$, $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top$.

Differentiate the duality identity $h(p, \bar{u}) = x(p, e(p, \bar{u}))$ with respect to p :

$$D_p h(p, \bar{u}) = D_p x(p, e(p, \bar{u})) + D_w x(p, e(p, \bar{u})) \nabla_p e(p, \bar{u})^\top$$

By Shephard's lemma and the duality identities,

$$D_p h(p, v(p, w)) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top = S(p, w).$$

2.10 Workhorse Demand through Duality

2.10.1 CES Expenditure, Compensation, and Substitution

Keep Chapter 1's degree-one CES normalization. Its unit expenditure index $\mathcal{P}(p) := e(p, 1)$ records the minimum cost of one unit of that utility index. Substituting the CES demand into utility, and then using duality, gives

$$\mathcal{P}(p) = \begin{cases} (\sum_{\ell} a_{\ell}^{\sigma} p_{\ell}^{1-\sigma})^{1/(1-\sigma)}, & \sigma \neq 1, \\ \prod_{\ell} (p_{\ell}/a_{\ell})^{a_{\ell}}, & \sigma = 1, \end{cases}$$

$$v(p, w) = \frac{w}{\mathcal{P}(p)}, \quad e(p, \bar{u}) = \bar{u} \mathcal{P}(p), \quad h_{\ell}(p, \bar{u}) = \bar{u} a_{\ell}^{\sigma} p_{\ell}^{-\sigma} \mathcal{P}(p)^{\sigma}.$$

Here $\bar{u} > 0$. The Cobb-Douglas factors a_{ℓ} in the unit expenditure formula follow from the chosen utility normalization; omitting them changes the utility units. Marshallian demand itself is unaffected by a positive rescaling of utility. Differentiating Hicksian demand gives own-price elasticity $-\sigma(1 - b_{\ell})$ and, for $k \neq \ell$, cross-price elasticity $\sigma b_k > 0$. CES goods are compensated substitutes even when $\sigma < 1$ makes them gross complements at fixed wealth.

Example 2.11. Exact compensation after a price increase. With equal weights, $\sigma = 2$, $p^0 = (1, 2)$, and $w = 120$, Chapter 1 found $x^0 = (80, 20)$. Here $\mathcal{P}(p^0) = 8/3$ and initial utility is 45. If $p^1 = (2, 2)$, then $\mathcal{P}(p^1) = 4$. Maintaining utility 45 requires wealth 180, so compensation is 60 and the compensated bundle is $(45, 45)$. At unchanged wealth 120, the final bundle is $(30, 30)$. For good 1, the total change -50 is the compensated change -35 plus the remaining wealth effect -15 . This is a finite-change Hicks decomposition; the Slutsky equation describes the corresponding local derivative.

2.10.2 A Homothetic Translog Expenditure Model

The translog family provides a flexible quadratic approximation in logarithms. We use a homothetic expenditure specification; richer translog families also permit wealth-dependent shares. Prices are measured relative to fixed reference units, so each logarithm is dimensionless. Write

$$\log \mathcal{P}_{\text{TL}}(p) = \alpha_0 + \sum_{\ell=1}^L \alpha_{\ell} \log p_{\ell} + \frac{1}{2} \sum_{\ell=1}^L \sum_{k=1}^L \gamma_{\ell k} \log p_{\ell} \log p_k.$$

The intercept is α_0 , the linear coefficients are α_{ℓ} , and the symmetric curvature matrix is

$\Gamma = (\gamma_{\ell k})$. Impose $\sum_{\ell} \alpha_{\ell} = 1$ and $\Gamma \mathbf{1}_L = 0$. These restrictions make $\mathcal{P}_{\text{TL}}$ homogeneous of degree one in prices. The matrix Γ here is unrelated to the subsistence vector γ in Stone-Geary utility. The translog approach originates with Christensen, Jorgenson, and Lau; the expenditure formulation and curvature calculation below also follow Diewert's treatment.

[Christensen et al., 1975, Diewert, 2023]

Suppose the price region under study is regular, as specified below. With $e(p, \bar{u}) = \bar{u} \mathcal{P}_{\text{TL}}(p)$ and $v(p, w) = w / \mathcal{P}_{\text{TL}}(p)$, Shephard's lemma gives

$$b_{\ell}(p) = \frac{\partial \log \mathcal{P}_{\text{TL}}(p)}{\partial \log p_{\ell}} = \alpha_{\ell} + \sum_k \gamma_{\ell k} \log p_k,$$

$$h_{\ell}(p, \bar{u}) = \frac{\bar{u} \mathcal{P}_{\text{TL}}(p)}{p_{\ell}} b_{\ell}(p), \quad x_{\ell}(p, w) = \frac{w}{p_{\ell}} b_{\ell}(p).$$

Adding up and homogeneity follow from the coefficient restrictions, but nonnegative demand and substitution restrictions require more. On an open convex cone $\mathcal{D} \subseteq \mathbb{R}_{++}^L$, require $b_{\ell}(p) > 0$ and the following matrix to be negative semidefinite at every price in $\mathcal{D}$:

$$H(p) := \Gamma + \mathbf{b}(p) \mathbf{b}(p)^{\top} - \text{diag}(\mathbf{b}(p)).$$

To see the role of this condition, let $D = \text{diag}(p_1, \dots, p_L)$. Differentiating $\partial_{p_{\ell}} \mathcal{P}_{\text{TL}} = \mathcal{P}_{\text{TL}} b_{\ell} / p_{\ell}$ yields

$$\nabla_p^2 \mathcal{P}_{\text{TL}}(p) = \mathcal{P}_{\text{TL}}(p) D^{-1} H(p) D^{-1}.$$

Thus $H(p) \preceq 0$ is exactly the local curvature restriction. Requiring it throughout a convex price domain establishes concavity on that domain. Positive fitted shares alone do not establish rationalizability, and checking curvature at one observation does not justify a distant counterfactual. The Slutsky matrix is $S(p, w) = w D^{-1} H(p) D^{-1}$, so symmetry, negativity, and the zero price direction are explicit.

There is a direct optimizing model behind these formulas. Define, for $x \in \mathbb{R}_+^L$,

$$u_{\mathcal{D}}(x) = \inf_{r \in \mathcal{D}} \frac{r \cdot x}{\mathcal{P}_{\text{TL}}(r)}.$$

This is a nonnegative, concave, homogeneous, nondecreasing utility. At $p \in \mathcal{D}$, concavity and homogeneity of the price index imply $r \cdot \nabla \mathcal{P}_{\text{TL}}(p) \geq \mathcal{P}_{\text{TL}}(r)$ for every $r \in \mathcal{D}$, with equality at $r = p$. Hence $u_{\mathcal{D}}(\nabla \mathcal{P}_{\text{TL}}(p)) = 1$. Scaling that bundle attains target utility $\bar{u} > 0$ at expenditure $\bar{u} \mathcal{P}_{\text{TL}}(p)$; any bundle reaching the target costs at least that much. This proves the expenditure

formula, and therefore the displayed demand, at prices in $\mathcal{D}$. It does not extend the translog formula to prices outside its regular region.

The compensated and uncompensated elasticities make the extra flexibility visible. Let $\delta_{\ell k}$ equal one when $\ell = k$ and zero otherwise. Then

$$\frac{\partial \log x_\ell}{\partial \log p_k} = \frac{\gamma_{\ell k}}{b_\ell} - \delta_{\ell k}, \quad \frac{\partial \log h_\ell}{\partial \log p_k} = \frac{\gamma_{\ell k}}{b_\ell} + b_k - \delta_{\ell k}.$$

Wealth elasticities are one in this homothetic version, while price elasticities vary with the shares. For two goods, the elasticity of substitution is $1 + \gamma_{12}/(b_1 b_2)$ on the regular region. With more goods the same expression is the Allen-Uzawa pairwise elasticity, rather than a universal elasticity of every quantity ratio. Setting $\Gamma = 0$ recovers constant shares and a Cobb-Douglas price index. General translog utility models need not be homothetic.

Example 2.12. A translog model with changing shares. Let $L = 2$, $\alpha_0 = 0$, $\alpha_1 = \alpha_2 = 1/2$, and $\Gamma = \frac{1}{10} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Writing $r = p_1/p_2$ gives

$$\mathcal{P}_{\text{TL}}(p) = \sqrt{p_1 p_2} \exp\left(\frac{(\log r)^2}{20}\right), \quad b_1 = \frac{1}{2} + \frac{\log r}{10}, \quad b_2 = 1 - b_1.$$

At $p = (1, 1)$ and $w = 100$, demand is $(50, 50)$. At $p = (2, 1)$ and the same wealth, shares are approximately $(0.569315, 0.430685)$ and demand is $(28.4657, 43.0685)$. The first quantity falls while its budget share rises. The substitution elasticity changes from 0.6 at equal prices to approximately 0.592162 at $(2, 1)$; a single CES elasticity cannot reproduce that variation throughout the region.

Example 2.13. Positive shares can conceal invalid curvature. In the preceding model, $H = (1/10 - b_1 b_2) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Shares are positive when $|\log r| < 5$, but curvature holds only when $|\log r| \leq \sqrt{15}$. Thus $|\log r| < \sqrt{15}$ defines an open convex cone on which the construction above is valid. At $r = \exp(4)$ the shares $(0.9, 0.1)$ still look admissible, yet $1/10 - b_1 b_2 = 0.01 > 0$: compensated own-price responses have the wrong sign. This counterfactual must be rejected for this parameterization.

2.11 What Utility Maximization Implies

Proposition 2.14. *Suppose demand $x(p, w)$ is generated by maximizing a complete, transitive, continuous, and locally nonsatiated preference relation represented by u on each linear budget;*

that is, $x(p, w) = \arg \max_{y \in B(p, w)} u(y)$ for every $(p, w) \in \mathbf{P} \times \mathbf{W}$. Then demand satisfies Walras' law and is homogeneous of degree zero.

If, in addition, demand is single-valued and continuously differentiable and the expenditure function is twice continuously differentiable, then the Slutsky matrix $S(p, w)$ is symmetric and negative semidefinite, and it is singular in the price direction: $S(p, w)p = 0$ for every $(p, w) \in \mathbf{P} \times \mathbf{W}$.

This proposition summarizes our previous results.

2.12 Welfare Evaluation

We will compare the wealth needed to achieve a certain level of utility at a given price-wealth pair.

$$u^1 = v(p^1, w)$$

$$u^0 = v(p^0, w).$$

We are interested in comparing the expenditure needed to achieve u^1 or u^0 .

For a fixed reference price and a given utility level u , the needed expenditure to reach it is given by

$$e(p^{ref}, u).$$

Thus the comparison in terms of expenditure is given by

$$e(p^{ref}, u^1) - e(p^{ref}, u^0)$$

$$e(p^{ref}, v(p^1, w)) - e(p^{ref}, v(p^0, w)).$$

This expression will be positive when it takes more wealth to achieve u^1 at prices p^{ref} than to achieve u^0 . This difference is measured in dollar terms. That is why $e(p^{ref}, v(p, w))$ is often called money metric indirect utility function.

We can construct many money metric indirect utility functions with different reference price.

When $p^{ref} = p^0$ is the original price situation, the expenditure change equals the wealth change that makes the consumer indifferent between the new price with the old wealth and the old price with the new wealth. This asks what wealth change is equivalent to the price change.

Equivalent variation is

$$EV(p^0, p^1, w) = e(p^0, v(p^1, w)) - e(p^0, v(p^0, w)) = e(p^0, v(p^1, w)) - w,$$

since $e(p^0, v(p^0, w)) = w$.

With this sign convention, equivalent variation is the money change at the reference price that is equivalent to the price change: it is willingness to pay for an improvement, and its negative is willingness to pay to avoid a worsening.

The other case considered is the one where the new price is taken as the reference price, when $p^{ref} = p^1$, the expenditure change equals the wealth change that makes the consumer indifferent between the original situation (p^0, w) and the new situation $(p^1, w + \Delta w)$.

Compensating variation is

$$CV(p^0, p^1, w) = e(p^1, v(p^1, w)) - e(p^1, v(p^0, w)) = w - e(p^1, v(p^0, w)).$$

The sign convention for compensating variation is analogous: a negative value for a worsening is the compensation required to restore the initial utility.

2.13 Problems

Part I: By hand

Problem 2.1 [Core]. Cobb-Douglas duality.

For $u(x) = x_1^\alpha x_2^{1-\alpha}$ with $0 < \alpha < 1$, $p \gg 0$, $w > 0$, and $\bar{u} > 0$, derive the Marshallian demand $x(p, w)$, indirect utility $v(p, w)$, Hicksian demand $h(p, \bar{u})$, and expenditure function $e(p, \bar{u})$.

- (a) Verify Walras' law and the identity $e(p, v(p, w)) = w$.
- (b) Verify $h(p, v(p, w)) = x(p, w)$.
- (c) Verify Roy's identity and Shephard's lemma.

Problem 2.5 [Core]. Translog shares and the regular price region.

Consider a homothetic translog index with $\alpha_0 = 0$, $\alpha = (2/5, 3/5)$, and $\Gamma = -\frac{1}{8} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$.

- (a) Derive shares and demands as functions of relative prices and wealth. Find the open interval of p_1/p_2 for which both shares are positive.

- (b) Compute $H = \Gamma + \mathbf{b}\mathbf{b}^\top - \text{diag}(\mathbf{b})$. Prove negative semidefiniteness throughout that interval.
- (c) At $p = (2, 1)$ and $w = 100$, calculate demand, Marshallian elasticities, and compensated elasticities. Verify adding up and homogeneity.
- (d) Explain why setting a negative predicted share to zero while retaining the other formula does not generally give a valid extension.

Part II: Local CES and translog welfare approximation — chapter use case and verified scale-up

Problem 2.2 [Proof]. Differential restrictions on demand.

Let $x : \mathbb{R}_{++}^L \times \mathbb{R}_{++} \rightarrow \mathbb{R}_+^L$ be continuously differentiable and satisfy Walras' law.

- (a) Differentiate Walras' law with respect to wealth and prices.
- (b) Define the Slutsky matrix by $S(p, w) = D_p x(p, w) + D_w x(p, w)x(p, w)^\top$. Prove that $p^\top S(p, w) = 0$.
- (c) Explain which additional restrictions on S follow from utility maximization and which of them do not follow from Walras' law alone.

Problem 2.3 [Proof]. The Slutsky matrix for Cobb-Douglas demand.

Using the Marshallian demand from Problem 2.1, compute $S(p, w) = D_p x + D_w x x^\top$.

- (a) Show that S is symmetric.
- (b) Show directly that $z^\top S z \leq 0$ for every $z \in \mathbb{R}^2$.
- (c) Find the null space of S and interpret it using homogeneity.

Problem 2.4 [Proof]. CES duality and compensated substitution.

Use Chapter 1's CES utility, with positive normalized weights and $\sigma > 0$.

- (a) Derive $\mathcal{P}(p)$, expenditure, and Hicksian demand. Treat $\sigma = 1$ separately and retain the constants implied by the utility normalization.
- (b) Show that the Slutsky matrix has off-diagonal entries $S_{\ell k} = \sigma x_\ell x_k / w$ and diagonal entries $S_{\ell \ell} = -\sigma x_\ell (1 - b_\ell) / p_\ell$. Prove symmetry and negative semidefiniteness using a weighted-variance argument. Identify its null space for strictly positive demand.
- (c) Explain how compensated cross-price effects can be positive when uncompensated cross-price effects are negative.
- (d) For equal weights, $\sigma = 2$, initial prices $(1, 3)$, wealth 90, and final prices $(2, 3)$, calculate compensation to retain initial utility. Decompose the first quantity's change into a compensated change and the remaining wealth effect.

Chapter use case

Seed: Problems 2.4, 2.5. Fixed: all unlisted primitives and assumptions. Scaled: log relative-price distance from one reference price. Tool: Codex + Python. Verification: Match the CES value, gradient, and Hessian at the reference price; enforce positive shares and negative semidefiniteness on the translog regular region; report welfare comparisons only on that region.

Problem 2.6 [Applied]. A translog approximation to CES demand.

Fix a two-good CES model, a positive reference price vector p^0 , and substitution elasticity $\sigma > 0$. Write b_1^0, b_2^0 for the reference shares and use normalized prices $r_\ell = p_\ell/p_\ell^0$.

- (a) Compute the gradient and Hessian of $\log \mathcal{P}_{\text{CES}}$ with respect to log prices at p^0 .
- (b) Choose translog coefficients in $\log r$ so the log price index agrees with CES through second order at p^0 .
- (c) Verify the restrictions for homogeneity. Show that demand levels and local price responses agree at the reference observation.
- (d) Explain why predictions can disagree for a large price change. State the regularity checks needed before making welfare comparisons.

Chapter 3

Revealed Preference and Recoverability

Question: Could observed behavior have been generated by the benchmark model, and what can the data identify?

The direction of analysis now reverses. Instead of deriving choices from u , we ask whether observed demand could have been generated by the benchmark model and what features of that model the data recover. Local restrictions concern derivatives of a demand function; finite-data restrictions concern revealed-preference comparisons across observed budgets. GARP characterizes classical utility maximization, while WGARP admits the broader coalitional multi-utility model introduced after Afriat's theorem.

3.1 Observed Demand and the Testing Question

After specifying the consumer's environment, we take demand as the observable primitive. A demand function has the typed signature

$$x : \mathbb{R}_{++}^L \times \mathbb{R}_{++} \rightarrow X.$$

It must satisfy $x(p, w) \in B(p, w)$ for every price--wealth pair; equivalently, $p \cdot x(p, w) \leq w$.

At prices and wealth (p, w) , the coordinate $x_\ell(p, w)$ is the quantity consumed of commodity ℓ .

Demand describes behavior; it does not yet explain that behavior through preferences or optimization. Treating the entire function as observed is an idealization, because an empirical dataset

contains only finitely many price--wealth observations. We therefore distinguish restrictions on a full demand function from tests that can be implemented on finite data.

Example 3.1. Cobb-Douglas Demand Function. $x(p, w) = (\frac{\alpha w}{p_1}, \frac{(1-\alpha)w}{p_2})^\top$ for $\alpha \in (0, 1)$.

3.2 Walras' Law, Homogeneity, and WARP

With the environment and demand primitive in place, we turn to consistency in consumption choices. This approach restricts observable behavior before imposing a specific preference representation.

We will impose a set of restrictions or axioms over the Marshallian demand function that we expect to hold in all situations irrespective of the details. Here we understand modeling consumption as restricting the set of allowable behaviors.

These axioms are minimal theoretical requirements, not empirical laws that every dataset must satisfy. Their value is that they make the set of admissible behavior explicit and provide a baseline against which richer models can be compared.

Axiom 3.2. (*Walras' law*) A demand function x is said to satisfy Walras' law if and only if $p \cdot x(p, w) = w$ for all price-wealth pairs.

Walras' law imposes budget exhaustion; it rules out choices that leave unspent wealth when a locally nonsatiated consumer could purchase more.

Axiom 3.3. (*Homogeneity of degree zero, HD0.*) A demand function x is homogeneous of degree zero if, for every $\alpha > 0$, $x(\alpha p, \alpha w) = x(p, w)$.

HD0 says that multiplying all prices and wealth by the same positive scalar does not change demand. It is a normalization of units, not a behavioral claim about intelligence.

Axiom 3.4. (*Weak axiom of revealed preference, WARP.*) A demand function x satisfies WARP if, for any two price--wealth pairs (p, w) and (p', w') , whenever $x(p, w) \neq x(p', w')$ and $p \cdot x(p', w') \leq w$, then $p' \cdot x(p, w) > w'$.

WARP excludes a two-observation reversal: if the bundle chosen at the second budget was affordable at the first, then the distinct bundle chosen at the first must be unaffordable at the second.

Let the observation-index set be $\mathbb{T} = \{1, \dots, T\}$ and consider the dataset $\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$, where $s, t \in \mathbb{T}$ index observations, $p^t \gg 0$, $x^t \in X \setminus \{0\}$, and observed expenditure is $w^t := p^t \cdot x^t$.

We define direct weak revealed preference by $\succeq^{R,D}$. Thus $x^t \succeq^{R,D} y$ means that the observed bundle cost at least as much as y at the prices under which x^t was chosen. the compositional superscripts and the distinction between WARP and WGARP follow [Aguiar et al., 2026]. Formally,

$$x^t \succeq^{R,D} y \iff p^t \cdot x^t \geq p^t \cdot y.$$

For distinct chosen bundles z and y , WARP is equivalently $z \succeq^{R,D} y \implies \neg(y \succeq^{R,D} z)$. This is a restriction on observed choices; it need not be interpreted as a primitive psychological preference relation.

3.3 Local Implications of WARP

After establishing the axioms of behavior, the next step is to study their implications for the parameters of the model. The parameters of the model in our study of consumer behavior are prices and wealth pairs. Assume throughout this section that the demand function is continuously differentiable on an open domain of strictly positive prices and wealth.

- The price Jacobian is $D_p x(p, w) \in \mathbb{R}^{L \times L}$; its (ℓ, k) entry is $\partial_{p_k} x_\ell(p, w)$.
- We let $D_w x(p, w) = (\partial_w x_\ell(p, w))_{\ell=1}^L$ be the vector of wealth effects.
- The Slutsky matrix is $S(p, w) = D_p x(p, w) + D_w x(p, w)x(p, w)^\top$. It measures the derivative of demand under a price change that adjusts wealth just enough to keep the previously chosen bundle affordable. Its (ℓ, k) entry is $S_{\ell k} = \partial_{p_k} x_\ell(p, w) + \partial_w x_\ell(p, w)x_k$.

Fix a reference bundle q and consider the compensated demand $x(p, p \cdot q)$. The chain rule gives $D_p[x(p, p \cdot q)] = D_p x(p, p \cdot q) + D_w x(p, p \cdot q)q^\top$. At $q = x(p, w)$, Walras' law gives $p \cdot q = w$, so this derivative is exactly the Slutsky matrix: $S(p, w) = D_p x(p, w) + D_w x(p, w)x(p, w)^\top$.

Implications of Walras' Law for the Slutsky Matrix

Fact 3.5. *If x satisfies Walras' law, then (i) $p^\top D_w x(p, w) = 1$, (ii) $p^\top D_p x(p, w) = -x(p, w)^\top$, (iii) $p^\top S(p, w) = 0$.*

Part (i) implies that at least one good has a positive wealth effect. Because prices are strictly positive and $\sum_\ell p_\ell \partial_w x_\ell(p, w) = 1$, there must be some ℓ such that $\partial_w x_\ell(p, w) > 0$. In general, Walras' law restricts the wealth effects, while part (iii) restricts consumption changes under Slutsky-compensated price changes.

Implications of Homogeneity of Degree Zero.

Claim 3.6. If x satisfies Walras' law and homogeneity of degree zero, then $S(p, w)p = 0$.

Proof. For fixed (p, w) , homogeneity gives $x(\lambda p, \lambda w) = x(p, w)$ for all $\lambda > 0$. Differentiating with respect to λ at $\lambda = 1$ yields

$$D_p x(p, w)p + D_w x(p, w)w = 0.$$

Walras' law gives $w = p \cdot x(p, w) = x(p, w)^\top p$; therefore the left-hand side is $S(p, w)p$, as required. $\square$

Implications of WARP.

The principal two-budget implication of WARP under Walras' law is the compensated law of demand (CLD).

Definition 3.7. (CLD) We say that x satisfies the Compensated Law of Demand if and only if $(p^* - p) \cdot (x(p^*, w^*) - x(p, w)) < 0$ when $x(p^*, w^*) \neq x(p, w)$ and $w^* = p^* \cdot x(p, w)$.

We next establish that WARP implies the compensated law of demand.

Proposition 3.8. *Let x satisfy Walras' law and WARP. Then x satisfies the CLD.*

Proof. Fix two price-wealth pairs satisfying $w^* = p^* \cdot x(p, w)$ and $x(p^*, w^*) \neq x(p, w)$. Set $z = x(p, w)$ and $y = x(p^*, w^*)$.

Walras' law and the compensation condition give $p \cdot z = w$ and $p^* \cdot z = p^* \cdot y = w^*$.

Because $p^* \cdot z \leq w^*$, WARP and $y \neq z$ imply $p \cdot y > w$, hence $p \cdot (y - z) > 0$.

Combining this strict inequality with $p^* \cdot (y - z) = 0$ gives

$$(p^* - p) \cdot (y - z) < 0,$$

$\square$

The differential version yields the following corollary.

Corollary 3.9. *Suppose x is continuously differentiable and satisfies Walras' law and WARP. Then, for every $a \in \mathbb{R}^L$ and every price-wealth pair, $a^\top S(p, w)a \leq 0$. Equivalently, the symmetric part $[S(p, w) + S(p, w)^\top]/2$ is negative semidefinite.*

Proof. Fix a price-wealth pair (p, w) and a direction $a \in \mathbb{R}^L$. For sufficiently small t , define $p(t) = p + ta$ and compensate wealth by $w(t) = p(t) \cdot x(p, w)$.

$$ta \cdot [x(p(t), w(t)) - x(p, w)] \leq 0.$$

$$\frac{a \cdot [x(p(t), w(t)) - x(p, w)]}{t} \leq 0$$

for every nonzero sufficiently small t after dividing the revealed-preference inequality by $t^2 > 0$.

Continuous differentiability and the chain rule imply

$$\lim_{t \rightarrow 0} \frac{x(p(t), w(t)) - x(p, w)}{t} = D_p x(p, w)a + D_w x(p, w) x(p, w)^\top a = S(p, w)a.$$

Taking limits gives $a^\top S(p, w)a \leq 0$.

□

Marshallian and compensated price effects

The Slutsky decomposition separates the total response of demand to a price change from the response at fixed purchasing power. For a differentiable demand function,

$$\partial_{p_k} x_\ell(p, w) = S_{\ell k}(p, w) - x_k(p, w) \partial_w x_\ell(p, w).$$

Under WARP and Walras' law the compensated matrix has a negative-semidefinite symmetric part. Thus a positive Marshallian own-price derivative $\partial_{p_\ell} x_\ell > 0$ is possible only when the income effect is sufficiently adverse: the good is inferior ($\partial_w x_\ell < 0$), and the income effect dominates the nonpositive compensated effect. This is the Giffen mechanism.

The possibility of a Giffen response does not by itself establish a utility representation. Two distinct rationalizability routes are developed later. For a differentiable demand system, local integrability uses Walras' law, homogeneity, Slutsky symmetry and negative semidefiniteness together with explicit domain and regularity hypotheses. For a finite budget dataset, observed expenditure exhaustion and GARP provide the Afriat certificate. The compensated response, rather than the Marshallian derivative alone, is the object restricted by revealed preference.

3.4 Integrability

We now ask for sufficient conditions under which an observed demand function is generated by utility maximization. In the abstract choice setting, WARP and GARP restrict rationalizability. Competitive budget sets are a smaller domain than arbitrary menus, so differentiable demand requires additional restrictions. The relevant restrictions are stated in terms of the Slutsky matrix.

Assume throughout this subsection that the demand function x is continuously differentiable.

Recall that the Slutsky matrix function is $S(p, w) = D_p x(p, w) + D_w x(p, w)x(p, w)^\top$, or in the scalar version $[S(p, w)]_{\ell k} = \partial_{p_k} x_\ell(p, w) + \partial_w x_\ell(p, w)x_k(p, w)$ for all $\ell, k \in \{1, \dots, L\}$.

Remark 3.10. Integrability roadmap. On an open domain, Walras' law, homogeneity, and a symmetric negative-semidefinite Slutsky matrix are the central differential restrictions. They are necessary for every regular utility-generated demand. A sufficiency theorem additionally specifies rank or nondegeneracy conditions, the topology of the price-wealth domain, and boundary behavior for the associated differential system. Because alternative integrability theorems impose different global hypotheses, this text does not present the preceding checklist as a stand-alone global theorem; applications must cite and verify a precise version.

The qualifications are essential. Symmetry and negative semidefiniteness alone, stated on an unspecified domain, do not guarantee a global representation: one must control the domain, boundary behavior, and the differential equation's existence and uniqueness. Conversely, every continuously differentiable demand generated by a regular locally nonsatiated utility satisfies Walras' law, homogeneity, symmetry, and negative semidefiniteness of the Slutsky matrix.

Remark. Some versions of the global integrability theorem impose a bound on wealth elasticity, while others replace it with different domain or boundary conditions. No such bound is a universal consequence of local Slutsky restrictions; the hypotheses of the particular global theorem being used must be checked.

3.5 Examples and Applications of Demand Integrability

Consider two commodities, $L = 2$, and a parameter $0 < \alpha < 1$. Suppose observed demand for good 1 is $x_1(p, w) = \frac{\alpha w}{p_1}$. Normalize $p_2 = 1$, and impose Walras' law. Demand for good 2 is then the budget residual, $x_2 = w - p_1 x_1 = (1 - \alpha)w$. The following calculation recovers a normalized indirect utility and then a direct utility representation.

Fix an initial price-wealth pair $(p_1^0, w^0) \in \mathbb{R}_{++}^2$. For $\xi > 0$, let $\chi(\xi; p_1^0, w^0)$ denote the wealth required at prices $(\xi, 1)$ to remain on the indifference curve through the initial pair. The integrability differential equation and its initial condition are

$$\frac{\partial \chi(\xi; p_1^0, w^0)}{\partial \xi} = x_1((\xi, 1), \chi(\xi; p_1^0, w^0)) = \frac{\alpha \chi(\xi; p_1^0, w^0)}{\xi}, \quad \chi(p_1^0; p_1^0, w^0) = w^0.$$

The unique positive solution is

$$\chi(\xi; p_1^0, w^0) = w^0 \left(\frac{\xi}{p_1^0} \right)^\alpha.$$

Using $(1, 1)$ as the reference price vector gives the normalization

$$v(p_1, w) := \chi(1; p_1, w) = \frac{w}{p_1^\alpha}.$$

For any bundle $(x_1, x_2) \gg 0$, invert the two demand equations to recover the normalized price and wealth that generate it:

$$x_1(p, w) = \frac{\alpha w}{p_1}; \quad x_2(p, w) = (1 - \alpha)w$$

$$\frac{x_1}{x_2} = \frac{\alpha}{(1 - \alpha)} \frac{1}{p_1}$$

$$p_1 = \frac{\alpha}{1 - \alpha} \frac{x_2}{x_1}, \quad w = \frac{x_2}{1 - \alpha}.$$

Substituting these inverse-demand expressions into the normalized indirect utility gives

$$u(x_1, x_2) := v\left(\frac{\alpha}{1 - \alpha} \frac{x_2}{x_1}, \frac{x_2}{1 - \alpha}\right).$$

$$u(x_1, x_2) = \frac{\frac{x_2}{1 - \alpha}}{\left(\frac{\alpha}{1 - \alpha} \frac{x_2}{x_1}\right)^\alpha} = \left(\frac{x_2}{1 - \alpha}\right)^{1 - \alpha} \left(\frac{x_1}{\alpha}\right)^\alpha$$

$$u(x_1, x_2) = \hat{c} x_1^\alpha x_2^{1 - \alpha}, \quad \hat{c} = \frac{1}{\alpha^\alpha (1 - \alpha)^{1 - \alpha}}.$$

This is a Cobb-Douglas utility function.

3.6 GARP and Afriat's Theorem

Retain the expenditure-exhausting finite dataset $\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$ and direct revealed preference $\succeq^{R,D}$ from the preceding section. Let $\succeq^R$ be its transitive closure on the observed bundles. Thus $x^t \succeq^R x^s$ when there is a finite chain of direct weak comparisons from x^t to x^s . Define $x^t \succ^{R,D} x^s$ when $p^t \cdot x^t > p^t \cdot x^s$.

Definition 3.11. Utility rationalization of budget data. A utility function $u : X \rightarrow \mathbb{R}$ rationalizes $\mathcal{O}$ if, for every $t \in \mathbb{T}$, $u(x^t) \geq u(x)$ whenever $p^t \cdot x \leq p^t \cdot x^t$. Ties are allowed, so the observed bundle need not be the unique maximizer.

A finite demand dataset $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$ satisfies the generalized axiom of revealed preference (GARP) if

$$x^t \succeq^R x^s \implies \neg(x^s \succ^{R,D} x^t)$$

for all observations s, t , where $\succeq^R$ is the transitive closure of direct revealed preference and

$$x^s \succ^{R,D} x^t \iff p^s \cdot x^s > p^s \cdot x^t.$$

Equivalently, a chain of weak revealed-preference comparisons may not be closed by a strict direct comparison in the opposite direction.

Proposition 3.12. (*Afriat's theorem.*) Let $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$ be a finite dataset with $p^t \gg 0$ and $x^t \in \mathbb{R}_+^L$ for every observation t . The following are equivalent:

1. There exists a locally nonsatiated utility function u that rationalizes the data in the sense that for all t , $u(x^t) \geq u(x)$ for all x such that $p^t \cdot x^t \geq p^t \cdot x$.
2. The dataset $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$ satisfies GARP.
3. There exist numbers $(U^t, \lambda^t) \in \mathbb{R} \times \mathbb{R}_{++}$ satisfying the Afriat inequalities

$$U^t \leq U^s + \lambda^s p^s \cdot (x^t - x^s) \quad \forall s, t \in \{1, \dots, T\}$$

4. There exists a continuous, concave, and strictly increasing utility function u that rationalizes the data in the sense that for all t , $u(x^t) \geq u(x)$ for all x such that $p^t \cdot x^t \geq p^t \cdot x$.

Proof. (1) $\implies$ (2), (2) $\implies$ (3), (3) $\implies$ (4), (4) $\implies$ (1).

(1) implies (2),

Assume that $\mathcal{O}$ is rationalized by a locally nonsatiated utility function and suppose, toward a contradiction, that GARP fails. Then there are observation indices $t_0, \dots, t_m$ such that

$x^{t_{k-1}} \succeq^{R,D} x^{t_k}$ for $k = 1, \dots, m$ and $x^{t_m} \succ^{R,D} x^{t_0}$. Rationalization gives $u(x^{t_0}) \geq \dots \geq u(x^{t_m})$. The strict comparison implies $p^{t_m} \cdot x^{t_0} < p^{t_m} \cdot x^{t_m}$. By local nonsatiation, there is a bundle y with $\|y - x^{t_0}\|_2 < \varepsilon$ and $u(y) > u(x^{t_0})$; choosing $\varepsilon > 0$ sufficiently small preserves strict affordability at prices p^{t_m} . Thus y is affordable at observation t_m and satisfies $u(y) > u(x^{t_m})$, contradicting rationalization.

(2) implies (3)

The implication from GARP to the Afriat inequalities is the finite Afriat feasibility lemma. It is a theorem of alternatives for the displayed linear system, not an immediate consequence of naming Farkas' lemma. We cite the lemma here: for a finite, expenditure-exhausting dataset with strictly positive prices, GARP holds if and only if there exist $(U^1, \dots, U^T)$ and positive $(\lambda^1, \dots, \lambda^T)$ satisfying every Afriat inequality. The converse can be seen directly by summing the inequalities around a revealed-preference cycle; the forward direction uses the full finite feasibility lemma.

(3) implies (4)

Let $u(x) = \min_{t \in \{1, \dots, T\}} \{U^t + \lambda^t p^t \cdot (x - x^t)\}$. As the pointwise minimum of finitely many affine functions, u is continuous and concave. It is strictly increasing in the product order because $\lambda^t > 0$ and $p^t \gg 0$ for every observation t .

To verify rationalization, first observe that $u(x^t) = U^t$ for every observation t . Indeed, the Afriat inequality gives

$$U^t \leq U^m + \lambda^m p^m \cdot (x^t - x^m)$$

for every m , while the term with $m = t$ equals U^t . Therefore the minimum of these affine terms is exactly U^t .

It remains to show that $u(x) \leq u(x^s)$, if $p^s \cdot x^s \geq p^s \cdot x$

If $p^s \cdot x^s \geq p^s \cdot x$, or $p^s \cdot (x^s - x) \geq 0$, then $u(x) = \min_t \{U^t + \lambda^t p^t \cdot (x - x^t)\} \leq U^s + \lambda^s p^s \cdot (x - x^s) \leq U^s = u(x^s)$.

Thus $u(x^s) \geq u(x)$ when $p^s \cdot x^s \geq p^s \cdot x$.

(4) implies (1).

If u is strictly increasing, then it is locally nonsatiated.

□

3.7 WGARP and Coalitional Multi-Utility

Afriat's theorem asks whether all observations can be generated by maximization of one utility function. Its answer is GARP. A failure of GARP rejects that joint transitive explanation, but it need not make every pair of observations inconsistent. WGARP asks the weaker pairwise question: can the data avoid every strict two-observation reversal even when a longer revealed-preference cycle remains? Coalitional multi-utility (CMU) is the optimizing model behind that condition [Aguiar et al., 2026]. Crucially, CMU extends the benchmark model: ordinary utility maximization is exactly its singleton special case.

Definition 3.13. Budget WGARP and WARP. The finite dataset $\mathcal{O}$ satisfies WGARP if, for all $s, t \in \mathbb{T}$,

$$p^t \cdot x^t \geq p^t \cdot x^s \implies \neg(p^s \cdot x^s > p^s \cdot x^t).$$

It satisfies WARP if, whenever $x^t \neq x^s$,

$$p^t \cdot x^t \geq p^t \cdot x^s \implies p^s \cdot x^s < p^s \cdot x^t.$$

WGARP is the weak/strict condition appropriate for a demand correspondence. WARP also excludes mutual affordability at equality and corresponds to a strict, single-valued version of the model. GARP and SARP apply the analogous prohibitions after taking revealed-preference closure.

Preference-function demand. A $r : X \times X \rightarrow \mathbb{R}$ is a numerical comparison rule, with $x \succeq_r y$ when $r(x, y) \geq 0$. Its demand correspondence is

$$x_r(p, w) = \{x \in B(p, w) : r(x, y) \geq 0 \text{ for every } y \in B(p, w)\}.$$

The function is asymmetric when $r(x, y) \geq 0 \Rightarrow r(y, x) \leq 0$ and $r(x, y) > 0 \Rightarrow r(y, x) < 0$. It is strictly increasing when $x > z \Rightarrow r(x, y) > r(z, y)$ for every y . The strict part of $\succeq_r$ remains the relational strict part; it is not defined merely by $r(x, y) > 0$.

Coherent CMU. Let $C(X)$ be the continuous real-valued functions on X with the compact-open topology. A compact coalition structure Ω is a nonempty compact collection of nonempty compact sets $U \subseteq C(X)$. Its preference function is

$$r_\Omega(x, y) = \max_{U \in \Omega} \min_{u \in U} [u(x) - u(y)].$$

The structure is coherent if $U \cap V \neq \emptyset$ for every $U, V \in \Omega$. Coherence is pairwise intersection; it does not require one utility to belong to every coalition. A CMU comparison holds when at least one coalition unanimously ranks the first bundle at least as high as the second.

Proposition 3.14. *Classical utility maximization is nested exactly. Let $u \in C(X)$ and set $\Omega_u = \{\{u\}\}$. Then Ω_u is compact and coherent, $r_{\Omega_u}(x, y) = u(x) - u(y)$, and*

$$x_{r_{\Omega_u}}(p, w) = \arg \max_{x \in B(p, w)} u(x).$$

Proof. Both optimization operators in the CMU formula collapse for the singleton coalition, so $r_{\Omega_u}(x, y) = u(x) - u(y)$. Therefore $x \in x_{r_{\Omega_u}}(p, w)$ exactly when $u(x) \geq u(y)$ for every feasible y , which is the ordinary utility-maximization condition. A singleton is compact and intersects itself. $\square$

Theorem 3.15. *Reader theorem ($\star$): WGARP and coherent CMU. For a finite expenditure-exhausting dataset with $L \geq 2$, strictly positive prices, and nonzero nonnegative choices, the following are equivalent: (i) WGARP; (ii) rationalization by an asymmetric, strictly increasing, continuous preference function; (iii) rationalization by a coherent, strictly increasing, continuous CMU; and (iv) rationalization by a skew-symmetric, coherent, continuous CMU whose coalitions contain strictly increasing concave utilities.*

This is an exact finite-data characterization, not only a necessary condition. The difficult direction uses the fact that every two-observation subdataset of a WGARP dataset satisfies GARP, constructs a pairwise Afriat utility for each pair, and assembles those utilities into a coherent coalition structure. Passing WGARP does not identify nontransitivity: every GARP dataset also passes WGARP and can admit both classical and CMU rationalizations.

Theorem 3.16. *Reader theorem ($\star$): WARP and strict coherent CMU. In the same finite environment, WARP is equivalent to rationalization by an asymmetric, strictly star-concave, continuous preference function; equivalently, by a coherent strictly star-concave CMU; equivalently, by a skew-symmetric coherent CMU whose coalitions contain strictly increasing strictly concave utilities. Here strict star-concavity requires $r(\alpha x + (1 - \alpha)y, x) > \alpha r(x, x) + (1 - \alpha)r(y, x)$ for $x \neq y$ and $\alpha \in (0, 1)$.*

The WARP theorem is separate from the WGARP theorem. WARP is not the finite-data condition for maximization of one ordinary utility function: GARP is the classical weak-

rationalizability condition, and SARP is the corresponding strict acyclicity condition. The strict CMU theorem yields at most one demanded bundle on a budget when demand is nonempty, but it does not guarantee nonempty demand on every counterfactual budget.

Example 3.17. A three-good WGARP dataset that violates GARP. Let

$$\begin{aligned} p^1 &= (4, 1, 5), & p^2 &= (5, 4, 1), & p^3 &= (1, 5, 4), \\ x^1 &= (4, 1, 1), & x^2 &= (1, 4, 1), & x^3 &= (1, 1, 4). \end{aligned}$$

Then

$$[p^t \cdot x^s]_{t,s=1}^3 = \begin{pmatrix} 22 & 13 & 25 \\ 25 & 22 & 13 \\ 13 & 25 & 22 \end{pmatrix}.$$

Hence $x^1 \succ^{R,D} x^2 \succ^{R,D} x^3 \succ^{R,D} x^1$, so GARP fails. Each reverse-budget cost is $25 > 22$, however, so no pair creates a WGARP reversal. This dataset also satisfies WARP because the chosen bundles are distinct and no pair is mutually affordable. With two goods, WGARP and GARP coincide; three goods are essential for this separation under linear budgets.

Proposition 3.18. *Compensated law of demand for preference functions. Let r be asymmetric and strictly increasing, let $x \in x_r(p, w)$, and set $w' = p' \cdot x$. If $x' \in x_r(p', w')$, then*

$$(p' - p) \cdot (x' - x) \leq 0.$$

Proof. First note that strict increase and asymmetry imply budget exhaustion at every nonempty demand. If a demanded bundle had slack, a strictly larger feasible bundle z would satisfy $r(x, z) \geq 0$ by maximality, while strict increase in the first argument would give $r(z, z) > r(x, z) \geq 0$, contradicting asymmetry at (z, z) . Hence $p' \cdot (x' - x) = 0$. Suppose, toward a contradiction, that $p \cdot x' < p \cdot x$. Then some $z > x'$ remains affordable at the original budget. Since x' defeats x at the compensated budget, $r(x', x) \geq 0$. Strict increase gives $r(z, x) > 0$, and asymmetry then gives $r(x, z) < 0$, contradicting $x \in x_r(p, w)$. Therefore $p \cdot (x' - x) \geq 0$, and subtracting this inequality from $p' \cdot (x' - x) = 0$ yields the result. The claim is conditional on both demand sets being nonempty. $\square$

Remark 3.19. What survives, and what does not. Singleton CMU reproduces the UMP exactly. Under asymmetry and strict increase, nonempty CMU demand satisfies budget exhaustion,

homogeneity of degree zero, WGARP, and the compensated inequality above. WGARP or coherence alone does not imply one scalar utility, transitivity, GARP, convex preferences, nonempty counterfactual demand, Roy's identity, Shephard's lemma, Slutsky symmetry, or the standard scalar-utility welfare measures. Concavity of the component utilities does not make the induced CMU relation globally convex. An arbitrary noncoherent CMU need not satisfy WGARP.

The star marks a compressed reader theorem, not a weaker truth standard and not a Lean-verification badge. The six-way WGARP statement, the distinct WARP theorem, topology, proof boundary, and formalization record appear in Appendix B.

3.8 SARP and Strict Rationalization

The preceding section used GARP to characterize weak rationalization, allowing ties in observed choice. SARP is the acyclicity condition associated with strict or unique rationalization. The exact equivalence depends on how duplicate observed bundles and strict choice are treated, so this section states the condition without claiming that merely replacing every Afriat inequality by a strict one is sufficient.

Definition 3.20. (Strong axiom of revealed preference, SARP.) The dataset satisfies SARP if the reflexive-transitive revealed relation $\succeq^R$ has no nontrivial cycle: whenever $x^t \succeq^R x^s$ and $x^t \neq x^s$, it is not the case that $x^s \succeq^R x^t$.

SARP is the natural condition when observed choice is single-valued and ties are ruled out. The weak Afriat theorem above is the maintained finite-data result in this book; strict rationalization results require additional assumptions, including care with duplicate observed bundles.

3.9 Forecasting with Varian's Support Set

Definition 3.21. Given a finite price-choice dataset $\mathcal{O}$ and a new price-wealth pair (p^{T+1}, w^{T+1}) , define the augmented dataset associated with a candidate bundle by $\mathcal{O}^+(x) = \mathcal{O} \cup \{(p^{T+1}, x)\}$.

Varian's support set is

$$\mathcal{S}_{\mathcal{O}}(p^{T+1}, w^{T+1}) = \left\{ x \in X : \begin{array}{l} \mathcal{O}^+(x) \text{ satisfies GARP,} \\ p^{T+1} \cdot x = w^{T+1} \end{array} \right\}.$$

The definition already gives the exact criterion: a candidate bundle belongs to the support set if it exhausts the new budget and the augmented dataset satisfies GARP. The transitive closure must be checked, not only one-step comparisons. Whenever an old bundle x^t is indirectly revealed preferred to the candidate, GARP requires the inequality

$$p^{T+1} \cdot x \leq p^{T+1} \cdot x^t.$$

3.10 Upper Bounds for Welfare Analysis

Definition 3.22. Money-metric utility at reference prices p is defined by $\mu_p(x) = e(p, u(x))$ where e is the expenditure function associated with u , $e(p, \bar{u}) = \min_{z \in X} \{p \cdot z : u(z) \geq \bar{u}\}$.

For fixed p , $\mu_p(x)$ is a utility function that represents the same preferences as u but is expressed in money units.

For a counterfactual bundle x , extend direct revealed preference to the set $\{x, x^1, \dots, x^T\}$ by declaring

$$x^t \succeq_{\mathcal{O}, x}^{R, D} y \iff p^t \cdot x^t \geq p^t \cdot y, \quad y \in \{x, x^1, \dots, x^T\},$$

and let $\succeq_{\mathcal{O}, x}^R$ be its transitive closure. The data-dependent upper bound is

$$m_{\mathcal{O}}^+(p, x) = \min\{p \cdot x^t : x^t \succeq_{\mathcal{O}, x}^R x, \quad t = 1, \dots, T\},$$

when the index set is nonempty, and $m_{\mathcal{O}}^+(p, x) = +\infty$ otherwise. If u rationalizes $\mathcal{O}$, every revealed chain in the extended relation implies $u(x^t) \geq u(x)$. Monotonicity of expenditure in its utility target and feasibility of x^t then give

$$\mu_p(x) = e(p, u(x)) \leq e(p, u(x^t)) \leq p \cdot x^t.$$

Taking the minimum proves $\mu_p(x) \leq m_{\mathcal{O}}^+(p, x)$ whenever the bound is finite.

3.11 Experimental Tests of GARP

Choi, Fisman, Gale, and Kariv study 93 participants recruited from undergraduate classes and staff at the University of California, Berkeley. Each participant completed 50 portfolio-allocation problems. In every problem the participant chose a point on a linear budget over two Arrow securities, one paying in each of two states. A chosen portfolio can therefore be interpreted as a

bundle of state-contingent consumption. The 47 participants in the symmetric treatment faced $\pi_1 = \pi_2 = 1/2$; the remaining 46 faced $\{\pi_1, \pi_2\} = \{1/3, 2/3\}$. The computer varied the budget line across rounds, and each line may be normalized as $p^t \cdot x = 1$. Thus participant i generated a panel $\mathcal{O}_i = \{(p_i^t, x_i^t)\}_{t=1}^{50}$ to which GARP can be applied [Choi et al., 2007].

The exact-GARP tabulation in [Polisson et al., 2020] reports that 12 of 47 participants in the symmetric treatment and 4 of 46 in the asymmetric treatment satisfy GARP. Overall, 16 of 93 participants---approximately 17 percent---pass the exact test. The interpretation is precise but limited: a recorded panel that fails cannot be rationalized exactly by one locally nonsatiated utility function under the maintained budget data. The result alone does not distinguish preference instability from optimization error, recording error, or price misperception. Nor does a binary pass rate measure the economic magnitude of a violation: one strict revealed-preference cycle is sufficient for rejection. The critical cost-efficiency index introduced next provides a graded measure, while a random-choice benchmark helps assess the power of the design.

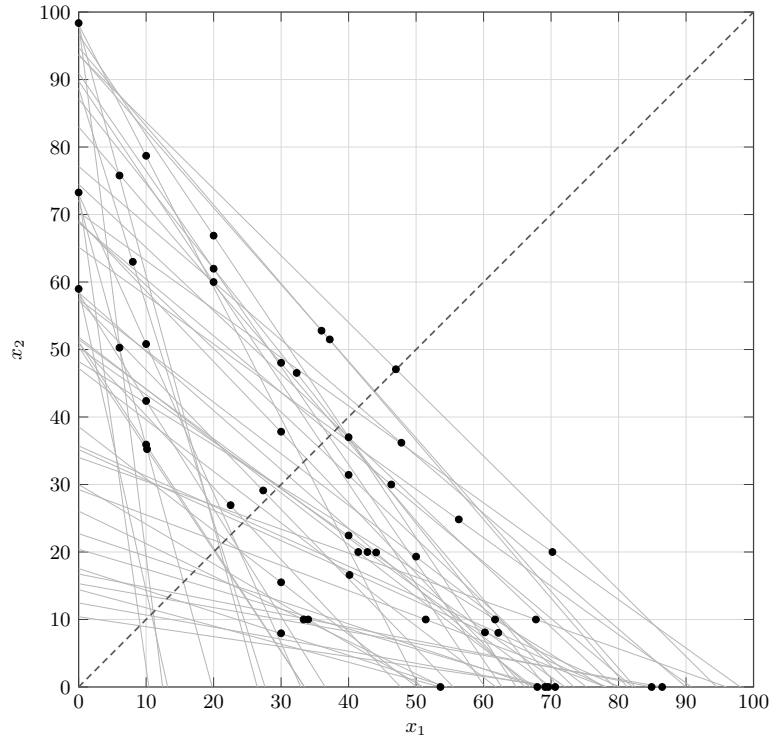


Figure 3.1: Observed portfolio choices of Subject 216 in the symmetric treatment. Gray lines are budget lines, black points are chosen portfolios, and the dashed 45-degree line marks equal consumption in the two states. Redrawn from Quah's survey using the experiment of [Quah, 2015, Choi et al., 2007].

3.12 The Critical Cost-Efficiency Index

When a dataset $\mathcal{O}$ fails GARP, the critical cost-efficiency index measures the size of the revealed-preference violation. For an efficiency level $\alpha \in [0, 1]$, write the efficiency-adjusted relations as $\succeq_\alpha^{R,D}, \succ_\alpha^{R,D}, \prec_\alpha^R$. The direct relations are defined by

$$x^t \succeq_\alpha^{R,D} x^s \iff \alpha p^t \cdot x^t \geq p^t \cdot x^s,$$

$$x^t \succ_\alpha^{R,D} x^s \iff \alpha p^t \cdot x^t > p^t \cdot x^s.$$

Let $\succeq_\alpha^R$ be the transitive closure of $\succeq_\alpha^{R,D}$. The dataset satisfies α -GARP when every pair of observations s, t obeys

$$x^t \succeq_\alpha^R x^s \implies \neg(x^s \succ_\alpha^{R,D} x^t).$$

Definition 3.23. The Critical Cost Efficiency Index (CCEI) of a dataset $\mathcal{O}$ is $\text{CCEI}(\mathcal{O}) = \sup\{\alpha \in [0, 1] : \mathcal{O} \text{ satisfies } \alpha\text{-GARP}\}$. A value of one means that the original data satisfy GARP; a smaller value is the largest efficiency level at which the adjusted revealed-preference test remains free of GARP violations. This is an expenditure-efficiency normalization, not a claim that the recorded bundle was literally purchased after discarding the remaining expenditure.

The CCEI distribution for the experiment is shown below.

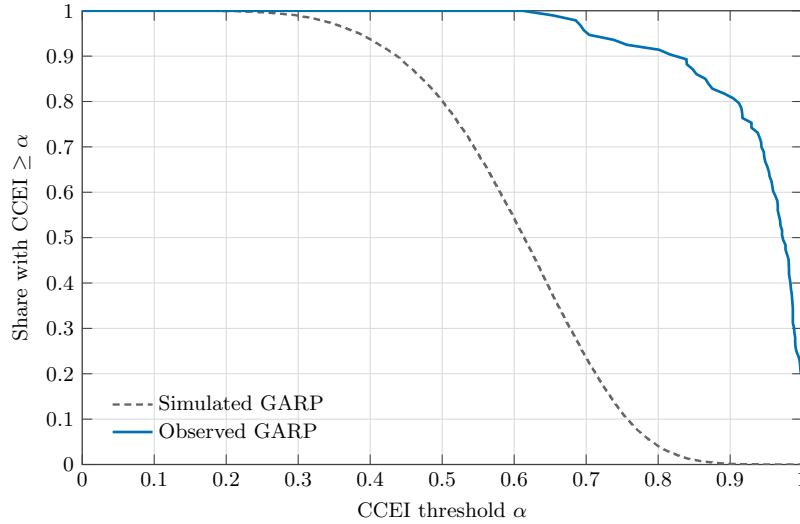


Figure 3.2: Survival functions $\alpha \mapsto \Pr\{\text{CCEI}_i \geq \alpha\}$ for the experimental participants and for Bronars-style simulated participants who choose uniformly on each budget line. Redrawn from Quah's survey; experiment and simulation design from Choi et al., with subsequent tabulation in Polisson, Quah, and Renou[Quah, 2015, Choi et al., 2007, Polisson et al., 2020].

3.13 Measurement Error and Statistical Revealed Preference

Ahn, Choi, Gale, and Kariv collect 50 portfolio choices from each of 154 participants. The assets are three Arrow securities corresponding to three states: one state has a known probability, whereas the other two states are ambiguous. This is a three-state contingent-consumption experiment, not an ordinary three-good demand experiment. Applied directly to the recorded choices and budgets, the deterministic GARP test is passed by 12.99 percent of participants[Ahn et al., 2014, Aguiar and Kashaev, 2021].

A deterministic revealed-preference test treats recorded prices and choices as the economically relevant data. Aguiar and Kashaev instead ask whether the population distribution of recorded panels can be generated by latent utility-maximizing behavior together with a disciplined measurement process[Aguiar and Kashaev, 2021]. For participant i and task t , let (p_i^t, x_i^t) denote the recorded pair and (p_i^{*t}, x_i^{*t}) the latent pair entering the maintained optimizing model. An additive relation such as $x_i^t = x_i^{*t} + \varepsilon_{x,i}^t$ or $p_i^t = p_i^{*t} + \varepsilon_{p,i}^t$ is a schematic illustration, not the identifying content of the framework. Empirical content comes from restrictions linking observed and latent variables---especially population centering conditions---combined with utility-maximization restrictions. If both latent data and measurement error are unrestricted, observed violations can be rationalized too freely for the exercise to be informative.

In their application, Aguiar and Kashaev study two measurement channels separately. Under the price-misperception specification, recorded choices are exact and perceived prices may differ from displayed prices, subject to agreement between perceived and displayed price vectors on average for each task. Under the trembling-hand specification, displayed prices are exact and chosen allocations may contain mean-zero error. The validation evidence is asymmetric: the paper reports direct evidence for the centering condition on hand errors, whereas the evidence for price misperception is indirect and tied to the graphical design.

The empirical comparison is therefore conditional. At the 5 percent significance level, the authors do not reject the joint null of static utility maximization and mean-centered price misperception. They strongly reject the joint null when the only discrepancy is a centered trembling-hand error in consumption. The data thus favor the price-misperception specification over that particular consumption-error specification. They do not establish that every participant is deterministically GARP-consistent, that price misperception is the unique source of every violation, or that utility maximization has been confirmed. Failure to reject is evidence about a

joint model of preferences and measurement, not proof of rationality.

3.14 Problems

Part I: By hand

Problem 3.1 [Core]. A two-observation WARP test.

A consumer chooses $x^1 = (2, 2)$ at $(p^1, w^1) = ((2, 1), 6)$ and $x^2 = (1, 5/2)$ at $(p^2, w^2) = ((1, 2), 6)$.

- Verify that each observed bundle exhausts its budget.
- Determine whether each bundle was affordable at the other observation.
- Does the dataset satisfy WARP for single-valued demand? Identify the two direct revealed-preference comparisons that determine your answer.

Problem 3.2 [Core]. Recovering a quasilinear utility.

Normalize $p_2 = 1$ and suppose an interior demand system satisfies $x_1(p_1, w) = \beta - \alpha p_1$, where $\alpha, \beta > 0$, and Walras' law.

- Derive $x_2(p_1, w)$ and state the domain on which both demands are nonnegative.
- Compute the normalized scalar Slutsky term for the non-numeraire demand.
- Recover a quasilinear utility representation $u(x_1, x_2) = g(x_1) + x_2$ that generates the demand on the interior domain. Distinguish general ordinal uniqueness up to an increasing transformation from the narrower normalization freedom after the coefficient on x_2 is fixed at one.

Problem 3.3 [Core]. Afriat inequalities by hand.

Consider two observations: $p^1 = (1, 1)$, $x^1 = (2, 1)$ and $p^2 = (2, 1)$, $x^2 = (1, 1)$.

- Compute observed expenditures and all direct revealed-preference comparisons.
- Check GARP.
- Verify that $U^1 = 1$, $U^2 = 0$, and $\lambda^1 = \lambda^2 = 1$ satisfy the Afriat inequalities.
- Write the corresponding piecewise-linear Afriat utility and verify that it assigns the required utility levels at the observations.

Problem 3.5 [Core]. Classical utility as singleton CMU.

Let $u : X \rightarrow \mathbb{R}$ be continuous, let $\Omega_u = \{\{u\}\}$, and define $r_{\Omega_u}(x, y) = \max_{U \in \Omega_u} \min_{v \in U} [v(x) - v(y)]$.

- Prove that $r_{\Omega_u}(x, y) = u(x) - u(y)$.

- (b) Prove that $x_{r_{\Omega_u}}(p, w) = \arg \max_{x \in B(p, w)} u(x)$.
- (c) Explain precisely which conclusions require continuity, strict monotonicity, or concavity of u , and which do not.

Part II: A counterfactual demand support set — chapter use case and verified scale-up

Problem 3.6 [Proof]. WGARP without GARP.

Consider prices $p^1 = (4, 1, 5)$, $p^2 = (5, 4, 1)$, $p^3 = (1, 5, 4)$ and choices $x^1 = (4, 1, 1)$, $x^2 = (1, 4, 1)$, $x^3 = (1, 1, 4)$.

- (a) Compute the 3×3 matrix with (t, s) entry $p^t \cdot x^s$.
- (b) Show that the data satisfy WGARP and WARP but violate GARP. Identify the revealed three-cycle and explain why no pair forms a reversal.
- (c) Suppose an asymmetric, strictly increasing preference function r rationalizes two choices $x \in x_r(p, w)$ and $x' \in x_r(p', p' \cdot x)$. Prove the compensated law of demand $(p' - p) \cdot (x' - x) \leq 0$. State where nonemptiness of demand is used.

Chapter use case

Seed: Problems 3.1, 3.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of observed budgets from two to T , with two goods and the counterfactual budget fixed. Tool: Codex + Python; Lean for candidate certificates. Verification: Recover the hand support interval and verify that added observations can only shrink it. Lean may check GARP or Afriat certificates for candidates; it does not yet certify the full support-set characterization.

Problem 3.4 [Applied]. Demand forecasting with Varian's support set.

A finite dataset $\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$ satisfies GARP. Fix a counterfactual budget (p^0, w^0) with $p^0 \gg 0$, and define $\mathcal{S}_{\mathcal{O}}(p^0, w^0) = \{x^0 : p^0 \cdot x^0 = w^0, \mathcal{O} \cup \{(p^0, x^0)\} \text{ satisfies GARP}\}$.

- (a) State a revealed-preference condition under which the augmented data imply $x^t \succeq^R x^0$.
- (b) Derive the associated linear restriction on $p^0 \cdot x^0$.
- (c) Explain why the restrictions in parts (a)–(b) are necessary but must be combined with every revealed-preference path into and out of the candidate observation to characterize $\mathcal{S}_{\mathcal{O}}(p^0, w^0)$. Explain why this set gives bounds rather than point identification.

Chapter 4

Choice Without Budget Geometry

Question: Which conclusions are properties of choice itself, rather than of linear budget geometry?

4.1 From Choice Data to Preference

The preceding chapters used the geometry of linear budgets. We now abstract from prices and wealth and retain only alternatives, feasible menus, and observed choices. This separation shows which conclusions follow from choice consistency itself and which depend on consumer-budget structure. It also supplies the preference and utility language that underlies the benchmark model used in Chapter 1.

4.2 Choice Environments

Let X be a nonempty set of alternatives and let $\mathcal{A} \subseteq 2^X \setminus \{\emptyset\}$ be the family of feasible menus.

Definition 4.1. Choice correspondence. A choice correspondence is a map $c : \mathcal{A} \rightrightarrows X$ satisfying $\emptyset \neq c(A) \subseteq A$ for every $A \in \mathcal{A}$. A choice structure is the pair $(\mathcal{A}, c)$.

Exact rationalization. A binary relation $\succeq$ rationalizes $(\mathcal{A}, c)$ if $c(A) = \{x \in A : x \succeq y \text{ for every } y \in A\}$ for every $A \in \mathcal{A}$. A rational preference relation is complete and transitive.

Remark 4.2. Replacing equality by the inclusion $c(A) \subseteq \{x \in A : x \succeq y \text{ for every } y \in A\}$ is only a selection-from-maximizers condition. By itself it has no revealed-preference content: universal indifference makes every feasible alternative greatest and therefore accommodates every nonempty

choice correspondence. Whenever the book uses weak rationalization for budget data, it instead imposes the structural conditions stated in Afriat's theorem.

4.3 Preference Relations

Definition 4.3. For a relation $\succeq \subseteq X \times X$, define its strict and symmetric parts by $x \succ y \iff x \succeq y$ and not $y \succeq x$ and $x \sim y \iff x \succeq y$ and $y \succeq x$. If $\succeq$ is complete and transitive, then $\succ$ is irreflexive and transitive, while $\sim$ is an equivalence relation. These elementary implications are the order-theoretic basis for utility representation.

Proposition 4.4. *Finite utility representation. If X is finite, a preference relation on X has a real-valued utility representation if and only if it is complete and transitive. A representation is unique up to a strictly increasing transformation of its range.*

For an infinite commodity or choice space, representation requires additional assumptions such as continuity and a suitable countability condition. The next theorem gives the continuous-utility result for the Euclidean commodity space used in Chapter 1.

4.4 Utility Representation

Definition 4.5. A preference relation $\succeq$ on $X \subseteq \mathbb{R}^L$ is continuous if its upper and lower contour sets are closed: for every $y \in X$,

$$\{x \in X : x \succeq y\} \quad \text{and} \quad \{x \in X : y \succeq x\}$$

are closed in X . Equivalently, if $x^n \rightarrow x$, then $x^n \succeq y$ for every n implies $x \succeq y$, and $y \succeq x^n$ for every n implies $y \succeq x$.

Lemma 4.6. *Every total preorder on a countable set has a real-valued utility representation.*

Proof. Let $I_\succeq$ be the symmetric part of the preorder. The quotient $X/I_\succeq$ is countable, and the relation induced on it is a linear order by Appendix A. Every countable linear order admits an order embedding into $\mathbb{Q}$: enumerate its elements and insert the image of each new element at a rational number between the images of its already embedded lower and upper neighbors. Composing this embedding with the quotient map gives $u : X \rightarrow \mathbb{Q} \subseteq \mathbb{R}$ such that $x \succeq y \iff u(x) \geq u(y)$. $\square$

Theorem 4.7. *(Continuous utility representation) If $\succeq$ is a continuous total preorder on $X = \mathbb{R}_+^L$,*

then there exists a continuous function $u : X \rightarrow \mathbb{R}$ such that, for every $x, y \in X$,

$$x \succeq y \iff u(x) \geq u(y).$$

Proof. Proof record. This is Debreu's continuous-utility theorem specialized to the Euclidean orthant; see [Debreu, 1959]. The orthant is second countable, separable, and connected, and a continuous total preorder on this space satisfies the theorem's topological hypotheses. The published proof constructs an order-dense countable family of indifference classes, assigns compatible real values to those classes, and extends the assignment continuously to the whole space. We rely on that theorem here rather than reproduce its topological construction. An arbitrary limit of a utility defined only on $\mathbb{Q}^L$ would not suffice, because existence and independence of the approximating sequence would still have to be proved. $\square$

The theorem guarantees existence, not uniqueness. If u represents $\succeq$ and f is strictly increasing on the range of u , then $f \circ u$ represents the same preference relation.

4.5 Revealed Choice and WGARP

Definition 4.8. Direct revealed preference. For a choice structure $(\mathcal{A}, c)$, write $x \succeq^C y$ if there is a menu $A \in \mathcal{A}$ with $x \in c(A)$ and $y \in A$. Write $x \succ^C y$ if, in addition, $y \notin c(A)$. Let $\succeq^{C*}$ be the reflexive-transitive closure of $\succeq^C$.

Weak Generalized Axiom of Revealed Preference (WGARP). The choice correspondence satisfies WGARP when $x \succeq^C y$ implies not $y \succ^C x$. Equivalently, if two menus contain both x and y , choosing x from one and strictly choosing y from the other is ruled out.

Proposition 4.9. *Necessity of WGARP. If a complete and transitive preference relation rationalizes a choice structure, then the structure satisfies WGARP.*

Proof. If $x \in c(A)$ and $y \in A$, rationalization gives $x \succeq y$. If a second menu strictly chose y over x , it would give $y \succ x$, contradicting the first comparison.

4.6 GARP in Abstract Choice and the Budget-Data Specialization

Definition 4.10. Generalized Axiom of Revealed Preference (GARP). A finite dataset satisfies GARP when $x \succeq^{C*} y$ never occurs together with $y \succ^C x$. Thus an indirect revealed-preference

chain may return to an alternative only when the return comparison is not strict.

Remark 4.11. Afriat preview. On finite linear-budget data with positive prices and expenditure exhaustion, GARP is equivalent to rationalization by a continuous, strictly increasing, concave utility function. Chapter 3 states the exact theorem, writes the Afriat inequalities, and constructs the rationalizing utility [Afriat, 1967, Varian, 1982]. This is a result about budget geometry and locally nonsatiated utility, not an equivalence with unrestricted selection from the greatest elements of an arbitrary relation.

WGARP is the one-step restriction for a possibly multivalued choice correspondence. GARP applies the no-strict-reversal requirement to the reflexive-transitive closure of direct revealed preference and is therefore the relevant condition when the data do not contain every pair of alternatives in a common menu. Exact single-valued rationalization can require additional tie-breaking assumptions, so the theorem above is stated for weak rationalization.

Example 4.12. Let $X = \{a, b, c\}$ and suppose $a \succ^C b$, $b \succ^C c$, and $c \succ^C a$. This strict cycle violates GARP and cannot be generated by a transitive preference relation.

The separation between the abstract choice language and the budget-set language is deliberate. In Chapter 3, prices and wealth impose the additional linear-budget structure, and the Afriat inequalities provide a finite certificate that can be checked mechanically.

4.7 Problems

Part I: By hand

Problem 4.1 [Core]. Incomplete preferences and multiple utilities.

Let $X = \{a, b, c\}$ and let R contain all reflexive pairs and the comparison aRb , but no other comparison between distinct alternatives.

- (a) Show that R is a preorder but is incomplete.
- (b) Prove that no real-valued utility function can represent R in the sense xRy if and only if $u(x) \geq u(y)$.
- (c) Construct two utility functions u_1, u_2 such that xRy if and only if $u_k(x) \geq u_k(y)$ for both $k = 1, 2$.

Part II: A menu-design rationality audit — chapter use case and verified scale-up

Problem 4.3 [Proof]. Maximization and WGARP.

Let R be complete and transitive on a finite set X . For each nonempty menu $A \subseteq X$, let $c(A)$ be the set of R -greatest elements of A .

- (a) Prove that $c(A)$ is nonempty.
- (b) Suppose $x \in c(A)$, $y \in c(B)$, and $x, y \in A \cap B$. Prove that xI_Ry , where xI_Ry means xRy and yRx .
- (c) Explain why this conclusion is WGARP, the appropriate weak axiom for a set-valued choice correspondence, and distinguish it from WARP for single-valued demand.

Chapter use case

Seed: Problem 4.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of observed menus on the fixed set $X = \{a, b, c\}$. Tool: Codex + Lean. Verification: Recover the three-menu cycle and the one-observation repair; return an exact cycle certificate or a complete transitive rationalization. The current pairwise-WARP theorem does not exclude this three-cycle; the general checker theorem is a new target.

Problem 4.2 [Core]. Choice cycles on finite menus.

Let $X = \{a, b, c\}$ and suppose a single-valued choice function satisfies $c(\{a, b\}) = a$, $c(\{b, c\}) = b$, and $c(\{a, c\}) = c$. Say that a preference relation R rationalizes these data when the observed alternative is the unique R -greatest element of each observed menu.

- (a) Write the direct revealed-preference comparisons generated by these choices.
- (b) Show that no transitive preference relation can rationalize the data.
- (c) State the smallest change to one observation that restores acyclicity, and exhibit a complete and transitive rationalization of the revised data.

Part II

Risk, Stochastic Choice, and Attention

Chapter 5

Expected Utility and Risk

Expected utility adds two ingredients to the choice framework: objective lotteries and a mixture operation. The chapter first gives the finite von Neumann--Morgenstern model and its affine uniqueness, then studies risk attitudes, stochastic dominance, portfolio choice, and revealed tests. Infinite lottery spaces are treated last because their topology and integrability assumptions are technical extensions rather than part of the finite core.

5.1 Primitives

Let $Z \subseteq \mathbb{R}$ be a finite set of monetary prizes. A lottery is a probability mass function $\mu : Z \rightarrow [0, 1]$ satisfying $\sum_{z \in Z} \mu(z) = 1$. Write $\Delta(Z)$ for the set of lotteries and δ_z for the degenerate lottery assigning probability one to z . Mixtures are defined pointwise by

$$[\alpha\mu + (1 - \alpha)\nu](z) = \alpha\mu(z) + (1 - \alpha)\nu(z), \quad \alpha \in [0, 1].$$

For a Bernoulli utility $u : Z \rightarrow \mathbb{R}$, expected utility is $U(\mu) = \sum_{z \in Z} u(z)\mu(z)$.

5.2 Expected-Utility Representation

We observe a choice correspondence on a family of subsets of $\Delta(Z)$ and represent the decision maker's preference by a relation $\succeq$ on lotteries.

Definition 5.1. Expected utility representation. The preference relation has an expected utility representation if there is a Bernoulli utility $u : Z \rightarrow \mathbb{R}$ such that $\mu \succeq \nu \iff \sum_{z \in Z} u(z)\mu(z) \geq \sum_{z \in Z} u(z)\nu(z)$.

5.3 Independence

Assume throughout this section that $\succeq$ is complete and transitive.

Axiom 5.2. (*Independence Axiom*) For all $\mu, \nu, \eta \in \Delta(Z)$ and $\alpha \in [0, 1]$, $\mu \succeq \nu$ implies $\alpha\mu + (1 - \alpha)\eta \succeq \alpha\nu + (1 - \alpha)\eta$.

The other axiom that we need is more technical

Axiom 5.3. (*Mixture continuity*) For all $\mu, \nu, \eta \in \Delta(Z)$, if $\mu \succ \nu \succ \eta$, then the set $\{\alpha \in [0, 1] | \alpha\mu + (1 - \alpha)\eta \succ \nu\}$ and $\{\alpha \in [0, 1] | \nu \succ \alpha\mu + (1 - \alpha)\eta\}$ are open in $[0, 1]$.

5.4 The Finite Expected-Utility Theorem

Theorem 5.4. The rational preference relation $\succeq$ satisfies the Independence Axiom and Mixture continuity if and only if there exists a (Bernoulli) utility function $u : Z \rightarrow \mathbb{R}$ such that: $\mu \succeq \nu$ if and only if $\sum_{z \in Z} u(z)\mu(z) \geq \sum_{z \in Z} u(z)\nu(z)$.

First we prove that there is an affine representation of $\succeq$.

Lemma 5.5. If the preference relation $\succeq$ satisfies the Independence Axiom and mixture continuity then there is an affine function $U : \Delta(Z) \rightarrow \mathbb{R}$ that represents $\succeq$.

Proof. Choose best and worst degenerate lotteries $\mu^+ = \delta_{z^+}$ and $\mu^- = \delta_{z^-}$ among the finitely many prizes. Repeated use of independence shows $\mu^+ \succeq \mu \succeq \mu^-$ for every lottery μ . If $\mu^+ \sim \mu^-$, all lotteries are indifferent and a constant affine representation works. Otherwise, independence implies

$$\beta\mu^+ + (1 - \beta)\mu^- \succ \alpha\mu^+ + (1 - \alpha)\mu^- \iff \beta > \alpha.$$

For a fixed μ , mixture continuity makes the two sets of mixture weights giving an outcome weakly above or weakly below μ closed. They cover the connected interval $[0, 1]$ and contain opposite endpoints, so they intersect. Strict monotonicity along the best--worst segment gives a unique α_μ with

$$\mu \sim \alpha_\mu\mu^+ + (1 - \alpha_\mu)\mu^-.$$

Thus $U(\mu) = \alpha_\mu$ represents $\succeq$. Finally, independence permits substitution of the equivalent best--worst mixtures for μ and ν , yielding

$$U(\beta\mu + (1 - \beta)\nu) = \beta U(\mu) + (1 - \beta)U(\nu).$$

Hence U is affine on the lottery simplex.

□

The second step is to show that an affine representation on the lottery simplex has the expected-utility form.

Lemma 5.6. *Let $U : \Delta(Z) \rightarrow \mathbb{R}$. Then U is affine if and only if there exists a $u : Z \rightarrow \mathbb{R}$ such that $U(\mu) = \sum_{z \in Z} u(z)\mu(z)$.*

For finite Z , no functional-analytic representation theorem is needed. Define $u(z) = U(\delta_z)$. Since every lottery has the finite decomposition $\mu = \sum_{z \in Z} \mu(z)\delta_z$, affinity of U gives $U(\mu) = \sum_{z \in Z} u(z)\mu(z)$. The converse follows immediately from distributivity.

Example 5.7. Let $Z = \{0, 10\}$, and let $\mu \in \Delta(Z)$ satisfy $\mu(0) = \mu(10) = \frac{1}{2}$. Then $U(\mu) = \frac{1}{2}u(10) + \frac{1}{2}u(0)$. Normalize the Bernoulli index by $u(0) = 0$ and $u(10) = 1$. The normalized expected utility is therefore $U(\mu) = 1/2$. Any positive affine transformation of this index represents the same preference.

5.5 Affine Uniqueness and Recoverability

Theorem 5.8. *Let $U : \Delta(Z) \rightarrow \mathbb{R}$ and $U' : \Delta(Z) \rightarrow \mathbb{R}$ be affine maps that represent $\succeq$ on $\Delta(Z)$, then there exist real numbers $\alpha > 0$ and b such that $U'(\mu) = \alpha U(\mu) + b$ for all $\mu \in \Delta(Z)$. If u, v are Bernoulli utilities such that $u \cdot \mu$ and $v \cdot \mu$ represent $\succeq$ then there are real numbers $\alpha > 0$ and b such that $v \cdot \mu = \alpha u \cdot \mu + b$.*

Affine uniqueness is stronger than ordinal uniqueness in deterministic choice. For an ordinary utility representation, every strictly increasing transformation preserves the ranking. Under expected utility, mixture linearity restricts admissible transformations to positive affine maps; a nonlinear increasing transformation generally destroys the independence representation.

5.6 The Allais Paradox

In this example, $Z = \{z_1, z_2, z_3\}$ with the ordered coordinate list $(z_1, z_2, z_3) = (2.5, 0.5, 0)$ million dollars and $N := |Z| = 3$. A lottery vector lists probabilities in this order, and $u_z := u(z)$.

First Prize	Second Prize	Third Prize
2.5 million	0.5 million	0

Two choice sets:

The first,

$$L_1 = (0, 1, 0) \text{ and } L'_1 = (0.10, 0.89, 0.01).$$

The second,

$$L_2 = (0, 0.11, 0.89) \text{ and } L'_2 = (0.10, 0, 0.90).$$

Some individuals choose $L_1 \succ L'_1$ and $L'_2 \succ L_2$.

This is inconsistent with expected utility.

If $L_1 \succ L'_1$ then

$$u_{0.5} > (0.10)u_{2.5} + (0.89)u_{0.5} + (0.01)u_0.$$

Subtracting $(0.89)u_{0.5}$ and adding $(0.89)u_0$ to both sides yields

$$(0.11)u_{0.5} + (0.89)u_0 > (0.1)u_{2.5} + (0.9)u_0,$$

and thus $L_2 \succ L'_2$.

5.7 Attitudes towards Risk

Assume that Bernoulli utility is defined on the convex hull $\text{co}(Z)$ of the finite monetary-prize set.

For a lottery $\mu \in \Delta(Z)$, its expected monetary value is $\bar{z}(\mu) = \sum_{z \in Z} z\mu(z) \in \text{co}(Z)$.

Definition 5.9. Risk aversion under expected utility. A decision maker is risk averse on this finite lottery domain if $u(\bar{z}(\mu)) \geq U(\mu)$ for every lottery $\mu \in \Delta(Z)$. By Jensen's inequality, concavity of the Bernoulli utility implies this condition:

$$\sum_{z \in Z} \mu(z)u(z) \leq u\left(\sum_{z \in Z} \mu(z)z\right).$$

If the same comparison is required for every finitely supported lottery with prizes in $\text{co}(Z)$, applying it to arbitrary two-point lotteries also makes concavity necessary.

Certainty equivalent and risk premium. If u is strictly increasing and its range contains $U(\mu)$, the certainty equivalent is the prize $\text{CE}(\mu; u) = u^{-1}(U(\mu))$ and the (monetary) risk premium is $\text{RP}(\mu; u) = \bar{z}(\mu) - \text{CE}(\mu; u)$. A risk-averse expected-utility decision maker has $\text{RP}(\mu; u) \geq 0$.

Probability premium. For prizes $x - \varepsilon, x, x + \varepsilon$ in the domain of u , $\text{PP}(x, \varepsilon; u)$ is defined by

$$u(x) = \left(\frac{1}{2} + \text{PP}(x, \varepsilon; u)\right)u(x + \varepsilon) + \left(\frac{1}{2} - \text{PP}(x, \varepsilon; u)\right)u(x - \varepsilon).$$

It is the change in probability needed to make the symmetric two-point lottery equivalent to the sure prize.

Proposition 5.10. *For a twice differentiable, strictly increasing Bernoulli utility, the local risk premium for a symmetric small spread satisfies*

$$\text{RP}\left(\frac{1}{2}\delta_{x-\varepsilon} + \frac{1}{2}\delta_{x+\varepsilon}; u\right) = -\frac{u''(x)}{2u'(x)}\varepsilon^2 + o(\varepsilon^2).$$

Thus local concavity corresponds to a nonnegative local risk premium.

5.8 First-Order Stochastic Dominance: Finite Support

Definition 5.11. First-order stochastic dominance. For lotteries $\mu, \nu \in \Delta(Z)$ with cumulative distribution functions $F_\mu(t) = \sum_{z \leq t} \mu(z)$ and $F_\nu(t) = \sum_{z \leq t} \nu(z)$, write $\mu \succeq^{\text{FOSD}} \nu$ if $F_\mu(t) \leq F_\nu(t)$ for every $t \in \mathbb{R}$. The relation is generally incomplete.

Theorem 5.12. *If $\mu \succeq^{\text{FOSD}} \nu$ and u is increasing, then $U(\mu) \geq U(\nu)$ for expected utility U . If the prizes are ordered as $Z = \{z_1 < \dots < z_N\}$, strict inequality follows whenever $F_\mu(z_k) < F_\nu(z_k)$ and $u(z_{k+1}) > u(z_k)$ for at least one cutoff $k < N$.*

5.9 Portfolio Choice: Demand for a Risky Asset

An asset is a divisible claim to a financial return in the future. Assume that u is continuously differentiable, strictly increasing, and concave on an interval containing every feasible terminal-wealth realization. Assume also that the return and marginal-utility expressions below are integrable.

Let α, β denote the amounts of money invested in a risky and safe asset respectively. The risky asset pays a random gross return $\tilde{R}$ dollars per dollar invested and the safe asset pays 1 dollar with certainty per dollar invested. The random return has cumulative distribution function $F_R(r)$. We assume that $\mathbb{E}[\tilde{R}] = \int r dF_R(r) > 1$, or that the risky asset expected return exceeds the return of the safe asset. Then the total return is $\alpha\tilde{R} + \beta$. Feasibility requires $\alpha + \beta = w$.

$$\max_{\alpha, \beta \geq 0} \int u(\alpha r + \beta) dF_R(r)$$

subject to $\alpha + \beta = w$.

Substituting $\beta = w - \alpha$ reduces the problem to

$$\max_{0 \leq \alpha \leq w} \int u(w + \alpha(r - 1)) dF_R(r).$$

Define the derivative of the objective on $[0, w]$ by

$$\phi(\alpha) = \int u'(w + \alpha[r - 1])[r - 1] dF_R(r).$$

The necessary and sufficient first-order conditions for an optimum α^* are

$$\begin{cases} \phi(\alpha^*) \leq 0 & \text{if } \alpha^* < w, \\ \phi(\alpha^*) \geq 0 & \text{if } \alpha^* > 0. \end{cases}$$

Because $u'(w) > 0$ and $\int r dF_R(r) > 1$, we have $\phi(0) = u'(w) \int (r - 1) dF_R(r) > 0$. Hence $\alpha^* = 0$ cannot satisfy the first-order conditions.

If a risk is actuarially favorable, then a risk-averse expected-utility maximizer will accept at least a sufficiently small amount under the stated differentiability and interior-wealth assumptions.

5.10 Revealed Expected Utility under Risk Aversion

Let $\mathcal{S}$ be a finite set of states with fixed probabilities $\varpi_s > 0$, satisfying $\sum_{s \in \mathcal{S}} \varpi_s = 1$, and let a contingent consumption plan be $x = (x_s)_{s \in \mathcal{S}} \in \mathbb{R}_{++}^{\mathcal{S}}$. At prices $p \in \mathbb{R}_{++}^{\mathcal{S}}$ and wealth $w > 0$, the consumer solves $\max_x \sum_{s \in \mathcal{S}} \varpi_s u(x_s)$ subject to $\sum_{s \in \mathcal{S}} p_s x_s \leq w$, where u is continuous, strictly increasing and concave, with positive marginal utility on the observed consumption range whenever derivatives are invoked.

Proposition 5.13. *Interior first-order conditions. If u is differentiable and an optimum is interior, then there is $\lambda > 0$ such that $\varpi_s u'(x_s) = \lambda p_s$ for every $s \in \mathcal{S}$. Concavity makes these conditions sufficient for a global optimum; boundary observations require the corresponding KKT inequalities.*

Definition 5.14. Risk-averse expected-utility rationalization. A finite dataset $\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$ is rationalized if there is one continuous, strictly increasing and concave Bernoulli utility with positive marginal utility on the observed range u such that each x^t maximizes $\sum_s \varpi_s u(x_s)$ on the budget set $\{x : p^t \cdot x \leq p^t \cdot x^t\}$.

Theorem 5.15. *Finite-data test. See [Polisson et al., 2020]. Assume all observations are interior*

and $\varpi_s > 0$. The dataset has a risk-averse expected-utility rationalization if and only if there are numbers $\beta_s^t > 0$ and $\lambda^t > 0$ that satisfy the price equations

$$\varpi_s \beta_s^t = \lambda^t p_s^t \quad \text{for every } s, t,$$

and the monotonicity inequalities

$$x_s^t > x_{s'}^{t'} \implies \beta_s^t \leq \beta_{s'}^{t'}$$

for every pair of observed state-contingent quantities. The monotonicity condition is the finite signature of a single decreasing marginal-utility function; the price equations are the common first-order conditions.

The theorem is an Afriat-style interpolation result for one Bernoulli utility across states and observations. It is stated for interior data to keep the notation transparent; zero quantities are handled by replacing equalities with KKT inequalities.

5.11 Experimental Evidence on Expected Utility

Polisson and Quah develop a revealed preferences test for expected utility. They apply their framework to the famous Choi, Fisman, Gale and Kariv (2007) dataset and find the following passing rates.

- GARP is the benchmark revealed-preference test.
- RDEU denotes rank-dependent expected utility.
- EU denotes expected utility.
- EU* imposes concavity of the Bernoulli utility function in addition to expected utility.

The “drop” column reports how many observations the test permits the researcher to discard.

Observations dropped	GARP	RDEU	EU	EU*
0	12/47 (26%)	2/47 (4%)	2/47 (4%)	0/47 (0%)
1	14/47 (30%)	7/47 (15%)	5/47 (11%)	0/47 (0%)
2	27/47 (57%)	9/47 (19%)	7/47 (15%)	2/47 (4%)
3	32/47 (68%)	12/47 (26%)	10/47 (21%)	2/47 (4%)

Figure 5.1: Share of 47 subjects passing each revealed-preference test after dropping up to three observations [Polisson and Quah, 2013].

5.12 Measuring Risk Aversion in the Lab

The Holt and Laury multiple-price-list task classifies each subject by the total number of safe (Option A) choices across ten decisions.

Assume constant relative risk aversion (CRRA), with coefficient r : $u(x) = x^{1-r}/(1-r)$. At $r = 1$, use $u(x) = \log x$. A monotone pattern with k initial Option A choices followed by Option B implies an interval for r .

Number of safe choices	CRRA interval for $U(x) = x^{1-r}/(1-r)$	Proportion of choice patterns	
		Low real ($\times 1$) ^a	Real ($\times 20$)
0–1	$r < -0.95$	.01	.01
2	$-0.95 < r < -0.49$	.01	.01
3	$-0.49 < r < -0.15$	.06	.04
4	$-0.15 < r < 0.15$	.26	.13
5	$0.15 < r < 0.41$	.26	.19
6	$0.41 < r < 0.68$	.23	.23
7	$0.68 < r < 0.97$	.13	.22
8	$0.97 < r < 1.37$	.03	.11
9–10	$1.37 < r$	.01	.06

^aAverage over the first and second low-real decisions.

Figure 5.2: Distribution of the number of safe choices in the Holt–Laury task. The final two columns give proportions of observed choice patterns classified by total number of Option A choices under low real ($\times 1$) and real $\times 20$ payoffs. The low-real column averages the first and second low-real decisions [Holt and Laury, 2002].

Individuals choose between A and B in each of the ten rows.

Each option has two monetary outcomes, A_1, A_2 and B_1, B_2 ; their common high-outcome probability rises by one tenth from one row to the next.

The expected payoff of A exceeds that of B in rows 1–4; the inequality reverses in rows 5–10.

- A risk-neutral expected-utility maximizer chooses A in rows 1, ..., 4 and B in rows 5, ..., 10.
- Option B has higher variance in each nondegenerate row (rows 1–9).
- Under expected utility and CRRA, five or more initial safe choices imply $r > 0$, and a later single switch to B corresponds to a larger risk-aversion parameter. Choosing A in row 10 violates monotonicity because both options are then certain and $3.85 > 2.00$.

One row is selected at random for payment after all choices are recorded. This random-lottery

incentive system makes every row potentially payoff relevant; interpreting the switching row as a risk-aversion measure additionally requires expected utility, monotonicity, and the stated payoff probabilities.

5.13 Expected Utility on Infinite Outcome Spaces

Let Z be a compact metric prize space and let Σ be its Borel σ -algebra. Recall that a collection of subsets is a σ -algebra when

- $\emptyset \in \Sigma$
- if $E \in \Sigma$ then $E^c \in \Sigma$
- If $\{A_n\}$ is a countable collection of sets such that $A_n \in \Sigma$ then $\cup_n A_n \in \Sigma$.

We say that (Z, Σ) is a measurable space whenever Σ is a σ -algebra.

Definition 5.16. We say that $\mu : \Sigma \rightarrow [0, 1]$ is a probability measure if:

- $\mu(Z) = 1$
- if $\{A_n\}$ is a countable collection of pairwise disjoint sets such that $A_n \in \Sigma$, then $\mu(\cup_n A_n) = \sum_n \mu(A_n)$.

We call the collection of probability measures $\Delta(Z, \Sigma)$ for short $\Delta(Z)$ when Σ is fixed.

$\alpha\mu + (1 - \alpha)\nu \in \Delta(Z)$ is defined as $(\alpha\mu + (1 - \alpha)\nu)(E) = \alpha\mu(E) + (1 - \alpha)\nu(E)$.

Equip this space with weak convergence: $\mu^n \Rightarrow \mu$ if $\int f d\mu^n \rightarrow \int f d\mu$ for continuous and bounded functions f .

Axiom 5.17. (*Weak continuity*) For any $\mu \in \Delta(Z)$ the sets $\{\nu \in \Delta(Z) : \nu \succ \mu\}$ and $\{\nu \in \Delta(Z) : \mu \succ \nu\}$ are open in the weak topology.

Theorem 5.18. Let Z be a compact metric prize space and $\Delta(Z)$ the Borel probability measures with the weak topology. A rational preference relation $\succeq$ on $\Delta(Z)$ satisfies the independence axiom and weak continuity if and only if there exists a $u : Z \rightarrow \mathbb{R}$ bounded and continuous such that $U(\mu) = \int u(z)d\mu(z)$ represents $\succeq$.

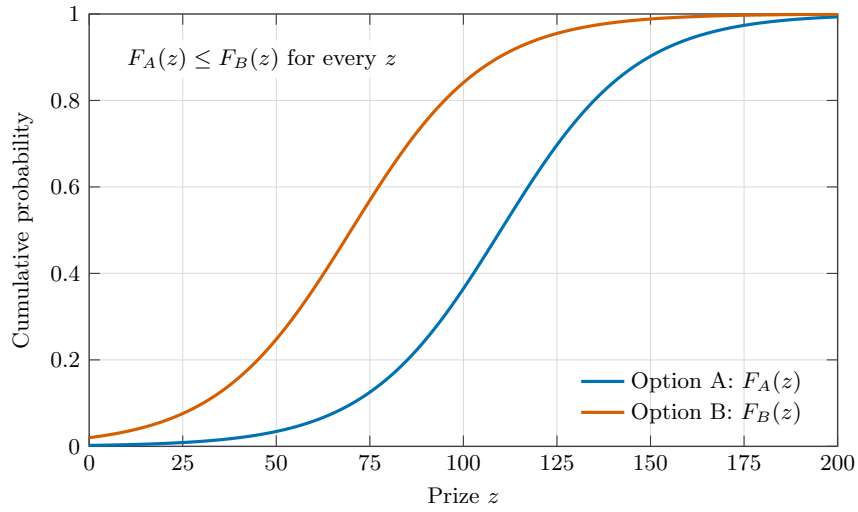


Figure 5.3: First-order stochastic dominance. Option A is a rightward location shift of Option B, so $F_A(z) \leq F_B(z)$ for every payoff z .

5.14 First-Order Stochastic Dominance: General Distributions

This section extends the finite-prize result proved earlier in the chapter to probability distributions with integrable utility.

Definition 5.19. Let F and G be cumulative distribution functions on $\mathbb{R}$. Write $F \succeq^{\text{FOSD}} G$ when $F(r) \leq G(r)$ for every $r \in \mathbb{R}$; strict dominance means that the inequality is strict at some continuity point.

Theorem 5.20. *First-order stochastic dominance. If $F \succeq^{\text{FOSD}} G$, then $\int u dF \geq \int u dG$ for every nondecreasing measurable Bernoulli utility u for which both expectations are finite. If $F \neq G$ and u is strictly increasing, the inequality is strict.*

5.15 Problems

Part I: By hand

Problem 5.1 [Core]. Affine expected-utility representations.

Let $Z = \{z_1, \dots, z_n\}$ and suppose $U(\mu) = \sum_j \mu_j u(z_j)$ represents a nontrivial preference on $\Delta(Z)$.

- (a) Show directly that U is affine in lotteries.

- (b) Prove that $U' = aU + b$ with $a > 0$ represents the same preference.
- (c) Show that if two expected-utility representations agree on preferences, their Bernoulli indices differ by a positive affine transformation.

Problem 5.2 [Core]. First-order stochastic dominance.

Lotteries μ and ν on prizes $\{0, 1, 2\}$ have probabilities $\mu = (0.2, 0.3, 0.5)$ and $\nu = (0.3, 0.4, 0.3)$.

- (a) Compare their cumulative distribution functions and determine which lottery first-order stochastically dominates the other.
- (b) Prove the expected-utility ranking for every increasing Bernoulli utility by discrete summation by parts.
- (c) Explain why strict dominance need not imply a strict preference for a weakly increasing utility.

Part II: A multi-asset portfolio under risk — chapter use case and verified scale-up

Problem 5.3 [Proof]. The Arrow-Pratt approximation.

An agent with twice continuously differentiable utility u and wealth w faces a zero-mean risk ε with variance σ^2 . Define the monetary risk premium RP by the indifference equation

$$u(w - \text{RP}) = \mathbb{E}[u(w + \varepsilon)].$$

- (a) Use second-order Taylor expansions to derive the approximation

$$\text{RP} \approx \frac{1}{2}A(w)\sigma^2,$$

where $A(w) = -u''(w)/u'(w)$.

- (b) State the regularity and small-risk qualifications needed for this approximation.
- (c) Explain why the formula is invariant to positive affine transformations of u .

Chapter use case

Seed: Problems 5.3, 5.4. Fixed: all unlisted primitives and assumptions. Scaled: the number of jointly normal risky assets from one to K , retaining CARA utility and unconstrained positions. Tool: Codex + Python. Verification: Recover the one-asset formula; verify the first-order condition and concavity using the covariance matrix; report singular-covariance cases explicitly and do not add portfolio constraints silently.

Problem 5.4 [Applied]. CARA-normal portfolio demand.

Terminal wealth is $W_T = wR_f + \theta(\tilde{R} - R_f)$, where $\tilde{R} \sim N(m_R, \sigma^2)$, R_f is risk free, $\theta \in \mathbb{R}$

(borrowing and short selling are unrestricted), and utility is $u(W_T) = -\exp(-aW_T)$ with $a > 0$.

- (a) Show that maximizing expected utility is equivalent to maximizing a certainty equivalent.
- (b) Derive optimal risky investment θ^* .
- (c) Give comparative statics with respect to $m_R - R_f$, a , and σ^2 .

Chapter 6

Stochastic Choice and Random Utility

Deterministic choice predicts a set of maximizers from each menu; stochastic choice records a probability distribution over that menu. We first define a coherent choice-probability system on a fixed finite universe, then ask when it is a mixture of deterministic rankings. Multinomial logit is presented afterward as a parametric special case, not as the definition of random utility.

6.1 Stochastic Choice on a Finite Universe

Let Y be a finite universe of alternatives and let $\mathcal{B} = 2^Y \setminus \{\emptyset\}$ be its collection of nonempty menus. A stochastic choice rule is a family $P = \{P_B\}_{B \in \mathcal{B}}$ with $P_B \in \Delta(B)$ for every menu. Thus $P_B(a) \geq 0$ for $a \in B$, $\sum_{a \in B} P_B(a) = 1$, and unavailable alternatives receive probability zero. The family, rather than a probability from only one menu, is the empirical object to be rationalized. A deterministic choice function is the special case in which every P_B is degenerate.

6.2 Random Utility Representation

Definition 6.1. Random-utility representation. Let Y be the finite universe, let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\mathbf{U} : \Omega \rightarrow \mathbb{R}^Y$ be a measurable random utility vector. For every nonempty menu $B \subseteq Y$, the induced choice probabilities are $P_B(a) = \mathbb{P}\{\mathbf{U}(a) > \mathbf{U}(b) \text{ for every } b \in B \setminus \{a\}\}$. Assume either that ties occur with probability zero or that a measurable tie-breaking rule is specified; under the zero-tie convention the displayed strict-max probabilities sum to one.

When choice probabilities are observed on every nonempty menu, the deterministic types form the finite set $\mathcal{R}_Y$ of strict linear orders on Y . Stacking the probabilities gives $\mathbf{P} = M\lambda$, where $\lambda \in \Delta(\mathcal{R}_Y)$ is a distribution over those orders and $M_{(a,B),r} = \mathbb{I}\{a \text{ is } r\text{-best in } B\}$. The finite feasibility condition $\mathbf{P} = M\lambda$ with $\lambda \geq 0$ and $\mathbf{1}_{|\mathcal{R}_Y|}^\top \lambda = 1$ is equivalent to the existence of such a distribution.

6.3 A Parametric Discrete-Choice Specification

Index alternatives by $j \in \{0, 1, \dots, J\}$, with $j = 0$ denoting an outside option, and consumers by i . Let x_j be observed attributes, p_j price, y_i income, V_j systematic utility, and ε_{ij} an unobserved preference shock. Conditional indirect utility is

$$U_{ij}^* = V_j(x_j, p_j, y_i) + \varepsilon_{ij}.$$

The corresponding probability is

$$P_{ij} = \Pr\{U_{ij}^* \geq U_{ik}^* \text{ for every available } k\}.$$

Ties occur with probability zero if every pairwise utility difference is atomless; a joint shock density is a familiar sufficient condition. A continuous marginal distribution for each shock alone is not sufficient when shocks may be dependent.

If the outside good has price p_0 and quantity z , the budget identity $p_j + p_0 z = y_i$ yields $z = (y_i - p_j)/p_0$ after alternative j is purchased. This reduction is valid only when the implied outside-good quantity is feasible.

6.4 Multinomial Logit and IIA

Suppose the shocks are independent and identically distributed Type-I extreme value with scale $\tau > 0$. Then the multinomial-logit probabilities are

$$P_{ij} = \frac{\exp(V_j/\tau)}{\sum_{k=0}^J \exp(V_k/\tau)}.$$

Proposition 6.2. *Independence of irrelevant alternatives. Under multinomial logit, for any two alternatives j, k that remain available, $\frac{P_{ij}}{P_{ik}} = \exp((V_j - V_k)/\tau)$ is independent of the other alternatives in the menu.*

Example 6.3. Red-bus/blue-bus. If walking and a red bus initially have equal probabilities, adding a blue bus that is nearly identical to the red bus should split the bus share while leaving walking's share close to its original value. Multinomial logit instead preserves the walking-to-red-bus odds ratio, illustrating the empirical content and limitation of IIA.

The multinomial probit model replaces the extreme-value shocks with a jointly normal vector. Its pairwise differences are generally correlated, so it does not impose IIA.

6.5 Random Expected Utility

Definition 6.4. Random expected utility. For a finite prize set Z and lotteries $\mu \in \Delta(Z)$, a random Bernoulli utility $\mathbf{u} : \Omega \rightarrow \mathbb{R}^Z$ induces $\mathbf{U}_\omega(\mu) = \sum_{z \in Z} \mathbf{u}_\omega(z) \mu(z)$. Choice probabilities are obtained by maximizing this random expected utility on each menu of lotteries.

The representation imposes linearity in mixtures and regularity under menu expansion. Extremeness and weak-star continuity are additional conditions in the Gul–Pesendorfer characterization; see[Gul and Pesendorfer, 2006].

6.6 Attributes, Choice Probabilities, and Consumer Surplus

With observed attributes $x = (x_a)_{a \in A}$, define the mean-utility vector $v(x) = (v(a, x_a))_{a \in A} \in \mathbb{R}^A$. A common specification is

$$U(a, x) = v(a, x_a) + \varepsilon_a,$$

where the distribution of $\varepsilon = (\varepsilon_a)_{a \in A}$ is independent of x . For a d -dimensional linear-index model, let $x_a, \beta \in \mathbb{R}^d$ and write $v(a, x_a) = x_a^\top \beta$. Independence is a maintained exogeneity assumption, not a consequence of the random-utility definition. For any mean-utility vector $v = (v(a))_{a \in A} \in \mathbb{R}^A$, define the conditional choice probability $P_A(a \mid v) := \mathbb{P}\{v(a) + \varepsilon_a > v(b) + \varepsilon_b \text{ for every } b \in A \setminus \{a\}\}$ under the same zero-tie convention as above. Conditional on attributes, $P_A(a \mid x)$ is shorthand for $P_A(a \mid v(x))$.

Definition 6.5. Expected maximum utility. The expected value as a function of the mean-utility vector is

$$\mathcal{V}(v) = \mathbb{E} \left[\max_{a \in A} \{v(a) + \varepsilon_a\} \right], \quad v \in \mathbb{R}^A.$$

If the deterministic utility contains $-\alpha p_a$ with $\alpha > 0$ as the price coefficient, the publication-safe money-metric object is the change between two scenarios, $\Delta CS = [\mathcal{V}(v^1) - \mathcal{V}(v^0)]/\alpha$. The level $\mathcal{V}(v)/\alpha$ depends on the additive utility normalization. If shocks are merely decision errors rather than preference shocks, this expression is not a welfare measure.

Proposition 6.6. *Williams--Daly--Zachary envelope result. See [McFadden, 1981]. If the shock distribution has finite first moments and ties occur with probability zero, then $\mathcal{V}$ is convex and differentiable in the vector of mean utilities, and $\frac{\partial \mathcal{V}(v)}{\partial v(a)} = P_A(a | v)$. If choice probabilities are differentiable in mean utilities (equivalently, if $\mathcal{V}$ is twice differentiable at the point), then the Jacobian of choice probabilities is the Hessian of $\mathcal{V}$, hence symmetric and positive semidefinite. Translation invariance implies that the rows and columns sum to zero.*

The envelope identity is the basis for likelihood and welfare calculations in discrete-choice models. Additional sign restrictions such as monotone substitution patterns require additional assumptions on the shock distribution and utility index; they do not follow from random utility alone.

6.7 Problems

Part I: By hand

Problem 6.1 [Core]. Logit odds and IIA.

Choice probabilities satisfy $P(j | A) = \exp(V_j/\tau) / \sum_{k \in A} \exp(V_k/\tau)$, where $\tau > 0$.

- (a) Derive the odds ratio $P(j | A)/P(k | A)$.
- (b) Prove the independence-of-irrelevant-alternatives property.
- (c) Explain precisely which part of IIA is an empirical restriction and which part is an algebraic implication of the model.

Problem 6.3 [Core]. A finite random-utility mixture.

There are alternatives a, b, c . A population consists of three strict preference types: $a \succ b \succ c$ with mass 0.5, $c \succ a \succ b$ with mass 0.2, and $b \succ a \succ c$ with mass 0.3.

- (a) Compute choice probabilities from menus $\{a, b\}$ and $\{a, b, c\}$.
- (b) Let the columns of M correspond to all six strict orders on $\{a, b, c\}$, with zero masses allowed. Stack the observed probabilities as $\mathbf{P}$ and write $\mathbf{P} = M\lambda$.
- (c) Explain why nonnegativity of λ and $\mathbf{1}_6^\top \lambda = 1$ are the central finite random-utility feasibility restrictions.

Part II: Red bus, blue bus, and duplicated products — chapter use case and verified scale-up

Seed: Problems 6.1, 6.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of observationally identical bus labels under independent-logit shocks. Tool: Codex + Python; Lean for the IIA core. Verification: Recover the one-bus and two-bus probabilities, verify that probabilities sum to one, and separate the verified logit odds result from any shared-shock comparison model.

Problem 6.2 [Applied]. The red-bus blue-bus problem.

Walking and a red bus initially have systematic utilities normalized so that each has probability one half under logit. A blue bus with the same systematic utility as the red bus is added.

- (a) Compute the new logit probabilities.
- (b) Compare the total bus share before and after entry.
- (c) Explain why a model with highly correlated bus-specific shocks would predict a different substitution pattern.

Chapter 7

Limited Attention and Costly Information

The benchmark chapters assume that the decision maker observes the relevant environment and optimizes without cognitive friction. This chapter separates two departures from that benchmark. Sparse-max demand takes distorted or incomplete price perception as given. Rational inattention instead makes information acquisition a choice: the decision maker trades the value of a more informative signal against its cost. Keeping these mechanisms distinct clarifies both their empirical content and their connection to the stochastic-choice models of the preceding chapter.

7.1 The Sparse-Max Decision Rule

Let $X = \mathbb{R}_+^L$, true prices $p \gg 0$, perceived prices $\hat{p} \gg 0$, and wealth $w > 0$. Assume u is differentiable and that the tangency system below has a solution. A sparse-max choice is obtained by first selecting a bundle that satisfies the perceived-price first-order condition

$$\nabla u(x) = \lambda \hat{p}, \quad \lambda > 0,$$

and then choosing λ so that the selected bundle satisfies the actual budget equation $p \cdot x = w$. At a boundary solution, replace the equality by the corresponding KKT inequalities. This two-stage rule is written $x^{\text{SM}}(p, \hat{p}, w)$. It is not the ordinary utility-maximization problem with $\hat{p}$ in the budget constraint.

Suppose instead that u is strictly concave and differentiable and that the relevant solutions

are interior. For a sparse-max tangency bundle x , set $\tilde{w} = \hat{p} \cdot x$. The perceived-price first-order condition is then sufficient for the unique global optimum, so

$$x = x^{\text{FI}}(\hat{p}, \tilde{w}).$$

Conversely, any full-information bundle $x^{\text{FI}}(\hat{p}, \tilde{w})$ satisfying $p \cdot x^{\text{FI}}(\hat{p}, \tilde{w}) = w$ obeys the sparse-max rule. If the map $\tilde{w} \mapsto p \cdot x^{\text{FI}}(\hat{p}, \tilde{w})$ is continuous, strictly increasing, and has w in its range, this $\tilde{w}$ is unique. This is a change of representation, not an assertion that the consumer faces $\hat{p}$ in the market.

7.2 Cobb--Douglas Sparse-Max Demand

Example 7.1. Let $u(x) = \sum_{\ell=1}^L \alpha_{\ell} \log x_{\ell}$ with $\alpha_{\ell} > 0$ and $\sum_{\ell} \alpha_{\ell} = 1$. The perceived-price first-order conditions give

$$x_{\ell} = \frac{\alpha_{\ell}}{\lambda \hat{p}_{\ell}}.$$

Imposing the actual budget yields $\lambda = \frac{1}{w} \sum_{k=1}^L \alpha_k \frac{p_k}{\hat{p}_k}$ and therefore

Example 7.2.

$$x_{\ell}^{\text{SM}}(p, \hat{p}, w) = \frac{\alpha_{\ell} w / \hat{p}_{\ell}}{\sum_{k=1}^L \alpha_k p_k / \hat{p}_k}.$$

Indeed, $\sum_{\ell} p_{\ell} x_{\ell}^{\text{SM}} = w$. If $\hat{p} = p$, this reduces to the ordinary Cobb--Douglas demand $x_{\ell} = \alpha_{\ell} w / p_{\ell}$.

7.3 Perceived Prices and Limited Attention

One parsimonious attention rule combines the true price with a default price $p^d \gg 0$:

$$\hat{p}_{\ell}(p) = a_{\ell} p_{\ell} + (1 - a_{\ell}) p_{\ell}^d, \quad a_{\ell} \in [0, 1].$$

The vector $a = (a_1, \dots, a_L)$ measures attention to price components. Full attention is $a_{\ell} = 1$ for every ℓ ; setting $a_{\ell} = 0$ makes that component equal to its default. With $A = \text{diag}(a_1, \dots, a_L)$, $\hat{p}(p) = Ap + (I_L - A)p^d$.

Proposition 7.3. *Comparative statics under perceived prices. Suppose the tangency bundle can be written as a differentiable map $r(\hat{p}, \lambda)$ and $x^{\text{SM}}(p, w) = r(\hat{p}(p), \lambda(p, w))$, where $\lambda(p, w)$ is*

determined by $p \cdot r(\hat{p}(p), \lambda) = w$. Then the chain rule gives

$$D_p x^{\text{SM}} = D_{\hat{p}} r A + \partial_{\lambda} r (\nabla_p \lambda)^{\top}, \quad D_w x^{\text{SM}} = \partial_{\lambda} r \partial_w \lambda.$$

The second terms are the response of the multiplier needed to restore the actual budget. Define the sparse-max and full-information Slutsky matrices by $S^{\text{SM}} = D_p x^{\text{SM}} + D_w x^{\text{SM}} (x^{\text{SM}})^{\top}$ and $S^{\text{FI}} = D_p x^{\text{FI}} + D_w x^{\text{FI}} (x^{\text{FI}})^{\top}$. Consequently, an identity such as $S^{\text{SM}} = S^{\text{FI}} A$ requires additional compensation and normalization assumptions; it is not a generic implication of the attention rule.

Because the default price is fixed, scaling (p, w) by a common positive factor generally changes $\hat{p}(p)$ and therefore can violate homogeneity of degree zero. WARP, Slutsky symmetry, and welfare theorems must likewise be checked for the specified behavioral demand rather than imported from the rational benchmark.

7.4 Rational Inattention: Choosing an Information Structure

Sparse-max demand begins with perceived prices. Rational inattention explains why a decision maker may choose to observe only a noisy summary of the state. Let $\mathcal{S}$ be a finite state space, let $\mathcal{A}$ be a finite action set, and let $\mu_0 \in \Delta(\mathcal{S})$ have full support. The payoff from action a in state s is $v(a, s)$. An information-and-decision strategy can be written directly as a stochastic choice rule $q(a \mid s)$, with $q(\cdot \mid s) \in \Delta(\mathcal{A})$ for every state. This direct formulation absorbs both the signal and the action chosen after each signal.

Definition 7.4. Shannon information cost. The unconditional action probability induced by q is $q(a) := \sum_{s \in \mathcal{S}} \mu_0(s) q(a \mid s)$. The mutual information between the state and the chosen action is

$$\mathcal{I}(q) = \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} \mu_0(s) q(a \mid s) \log \left(\frac{q(a \mid s)}{q(a)} \right),$$

The sum may equivalently be taken only over actions with $q(a) > 0$. A term with $q(a \mid s) = 0 < q(a)$ is defined as zero; because the prior has full support, $q(a) = 0$ implies $q(a \mid s) = 0$ for every state, so that inactive action contributes zero. The cost of the channel is $\kappa \mathcal{I}(q)$, where $\kappa > 0$ is the marginal cost of information.

The canonical rational-inattention problem is

$$\max_{q(\cdot|s) \in \Delta(\mathcal{A})} \left\{ \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} \mu_0(s) q(a|s) v(a, s) - \kappa \mathcal{I}(q) \right\}.$$

The first term is expected payoff and the second is the cost of making actions state-dependent. If $q(a|s) = q(a)$ for every (a, s) , then actions reveal no information about the state and $\mathcal{I}(q) = 0$. This formulation follows the Shannon-cost rational-inattention model [Sims, 2003, Matějka and McKay, 2015].

Proposition 7.5. *Rational-inattention choice probabilities. At an optimum, every action with $q(a) > 0$ satisfies*

$$q(a|s) = \frac{q(a) \exp\{v(a, s)/\kappa\}}{\sum_{b \in \mathcal{A}} q(b) \exp\{v(b, s)/\kappa\}} \quad \text{for every } s \in \mathcal{S}.$$

Conversely, a feasible channel satisfying these equations together with the consistency conditions $q(a) = \sum_s \mu_0(s) q(a|s)$ solves the concave program on its active support.

Proof. Attach a multiplier to $\sum_a q(a|s) = 1$ for each state and differentiate the objective with respect to a positive $q(a|s)$. Mutual information has derivative $\mu_0(s) \log(q(a|s)/q(a))$. The first-order condition therefore makes $\log(q(a|s)/q(a))$ equal to $v(a, s)/\kappa$ plus a state-specific normalizing constant. Exponentiating and imposing that conditional probabilities sum to one gives the displayed formula. Concavity supplies sufficiency on the active support. $\square$

The formula resembles multinomial logit, but its baseline weight $q(a)$ is endogenous and must satisfy a fixed-point condition. Thus random utility and rational inattention can produce similar choice probabilities for different structural reasons: the former randomizes tastes, whereas the latter optimally limits state information. The distinction matters for welfare and counterfactual information policies.

7.5 A Gaussian Precision-Choice Illustration

Let the unknown scalar state be $\theta \sim \mathcal{N}(\theta^d, \sigma_\theta^2)$ and let the signal be $Y = \theta + \varepsilon$ with independent $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. Under quadratic loss $\mathcal{L}(a, \theta) = \frac{1}{2}(a - \theta)^2$, the optimal action is the posterior mean

$$a^*(y) = \mathbb{E}[\theta | Y = y] = \gamma y + (1 - \gamma)\theta^d, \quad \gamma = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2} \in [0, 1].$$

The posterior variance is $(1 - \gamma)\sigma_\theta^2$, so expected loss at the optimal action is $\frac{1}{2}(1 - \gamma)\sigma_\theta^2$. If attention itself is costly, represent the cost by an increasing function $\mathcal{C}(\gamma)$ and solve

$$\max_{0 \leq \gamma \leq 1} -\frac{1}{2}(1 - \gamma)\sigma_\theta^2 - \mathcal{C}(\gamma).$$

For example, with $\mathcal{C}(\gamma) = \kappa\gamma^2/2$, the solution is $\gamma^* = \min\{1, \sigma_\theta^2/(2\kappa)\}$. Here $\gamma = 1$ denotes the limiting noiseless signal $\sigma_\varepsilon^2 = 0$. Cost specifications that diverge as $\gamma \uparrow 1$, such as the logarithmic specification below, instead guarantee an interior precision. The comparative statics therefore depend on the cost specification.

A capacity formulation is also transparent. If the information cost is $-\frac{1}{2}\log(1 - \gamma)$ and capacity is bounded by K , then $-\frac{1}{2}\log(1 - \gamma) \leq K$ implies $\gamma \leq 1 - e^{-2K}$. Since the value of information is increasing in γ , the capacity-constrained optimum attains this upper bound.

7.6 Group Thresholds under Noisy Signals

Consider a firm deciding whether to hire an applicant. Productivity is Q and hiring yields payoff Q , while rejecting yields 0. Before observing an applicant-specific signal, the firm observes $G \in \{0, 1\}$ and has prior $Q \mid G = g \sim \mathcal{N}(q_g, \sigma_q^2)$ with $q_0 < q_1$.

The firm observes $Y = Q + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is independent. The posterior mean is

$$\mu_g(y) = \mathbb{E}[Q \mid Y = y, G = g] = \gamma y + (1 - \gamma)q_g, \quad \gamma = \frac{\sigma_q^2}{\sigma_q^2 + \sigma_\varepsilon^2}.$$

Proposition 7.6. *Optimal hiring rule. Conditional on (y, g) , hire if and only if $\mu_g(y) > 0$; otherwise reject. For $\gamma > 0$ the signal threshold is $y_g = -(1 - \gamma)q_g/\gamma$.*

If both group means are negative, the lower-mean group has the higher signal threshold. Hiring rates must nevertheless be computed from the group-conditional distributions because $Y \mid G = g$ has mean q_g ; the groups share a noise variance, not a common signal distribution. If information is chosen, parameterize the decision by precision or by the informativeness index $\gamma \in [0, 1]$ and solve $\max_{0 \leq \gamma \leq 1} \mathbb{E}[\max\{\mu_g(Y), 0\}] - \mathcal{C}(\gamma)$ with an increasing attention cost $\mathcal{C}$. Greater noise variance means less attention, so a cost increasing in σ_ε^2 would reverse the intended economic tradeoff. Group-specific attention requires the group priors and a fully specified cost function.

7.7 Problems

Part I: By hand

Problem 7.1 [Core]. Sparse-max quasilinear demand.

Let $u(x_1, x_2) = \sqrt{x_1} + x_2$. The consumer uses perceived prices $\hat{p} \gg 0$ in the first-order condition but must satisfy the actual budget $p \cdot x = w$.

- (a) Derive the interior sparse-max choice.
- (b) State the condition under which the implied x_2 is nonnegative.
- (c) Show that ordinary Marshallian demand is recovered when $\hat{p} = p$.

Problem 7.2 [Core]. CES demand under perceived prices.

Let $u(x) = [\alpha x_1^\rho + (1 - \alpha)x_2^\rho]^{1/\rho}$, where $0 < \alpha < 1$ and $\rho < 1$, $\rho \neq 0$.

- (a) Use perceived prices $\hat{p}$ to derive the chosen ratio x_1/x_2 .
- (b) Scale the bundle so that it exhausts the actual budget $p \cdot x = w$.
- (c) Identify separately the perceived-price composition effect of changing $\hat{p}$ and the feasibility effect of changing p . Explain why the first is not automatically a Hicksian substitution effect.

Part II: Paying for information before choosing — chapter use case and verified scale-up

Seed: Problem 7.3. Fixed: all unlisted primitives and assumptions. Scaled: the common number N of equally likely states and matching actions, beginning at $N = 2$. Tool: Codex + Python. Verification: Recover the binary optimum, enforce conditional-probability and information-cost identities, and label any symmetry restriction used in the N -state extension. Problems 7.1 and 7.2 are comparison models, not derivations of rational inattention.

Problem 7.3 [Applied]. A binary rational-inattention problem.

There are two equally likely states and two actions. The decision maker receives payoff one from the correct action and zero from the incorrect action. A symmetric strategy is correct with probability $q \in [1/2, 1]$ and pays Shannon-information cost $\kappa[\log 2 - H(q)]$, where $H(q) = -q \log q - (1 - q) \log(1 - q)$ and $\kappa > 0$.

- (a) Write expected net payoff as a function of q .
- (b) Derive the interior optimum and show that $q^* = \exp(1/\kappa)/(1 + \exp(1/\kappa))$.
- (c) Describe the limits as $\kappa \downarrow 0$ and $\kappa \uparrow \infty$.

Part III

Aggregation and Markets

Chapter 8

Aggregate Demand and Representative Consumers

8.1 Three Questions About Aggregation

- When does aggregate demand depend on prices and total wealth rather than on the entire wealth distribution?
- When does aggregate demand satisfy a demand restriction such as WARP?
- When can aggregate demand be given a welfare interpretation?

Let consumers be indexed by $i = 1, \dots, I$. Consumer i has Marshallian demand $x_i(p, w_i)$ at common prices $p \in \mathbb{R}_{++}^L$ and wealth $w_i > 0$. Aggregate demand is

$$x^A(p, \mathbf{w}) = \sum_{i=1}^I x_i(p, w_i),$$

where $\mathbf{w} = (w_1, \dots, w_I)$. Total wealth is $W = \sum_i w_i$. Without additional structure, two wealth distributions with the same W can generate different aggregate bundles.

8.2 Gorman Aggregation

Definition 8.1. Gorman form. Consumer i has Gorman-form indirect utility if there are a consumer-specific intercept and a common positive wealth coefficient, $a_i(p), b(p)$ with $b(p) > 0$,

such that

$$v_i(p, w_i) = a_i(p) + b(p)w_i.$$

Proposition 8.2. *Gorman aggregation. See [Gorman, 1953, Mas-Colell et al., 1995]. Suppose each v_i has the Gorman form and is differentiable. Roy's identity gives*

$$x_i(p, w_i) = -\frac{\nabla_p a_i(p) + w_i \nabla_p b(p)}{b(p)}.$$

Therefore aggregate demand takes the form

$$x^A(p, \mathbf{w}) = A(p) + B(p)W,$$

where $A(p) = -\sum_i \nabla_p a_i(p)/b(p)$ and $B(p) = -\nabla_p b(p)/b(p)$. Aggregate demand consequently depends on (p, W) and not on the distribution of wealth.

Differentiating the displayed Gorman form yields the common wealth derivative

$$D_{w_i} x_i(p, w_i) = B(p) \quad \text{for every consumer } i.$$

This equality is necessary for the displayed Gorman representation; no global converse is asserted without separate integrability and domain hypotheses.

8.3 Aggregate Demand and WARP

Even if every individual demand satisfies WARP, aggregate demand need not do so. Aggregate WARP compares two price--total-wealth pairs while holding fixed the rule that assigns total wealth across consumers; the corresponding compensation need not leave each consumer's wealth unchanged.

Fix a wealth-allocation rule $\mathbf{w}(p, W) \in \mathbb{R}_+^I$ satisfying $\sum_i w_i(p, W) = W$, and define the induced price--total-wealth demand by $x^A(p, W) := x^A(p, \mathbf{w}(p, W))$. Every use of $x^A(p, W)$ below holds this rule fixed.

Definition 8.3. Aggregate WARP. A single-valued aggregate demand $x^A(p, W)$ satisfies WARP if, for distinct observations (p, W) and (p', W') , $p \cdot x^A(p', W') \leq W$ implies $p' \cdot x^A(p, W) > W'$ whenever $x^A(p, W) \neq x^A(p', W')$.

Proposition 8.4. *Sufficient condition for aggregate WARP. Suppose wealth shares are fixed and*

strictly positive, $w_i = \alpha_i W$, where $\alpha_i > 0$ and $\sum_i \alpha_i = 1$. Suppose every individual demand is single-valued, homogeneous of degree zero, and satisfies Walras' law. Assume also that, for every normalized price pair $q, q' \gg 0$,

$$(q - q') \cdot [x_i(q, \alpha_i) - x_i(q', \alpha_i)] \leq 0,$$

with strict inequality whenever the two individual demanded bundles differ. Then the aggregate demand formed with these fixed shares satisfies aggregate WARP.

Proof. Set $q = p/W$ and $X(q) = \sum_i x_i(q, \alpha_i)$. Homogeneity gives $x^A(p, W) = X(q)$, while Walras' law gives $q \cdot X(q) = 1$. Summing the individual inequalities yields

$$(q - q') \cdot [X(q) - X(q')] \leq 0,$$

and the inequality is strict whenever $X(q) \neq X(q')$. If $q \cdot X(q') \leq 1$ and the aggregate bundles differ, then

$$0 > (q - q') \cdot [X(q) - X(q')] = 2 - q \cdot X(q') - q' \cdot X(q),$$

so $q' \cdot X(q) > 2 - q \cdot X(q') \geq 1$. This is aggregate WARP on the normalized domain and hence on the original price--wealth domain. Fixed shares are essential: if shares change with prices or after compensation, the summation argument does not apply. $\square$

8.4 Representative Consumers and Welfare

Definition 8.5. Positive representative consumer. A preference relation $\succeq^{rep}$ is a positive representative consumer for x^A if its Walrasian demand equals $x^A(p, W)$ for every price--wealth pair in the domain. This is a statement about demand representation, not about welfare.

Normative representative consumer. Fix a social welfare function $\mathcal{W} : \mathbb{R}^I \rightarrow \mathbb{R}$ and individual indirect utilities v_i . A nonnegative wealth allocation $\mathbf{w}(p, W)$ is welfare-optimal if it solves

$$\max_{\mathbf{w} \in \mathbb{R}_+^I} \mathcal{W}(v_1(p, w_1), \dots, v_I(p, w_I)) \quad \text{subject to} \quad \sum_i w_i \leq W.$$

. A positive representative consumer is normative relative to $\mathcal{W}$ only when the wealth rule used to form x^A is welfare-optimal at every (p, W) .

Gorman aggregation can deliver a positive representative consumer, but it does not select a social welfare function or justify interpersonal utility comparisons. Normative claims require the

welfare criterion and the wealth-allocation rule to be stated explicitly.

8.5 Problems

Part I: By hand

Problem 8.1 [Core]. Gorman aggregation.

Suppose consumer i has indirect utility $v_i(p, w_i) = a_i(p) + b(p)w_i$, with the same positive function b for every consumer.

- (a) Apply Roy's identity to derive individual demand.
- (b) Show that aggregate demand depends on the wealth distribution only through total wealth $W = \sum_i w_i$.
- (c) Construct a representative indirect utility that generates aggregate demand.

Problem 8.2 [Core]. Aggregation under common homothetic preferences.

All consumers have the same homothetic preferences, strictly positive wealth $w_i > 0$, and a single-valued demand. Normalize the unit-wealth demand as $x(p, 1)$.

- (a) Show that $x_i(p, w_i) = w_i x(p, 1)$.
- (b) Derive aggregate demand and identify a representative consumer.
- (c) Explain why common homotheticity is sufficient but not necessary for exact aggregation.

Part II: Aggregation when household wealth moves with prices — chapter use case and verified scale-up

Seed: Problems 8.1, 8.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of consumers, with each wealth rule and preference class fixed. Tool: Codex + Python; Lean for the finite-sum core. Verification: Verify the finite aggregation identity and isolate every wealth-revaluation term. Do not describe the derivative or Slutsky bridge as Lean verified.

Problem 8.3 [Proof]. Why uncompensated aggregate price derivatives need not be symmetric.

Let $x^A(p) = \sum_i x_i(p, w_i(p))$, where individual wealth may vary with prices through endowments or profit shares.

- (a) Differentiate x^A with respect to prices.
- (b) Separate compensated substitution terms from wealth-revaluation terms.

(c) Explain why symmetry and negative semidefiniteness of each individual Slutsky matrix do not generally imply the same properties for the uncompensated derivative $D_p x^A$.

Part IV

Production and Equilibrium

Chapter 9

Theory of the Firm

This chapter treats technology as a feasible set of net-output vectors. Profit maximization and cost minimization are dual ways of valuing that set at prices; aggregation and efficiency follow from the same support-function geometry.

9.1 Production sets

There are $L \geq 1$ commodities. A production plan $y \in \mathbb{R}^L$ records net outputs: $y_\ell > 0$ is an output and $y_\ell < 0$ is an input. The production set $Y \subseteq \mathbb{R}^L$ contains exactly the technologically feasible plans.

The benchmark assumptions are: Y is nonempty and closed; $0 \in Y$ (inaction); and free disposal, $y \in Y$ and $y' \leq y$ imply $y' \in Y$. A no-free-lunch condition rules out $y \in Y \cap \mathbb{R}_+^L$ other than 0. Convexity of Y is the convex-technology assumption. Together with $0 \in Y$, it guarantees feasibility of proportional contractions, but it does not impose decreasing returns to scale for expansions.

If a differentiable transformation function satisfies $Y = \{y : F(y) \leq 0\}$ and $F(\bar{y}) = 0$ with $F_k(\bar{y}) \neq 0$, the implicit-function theorem gives the local slope of the transformation frontier:

$$\frac{dy_k}{dy_\ell} = -\frac{F_\ell(\bar{y})}{F_k(\bar{y})}.$$

The minus sign is essential: increasing one net output requires a compensating reduction in the other when the frontier is written as $F(y) = 0$.

9.2 Production functions and input substitution

With M outputs and $L - M$ inputs, write outputs as $q \in \mathbb{R}_+^M$ and input use as $\mathbf{z} \in \mathbb{R}_+^{L-M}$. In the single-output case a production function f describes free disposal through $q \leq f(\mathbf{z})$. If f is differentiable, the marginal rate of technical substitution between inputs ℓ and k is

$$\text{MRTS}_{\ell k}(\mathbf{z}) = \frac{f_\ell(\mathbf{z})}{f_k(\mathbf{z})},$$

the amount of input k needed to offset a marginal reduction in input ℓ while holding output fixed. Concavity of f makes the input requirement set $\{\mathbf{z} : f(\mathbf{z}) \geq q\}$ convex.

9.3 Profit maximization

Definition 9.1. For a price vector $p \in \mathbb{R}^L$, the profit function is

$$\pi_Y(p) = \sup_{y \in Y} p \cdot y,$$

and the supply correspondence is $\mathcal{Y}_Y(p) = \arg \max_{y \in Y} p \cdot y$ whenever the supremum is finite and attained. If Y is closed and unbounded, finiteness and attainment require separate assumptions.

Proposition 9.2. *Profit-function properties. See [Hotelling, 1932, Mas-Colell et al., 1995]. The function π_Y is convex and homogeneous of degree one in prices. The supply correspondence is homogeneous of degree zero; if Y is convex, it is convex-valued. At a differentiability point, Hotelling's lemma gives the unique supply vector $y_Y(p) = \nabla_p \pi_Y(p)$. If π_Y is twice differentiable, $D_p y_Y(p) = D_p^2 \pi_Y(p)$ is symmetric positive semidefinite and $D_p y_Y(p)p = 0$.*

The non-differentiable law of supply is the monotonicity inequality

$$(p - p') \cdot (y - y') \geq 0$$

for $y \in \mathcal{Y}_Y(p)$ and $y' \in \mathcal{Y}_Y(p')$. It follows by adding the two profit-maximization inequalities.

For a single-output technology, $q = f(\mathbf{z})$, let $p_q > 0$ be the scalar output price and $\mathbf{w} \gg 0$ the input-price vector. Profit maximization is $\max_{\mathbf{z} \geq 0} \{p_q f(\mathbf{z}) - \mathbf{w} \cdot \mathbf{z}\}$. At a differentiable interior optimum, $p_q \nabla f(\mathbf{z}^*) = \mathbf{w}$; with nonnegative inputs the KKT conditions are

$$p_q f_\ell(\mathbf{z}^*) \leq w_\ell, \quad z_\ell^* (w_\ell - p_q f_\ell(\mathbf{z}^*)) = 0.$$

9.4 Cost minimization

Definition 9.3. Given input prices $\mathbf{w} \gg 0$ and a target $q > 0$, the cost function is

$$C(\mathbf{w}, q) = \min_{\mathbf{z} \geq 0} \{\mathbf{w} \cdot \mathbf{z} : f(\mathbf{z}) \geq q\}.$$

Assume the minimum is finite and attained. The conditional factor-demand correspondence is the argmin set $\mathcal{Z}(\mathbf{w}, q)$. At a differentiable optimum satisfying a constraint qualification, the complete KKT system is

$$\begin{aligned} \mathbf{z}^* &\geq 0, \quad f(\mathbf{z}^*) \geq q, \quad \lambda \geq 0, \\ w_\ell - \lambda f_\ell(\mathbf{z}^*) &\geq 0, \quad z_\ell^*(w_\ell - \lambda f_\ell(\mathbf{z}^*)) = 0, \\ \lambda(f(\mathbf{z}^*) - q) &= 0. \end{aligned}$$

Proposition 9.4. *Cost-function properties. See [Shephard, 1953, Mas-Colell et al., 1995]. The cost function is homogeneous of degree one and concave in input prices, nondecreasing in the output target, and—when f is concave—convex in q . If the cost function is differentiable, Shephard's lemma gives the unique conditional factor demand $\mathbf{z}(\mathbf{w}, q) = \nabla_{\mathbf{w}} C(\mathbf{w}, q)$. If, in addition, the output constraint binds, the relevant constraint qualification holds, and the value function is differentiable in q , then the envelope theorem gives $\partial_q C(\mathbf{w}, q) = \lambda$. If twice differentiable, $D_{\mathbf{w}} \mathbf{z} = D_{\mathbf{w}}^2 C$ is symmetric negative semidefinite and $D_{\mathbf{w}} \mathbf{z} \mathbf{w} = 0$ by price homogeneity.*

The firm's output choice can be written as $\max_{q \geq 0} \{p_q q - C(\mathbf{w}, q)\}$. If the cost function is differentiable, an interior optimum satisfies $p_q = \partial_q C(\mathbf{w}, q)$; at $q = 0$ the corresponding one-sided inequality applies.

9.5 Aggregation and efficiency

For firms $j = 1, \dots, J$ with production sets Y_j , the aggregate technology is the Minkowski sum

$$Y = \sum_{j=1}^J Y_j = \left\{ \sum_{j=1}^J y_j : y_j \in Y_j \right\}.$$

Proposition 9.5. *If each firm's profit maximum is finite and attained, then*

$$\pi_Y(p) = \sum_{j=1}^J \pi_{Y_j}(p), \quad \mathcal{Y}_Y(p) = \sum_{j=1}^J \mathcal{Y}_{Y_j}(p)$$

where the second equality is understood as equality of attainable aggregate supply sets. The first equality follows directly from separability of the maximization over the Minkowski sum.

Definition 9.6. A feasible production plan $y \in Y$ is efficient if there is no $y' \in Y$ with $y' \geq y$ and $y' \neq y$.

Proposition 9.7. *If $p \gg 0$ and $y \in \mathcal{Y}_Y(p)$, then y is efficient. If Y is convex and closed and satisfies free disposal, every efficient plan is supported by some nonzero $p \geq 0$; this supporting-price statement is a separating-hyperplane theorem and may allow zero prices.*

9.6 Problems

Part I: By hand

Problem 9.1 [Core]. Cobb-Douglas production and cost.

A firm has production function $f(\mathbf{z}) = z_1^\alpha z_2^\beta$ with $\alpha, \beta > 0$.

- (a) Classify returns to scale as a function of $\alpha + \beta$.
- (b) Derive conditional factor demands and the cost function for output $q > 0$ and input prices $\mathbf{w} \gg 0$.
- (c) Verify that cost is homogeneous of degree one and concave in $\mathbf{w}$.

Problem 9.3 [Core]. Constant returns and unbounded profit.

Assume Y is a cone and $0 \in Y$.

- (a) Prove that either $\pi_Y(p) = 0$ or $\pi_Y(p) = +\infty$.
- (b) Characterize the price vectors for which profit is zero.
- (c) Explain why a competitive constant-returns firm can operate at any scale only when the zero-profit condition holds.

Part II: Input substitution in a multi-input firm — chapter use case and verified scale-up

Problem 9.2 [Proof]. Dual properties of the profit function.

Let $Y \subseteq \mathbb{R}^L$ be nonempty, closed, and convex. Assume that profit $\pi_Y(p) = \sup_{y \in Y} p \cdot y$ is finite on a neighborhood of the price under study and that the supremum is attained.

- (a) Prove that π_Y is convex and homogeneous of degree one in prices.
- (b) If supply is unique at p , use a subgradient argument to prove Hotelling's lemma and explain why the profit function is differentiable there.

- (c) State what changes when supply is set-valued.

Chapter use case

Seed: Problem 9.1. Fixed: all unlisted primitives and assumptions. Scaled: the input dimension from two to L , with the output target and technology family fixed. Tool: Codex + Python; Lean for finite algebra. Verification: Recover the two-input cost benchmark; check Shephard derivatives, Hessian symmetry, negative semidefiniteness, and the homogeneity null direction; state the analytic assumptions not encoded in Lean.

Problem 9.4 [Proof]. Shephard's lemma and substitution.

Suppose the minimum in $C(\mathbf{w}, q) = \min\{\mathbf{w} \cdot \mathbf{z} : f(\mathbf{z}) \geq q, \mathbf{z} \geq 0\}$ exists, conditional factor demand $\mathcal{Z}(\mathbf{w}, q)$ is single-valued, and C is twice differentiable in $\mathbf{w}$.

- (a) Prove Shephard's lemma.
- (b) Show that $D_{\mathbf{w}}\mathbf{z} = D_{\mathbf{w}\mathbf{w}}^2 C$ is symmetric and negative semidefinite.
- (c) Use homogeneity to show $D_{\mathbf{w}}\mathbf{z}(\mathbf{w}, q)\mathbf{w} = 0$.

Chapter 10

Competitive Markets and Partial Equilibrium

This chapter collects the competitive-equilibrium definitions used later and develops a transparent partial-equilibrium model with a numeraire. The partial-equilibrium model is deliberately specialized: its quasilinear preferences remove income effects for the good of interest.

10.1 Competitive equilibrium

There are I consumers, J firms, and L goods. Consumer i has consumption set $X_i \subseteq \mathbb{R}^L$, preference relation $\succeq_i$, and endowment $\omega_i \in \mathbb{R}^L$. Firm j has production set $Y_j \subseteq \mathbb{R}^L$, where positive coordinates are net outputs and negative coordinates are inputs. Let $\omega = \sum_i \omega_i$.

Definition 10.1. Feasibility. An allocation $(x_1, \dots, x_I; y_1, \dots, y_J)$ is feasible when $x_i \in X_i$, $y_j \in Y_j$, and

$$\sum_{i=1}^I x_i \leq \omega + \sum_{j=1}^J y_j.$$

Equality is imposed when all goods have strictly positive prices and free disposal is not used.

Competitive (Walrasian) equilibrium. A nonzero nonnegative price vector $p \geq 0$ and an allocation (x^*, y^*) form an equilibrium. Consumer i owns share $\theta_{ij} \geq 0$ of firm j , with $\sum_i \theta_{ij} = 1$ for every firm. Every firm solves

$$y_j^* \in \arg \max_{y_j \in Y_j} p \cdot y_j$$

and every consumer weakly prefers their assignment to every affordable bundle:

$$x_i^* \succeq_i x_i \quad \text{for every } x_i \in X_i \text{ with } p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*.$$

Markets clear exactly: $\sum_i x_i^* = \omega + \sum_j y_j^*$.

Proposition 10.2. *Price normalization and Walras' law. If (p, x^*, y^*) is a competitive equilibrium, then so is $(\lambda p, x^*, y^*)$ for every $\lambda > 0$. If consumers exhaust their budgets, summing the budget equalities and using market clearing gives*

$$p \cdot \left(\sum_i x_i^* - \omega - \sum_j y_j^* \right) = 0.$$

Therefore, if all but one market clear, the last market clears as well.

10.2 Partial equilibrium with a numeraire

There are two goods. Good x is the good of interest and m is a numeraire with price one.

Consumer i has

$$u_i(m_i, x_i) = m_i + \phi_i(x_i), \quad X_i = \mathbb{R} \times \mathbb{R}_+,$$

where ϕ_i is increasing, strictly concave, twice continuously differentiable, and bounded above.

Let $p > 0$ be the price of x and let ω_{mi} denote consumer i 's initial numeraire endowment..

Firm j produces $q_j \geq 0$ units of x at numeraire cost $c_j(q_j)$. Its net-output set is

$$Y_j = \{(-z_j, q_j) : q_j \geq 0, z_j \geq c_j(q_j)\}.$$

Assume c_j is increasing, strictly convex, twice continuously differentiable, and $c'_j(q) \rightarrow \infty$ as $q \rightarrow \infty$.

Definition 10.3. Partial competitive equilibrium. A price p and quantities $(x_1, \dots, x_I; q_1, \dots, q_J)$ form an equilibrium when

$$\phi'_i(x_i) \leq p, \quad x_i > 0 \Rightarrow \phi'_i(x_i) = p,$$

$$p \leq c'_j(q_j), \quad q_j > 0 \Rightarrow p = c'_j(q_j),$$

and the market for x clears:

$$\sum_i x_i = \sum_j q_j.$$

These are the KKT conditions for consumer and firm optimization plus market clearing.

Equivalently, the unique choices are characterized by

$$x_i(p) = \arg \max_{x \geq 0} \{\phi_i(x) - px\}, \quad q_j(p) = \arg \max_{q \geq 0} \{pq - c_j(q)\}.$$

The associated aggregate excess demand is $z(p) = \sum_i x_i(p) - \sum_j q_j(p)$.

Proposition 10.4. *Existence and uniqueness in the partial model. Suppose $0 < \min_j c'_j(0) < \max_i \phi'_i(0)$ and the preceding curvature and tail assumptions hold. Then z is continuous on the displayed positive-price interval, demand is nonincreasing, and supply is nondecreasing. A clearing price $p^* \in [\min_j c'_j(0), \max_i \phi'_i(0)]$ exists. The maintained strict curvature makes the aggregate crossing strict whenever a side is active, so the equilibrium price is unique.*

To see the sign change, at a price no larger than $\min_j c'_j(0)$ all firms choose zero output while at least one consumer demands a positive amount. At a price at least $\max_i \phi'_i(0)$ all consumers choose zero while at least one firm supplies a positive amount. The intermediate value theorem then gives a clearing price.

The numeraire allocation is recovered from the budget identity

$$m_i = \omega_{mi} + \sum_j \theta_{ij} (pq_j - c_j(q_j)) - px_i.$$

Partial equilibrium deliberately suppresses the wealth effects that would reappear with general preferences or additional markets.

10.3 Welfare in the partial model

Definition 10.5. Pareto efficiency. A feasible allocation is Pareto efficient if no other feasible allocation makes every consumer weakly better off and at least one consumer strictly better off.

Proposition 10.6. *First welfare theorem for the quasilinear model. See [Mas-Colell et al., 1995]. A competitive allocation is Pareto efficient under the stated convexity and local nonsatiation assumptions.*

The planner's problem is

$$\max_{x_i, q_j \geq 0} \left\{ \sum_i \phi_i(x_i) - \sum_j c_j(q_j) \right\} \quad \text{subject to} \quad \sum_i x_i \leq \sum_j q_j.$$

Its KKT multiplier μ satisfies $\phi'_i(x_i) \leq \mu$ with equality for $x_i > 0$ and $\mu \leq c'_j(q_j)$ with equality for $q_j > 0$. Thus a competitive equilibrium satisfies the planner conditions with $\mu = p$.

Proposition 10.7. *Second welfare theorem for the quasilinear model. See [Mas-Colell et al., 1995]. Under concavity, continuity, and the cheaper-bundle condition stated in the general theorem below, any solution of the planner's problem can be decentralized at a supporting price $p = \mu$ by lump-sum transfers of the numeraire whose sum is zero. The theorem concerns allocations or utility vectors attainable by the planner; an arbitrary utility vector need not be feasible.*

10.4 Problems

Part I: By hand

Problem 10.1 [Core]. Linear partial equilibrium.

Demand and supply are $Q^d(p) = a - bp$ and $Q^s(p) = d(p - c)$ on their economically relevant ranges, with $a, b, d > 0$, $c \geq 0$, and $a > bc$.

- (a) Solve for equilibrium price and quantity.
- (b) Compute consumer and producer surplus using the choke price a/b of inverse demand and the price intercept c of inverse supply, so that the usual triangular-surplus geometry applies.
- (c) Verify that the competitive quantity maximizes total surplus.

Part II: Music sales and concerts — chapter use case and verified scale-up

Problem 10.2 [Applied]. A per-unit tax.

Introduce a per-unit tax $t > 0$ paid by sellers into the market of Problem 10.1, so buyers pay p_b and sellers receive $p_s = p_b - t$. Assume t is small enough that equilibrium quantity remains nonnegative.

- (a) Solve for p_b , p_s , and quantity.
- (b) Derive tax incidence. Express the local burden shares using the demand and supply slopes, and interpret them using point elasticities evaluated at the common no-tax equilibrium.
- (c) Compute deadweight loss and its derivative with respect to t .

Problem 10.3 [Proof]. Planner and competitive allocation.

A representative consumer has quasilinear utility $u(q) + m$, with u increasing and strictly concave, and production cost $C(q)$ is increasing and convex.

- (a) State and solve the planner's problem using Kuhn-Tucker conditions.
- (b) Show that any interior competitive equilibrium satisfies the same condition.
- (c) State the assumptions required for the converse decentralization claim.

Chapter use case

Seed: Problems 1.2, 9.1, 10.1, 10.3. Fixed: all unlisted primitives and assumptions. Scaled: the music-volume or royalty share τ from the two benchmark values to a grid. Tool: Codex + Python. Verification: Reproduce the demand and cost formulas and both endpoint comparisons; bracket every stationary point, check both price boundaries, establish the global maximizer, and keep the volume-share and royalty-share scenarios distinct.

Problem 10.4 [Applied]. The Economics of Superstars: Music Sales and Concerts.

Taylor S. is a stylized superstar whose fans consume music services and concerts. This is the graduate version of the music-and-concerts case, with explicit preferences, technology, and market assumptions. All numbers are hypothetical; the model is not an empirical description of Taylor Swift's contracts. Quantities are divisible, and the mass of identical fans is one.

Part I. Fan demand. Write q_M, q_C for music and concert services, the two coordinates of consumption. A fan has wealth $w > 0$, prices $p_M, p_C > 0$, and utility $u(q_M, q_C) = (\frac{1}{2}q_M^\rho + \frac{1}{2}q_C^\rho)^{1/\rho}$, where $0 < \rho < 1$. The letters M, C are local commodity labels.

(a) Formulate the budget problem and derive demands using $\sigma = 1/(1 - \rho)$. For the remaining parts set $\rho = 1/2$, $w = 1$, and $p_M = 1$. Explain why music demand changes when the concert price changes.

Part II. Production. For $j \in \{M, C\}$ the artist's technology is $y_j = A_j \sqrt{n_j}$, where $n_j \geq 0$ is labor and $A_j > 0$ productivity. Labor costs $\nu > 0$ per hour, distinct from fan wealth w . The processes are independent. This stylized music-service cost does not describe the cost of creating an original recording as a function of its streams.

(b) Derive $C_j(\nu, y_j)$, average cost, and marginal cost. Explain decreasing returns. Set $A_M = A_C = \nu = 1$ in the numerical parts.

Part III. A share of volume and receipts. An exogenous $\tau \in (0, 1]$ is the fraction of music consumption attributable to this artist. Her output is $y_M = \tau q_M$ and her music receipts are $\tau p_M q_M$. Concert output is $y_C = q_C$. She takes p_M, τ as given and chooses $p_C > 0$. There is no shared labor constraint or fixed cost.

(c) Write total revenue and profit after substituting demand and costs. Distinguish a revenue derivative with respect to ticket price from marginal revenue with respect to output. Account for

changes in both revenue streams and production costs.

(d) Derive the first-order condition for the optimal concert price. Examine profit as $p_C \downarrow 0$ and $p_C \rightarrow \infty$. Establish global optimality and uniqueness, analytically or with a justified numerical method accounting for every stationary point and boundary.

(e) Compare $\tau = 1$ and $\tau = 1/2$. In a table report optimal concert prices, both fan quantities, both delivered outputs, revenues, costs, and profits. State numerical accuracy. Does the concert price rise or fall when the music share falls?

(f) Use the first-order condition or implicit differentiation to analyze the price response for $\tau \in [1/2, 1]$. Interpret the roles of substitution and production costs. Do not assume the same sign for every technology or parameter value.

Part IV. A royalty share. Now τ changes only the fraction of music revenue paid to the artist: music output is $y_M = q_M$, while receipts remain $\tau p_M q_M$. All other assumptions are unchanged.

(g) Write the revised profit function and repeat the comparison between $\tau = 1$ and $\tau = 1/2$. Identify the changed cost channel and determine whether the earlier price response survives.

(h) Explain what evidence distinguishes a fall in music volume from a fall in the royalty fraction. Discuss the roles of CES substitution, decreasing returns, and control of concert prices. Contrast the artist's decision with price-taking supply in the main chapter.

Chapter 11

General Equilibrium Theory

General equilibrium studies all markets simultaneously. Prices determine individual budget sets, firms' feasible transformations, and the wealth generated by ownership claims; an equilibrium is a fixed point at which every agent optimizes and all markets clear.

11.1 Pure exchange economies: the Edgeworth box

Consider two consumers $i = 1, 2$ and two goods $\ell = 1, 2$. Consumer i has consumption set $X_i = \mathbb{R}_+^2$, preference relation $\succeq_i$, and endowment $\omega_i = (\omega_{i1}, \omega_{i2}) \in \mathbb{R}_+^2$. Aggregate endowment is $\omega = \omega_1 + \omega_2 \gg 0$. An allocation $x = (x_1, x_2)$ is feasible when $x_1 + x_2 \leq \omega$ componentwise.

At prices $p \gg 0$, consumer i has wealth $p \cdot \omega_i$ and budget set

$$B_i(p) = \{x_i \in X_i : p \cdot x_i \leq p \cdot \omega_i\}.$$

Definition 11.1. Walrasian equilibrium in the exchange economy. A pair (p^*, x^*) with $p^* \gg 0$ is an equilibrium if markets clear, $x_1^* + x_2^* = \omega$, and

$$x_i^* \succeq_i x_i \quad \text{for every } x_i \in B_i(p^*), \quad i = 1, 2.$$

Prices are defined only up to multiplication by a positive scalar.

Example 11.2. Cobb--Douglas exchange. Suppose $u_i(x_{i1}, x_{i2}) = x_{i1}^\alpha x_{i2}^{1-\alpha}$ with $0 < \alpha < 1$, and endowments $\omega_1 = (1, 2)$ and $\omega_2 = (2, 1)$. The demands are

$$x_1(p) = \left(\frac{\alpha(p_1 + 2p_2)}{p_1}, \frac{(1-\alpha)(p_1 + 2p_2)}{p_2} \right), \quad x_2(p) = \left(\frac{\alpha(2p_1 + p_2)}{p_1}, \frac{(1-\alpha)(2p_1 + p_2)}{p_2} \right).$$

Clearing good 1 gives $p_1/p_2 = \alpha/(1-\alpha)$; one may normalize $p_2 = 1$ and recover the allocation from the two demands.

Definition 11.3. Pareto efficiency. A feasible allocation x is Pareto efficient if there is no feasible x' with $x'_i \succeq_i x_i$ for every $i \in \{1, 2\}$ and $x'_i \succ_i x_i$ for at least one $i \in \{1, 2\}$. The Pareto set is the set of all such allocations for the fixed aggregate endowment.

Proposition 11.4. *First welfare theorem for the exchange box. See[Mas-Colell et al., 1995]. If preferences are locally nonsatiated and (p^*, x^*) is a Walrasian equilibrium, then x^* is Pareto efficient.*

Second welfare theorem for the exchange box. See[Mas-Colell et al., 1995]. If preferences are continuous, convex, and strongly monotone, every Pareto-efficient allocation can be supported by a positive price vector after suitable lump-sum transfers $T_1 + T_2 = 0$. The transfers change wealth, not relative prices.

11.2 One consumer and one producer

There is one consumer with labor-time endowment $\bar{N}$ of time and no initial endowment of the consumption good. The consumption set is $X = [0, \bar{N}] \times \mathbb{R}_+$, and $u : X \rightarrow \mathbb{R}$ is the consumer's utility function. Let x_1 be leisure, x_2 consumption, and $f(z)$ output from labor $z \geq 0$. At output price $p > 0$ and wage $w > 0$, the firm solves

$$\pi_f(p, w) := \max_{z \geq 0} \{pf(z) - wz\}.$$

The consumer owns the firm, receives profit $\pi_f(p, w)$, and solves

$$\max_{(x_1, x_2) \in X} u(x_1, x_2) \quad \text{subject to} \quad px_2 \leq w(\bar{N} - x_1) + \pi_f(p, w).$$

Definition 11.5. A competitive equilibrium is a price pair (p^*, w^*) and choices (x^*, z^*) satisfying both optimization problems and the clearing conditions

$$x_2^* = f(z^*), \quad z^* = \bar{N} - x_1^*.$$

Proposition 11.6. *Under continuity, convexity, local nonsatiation, and the usual concavity and attainment assumptions, a competitive allocation solves the central-planner problem*

$$\max_{x_1, x_2} u(x_1, x_2) \quad \text{subject to} \quad x_2 \leq f(\bar{N} - x_1), \quad 0 \leq x_1 \leq \bar{N}.$$

Conversely, an interior planner solution can be supported by prices equal to the relevant marginal rates.

11.3 Production, ownership, and feasibility

Consider I consumers, J firms, and L commodities. Consumer i has consumption set X_i , preference $\succeq_i$ represented by $u_i : X_i \rightarrow \mathbb{R}$, and endowment $\omega_i \in \mathbb{R}^L$; firm j has production set $Y_j \subseteq \mathbb{R}^L$. Let $\omega = \sum_i \omega_i$. An allocation (x, y) is feasible when

$$\sum_{i=1}^I x_i \leq \omega + \sum_{j=1}^J y_j.$$

In a private-ownership economy, consumer i owns share $\theta_{ij} \in [0, 1]$ of firm j , with $\sum_i \theta_{ij} = 1$ for each j .

Definition 11.7. Walrasian equilibrium. A price vector $p \gg 0$ and allocation (x^*, y^*) form an equilibrium when, for every firm,

$$y_j^* \in \arg \max_{y_j \in Y_j} p \cdot y_j,$$

and, for every consumer,

$$x_i^* \in \arg \max_{x_i \in X_i} \{u_i(x_i) : p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*\},$$

with market clearing $\sum_i x_i^* = \omega + \sum_j y_j^*$.

Definition 11.8. Price equilibrium with transfers. Replace each consumer's ownership wealth by an assigned w_i satisfying $\sum_i w_i = p \cdot \omega + \sum_j p \cdot y_j^*$, and require $x_i^* \in \arg \max \{u_i(x_i) : p \cdot x_i \leq w_i\}$ while firms maximize profit and markets clear.

11.4 Welfare theorems

Proposition 11.9. *First Fundamental Welfare Theorem. See [Debreu, 1959, Mas-Colell et al., 1995]. If preferences are locally nonsatiated and (p, x^*, y^*) is a price equilibrium with transfers,*

then (x^*, y^*) is Pareto efficient. A Walrasian equilibrium is the special case in which wealth is generated by endowments and ownership shares.

The proof is the supporting-price argument. Consumer optimality and local nonsatiation imply that every bundle weakly preferred to x_i^* costs at least w_i ; a strictly preferred bundle costs strictly more. A Pareto improvement therefore has weakly greater expenditure for every consumer and strictly greater expenditure for at least one consumer. Summing and combining this strict inequality with firm profit maximization contradicts feasibility and market clearing.

Definition 11.10. Price quasi-equilibrium with transfers. A nonzero price vector, allocation, and assigned wealths form a price quasi-equilibrium with transfers when firms maximize profit, markets clear, $p \cdot x_i^* = w_i$ for every consumer, the assigned wealths add to aggregate endowment and profit value, and every bundle strictly preferred to x_i^* has cost at least w_i .

Proposition 11.11. *Second Fundamental Welfare Theorem. See [Debreu, 1959, Mas-Colell et al., 1995]. Suppose every production set is closed and convex, every preference is convex, continuous, and locally nonsatiated, and the aggregate attainable set is closed. Every Pareto-efficient allocation can be supported by a nonzero price vector as a price quasi-equilibrium with transfers. If, for every consumer, there is a bundle $\hat{x}_i \in X_i$ with $p \cdot \hat{x}_i < w_i$, continuity and convexity upgrade the quasi-equilibrium to a price equilibrium with transfers.*

The result follows by separating the convex aggregate set of bundles that make every consumer strictly better off from the convex aggregate production-plus-endowment set. The existence of a strictly cheaper feasible bundle for each consumer supplies the final upgrade from weak support to full budget maximization.

11.5 General equilibrium with inattention

An inattention model must distinguish true prices from the prices or signals used by a consumer's decision rule. Fix $\underline{m}_i \in (0, 1]$ and let the admissible attention class be $\mathcal{M}_i = \{\text{diag}(m_{i1}, \dots, m_{iL}) : m_{i\ell} \in [\underline{m}_i, 1]\}$. Thus $M_i p \gg 0$ whenever $M_i \in \mathcal{M}_i$ and $p \gg 0$. Write $D_i^B(p, M_i)$ for the demand correspondence generated by the fully specified behavioral model, while the unadorned $D_i(p, w_i)$ remains ordinary Walrasian demand at true prices and wealth. The model must state whether M_i changes perceived prices, the budget, utility, or information.

For a price-perception benchmark, $\tilde{p}_i = M_i p$ and a possible behavioral budget are written

explicitly, for example

$$D_i^B(p, M_i) = \arg \max_{x \in X_i} \{u_i(x) : \tilde{p}_i \cdot x \leq \tilde{p}_i \cdot \omega_i\}.$$

This is not a Walrasian budget unless $M_i = I_L$ or the model separately establishes equivalence with the true budget.

Definition 11.12. Inattention equilibrium. A pair (p, x) is an inattention equilibrium when each $x_i \in D_i^B(p, M_i)$ satisfies the model's actual feasibility restrictions and

$$\sum_{i=1}^I x_i = \sum_{i=1}^I \omega_i.$$

Proposition 11.13. *Compatibility with Pareto efficiency. Suppose preferences are differentiable, strictly increasing, and concave and x^* is an interior Pareto-efficient allocation. Then there is a common supporting price vector $q \gg 0$ and multipliers $\mu_i > 0$ such that $\nabla u_i(x_i^*) = \mu_i q$ for every i . If the behavioral first-order condition is $\nabla u_i(x_i^*) = \lambda_i M_i p$ with $\lambda_i > 0$, then each $M_i p$ must be collinear with the same q .*

This is a necessary compatibility condition, not an if-and-only-if statement about attention matrices. Identical matrices can support efficiency in a regular model, while different matrices can also be compatible at special allocations. Perceived-price optimality alone does not justify either welfare theorem.

11.6 Problems

Part I: By hand

Problem 11.1 [Core]. Cobb-Douglas exchange equilibrium.

There are two consumers with common utility $u_i(x_{i1}, x_{i2}) = x_{i1}^\alpha x_{i2}^{1-\alpha}$, $0 < \alpha < 1$, and endowments $\omega_1 = (1, 2)$, $\omega_2 = (2, 1)$.

- Derive individual Walrasian demands.
- Normalize $p_2 = 1$ and solve for the equilibrium relative price.
- Compute equilibrium allocations and verify both market-clearing equations.

Part II: A consumer-owned firm — chapter use case and verified scale-up

Problem 11.2 [Proof]. First Fundamental Welfare Theorem.

Consider a private-ownership economy with locally nonsatiated preferences and profit-maximizing firms.

- (a) State the definition of Pareto efficiency used in the theorem.
- (b) Prove that a competitive equilibrium allocation is Pareto efficient.
- (c) Identify exactly where local nonsatiation, budget feasibility, profit maximization, and market clearing enter the proof.

Problem 11.3 [Proof]. Supporting a Pareto optimum.

Suppose consumption sets and production sets are closed and convex, preferences are continuous, convex, and locally nonsatiated, and a feasible allocation is Pareto efficient. Assume the separation hypotheses stated in the chapter.

- (a) Formulate the aggregate set-separation argument and show how it supports the allocation as a price quasi-equilibrium with transfers.
- (b) Explain why the initial endowment distribution generally must be changed by lump-sum transfers.
- (c) State the positive-wealth or boundary condition needed to upgrade the quasi-equilibrium to a price equilibrium with transfers, and distinguish that object from a competitive equilibrium for the original ownership economy.

Chapter use case

Seed: Problems 11.1, 11.4. Fixed: all unlisted primitives and assumptions. Scaled: the production exponent from one-half to $\beta \in (0, 1)$. Tool: Codex + Python. Verification: Recover the square-root economy; verify planner and decentralized allocations coincide; check second-order conditions, profit finiteness, both market-clearing equations, and limiting behavior in β .

Problem 11.4 [Applied]. One consumer and one producer.

There are two goods. Good 1 can be consumed or used as input z to produce good 2 according to $y_2 = \sqrt{z}$. The consumer owns the firm and has an endowment $\bar{\omega}_1 > 0$ of good 1 and no good 2. Utility is $\log x_1 + \log x_2$.

- (a) Solve the planner's problem.
- (b) Find supporting relative prices from marginal conditions.

- (c) Verify that the consumer and firm choices clear both markets.

Chapter 12

Positive Theory of Equilibrium

The preceding chapter established welfare properties of competitive equilibrium. This chapter studies what those conditions imply—and what they leave undetermined—for aggregate excess demand, existence, local uniqueness, comparative statics, and the core.

12.1 Aggregate excess demand in pure exchange

Consider a pure-exchange economy with consumers $i = 1, \dots, I$, commodity space $\mathbb{R}_+^L$, endowments $\omega_i \in \mathbb{R}_+^L$, and aggregate endowment $\omega = \sum_i \omega_i \gg 0$. At a strictly positive price vector $p \gg 0$, consumer i has wealth $w_i = p \cdot \omega_i$ and demand correspondence

$$D_i(p, w_i) = \arg \max \{u_i(x) : x \geq 0, p \cdot x \leq w_i\}.$$

Definition 12.1. Aggregate excess demand. With possibly multivalued demand, define the Minkowski-sum correspondence

$$Z(p) = \sum_{i=1}^I D_i(p, p \cdot \omega_i) - \{\omega\}.$$

When every demand is single-valued, write its unique selection as $x_i(p)$ and the resulting excess-demand function as $z(p) = \sum_i x_i(p) - \omega$.

Pure-exchange equilibrium. A pair (p^*, x^*) with $p^* \gg 0$ is a Walrasian equilibrium when each $x_i^* \in D_i(p^*, p^* \cdot \omega_i)$ and $\sum_i x_i^* = \omega$. Prices are equivalent up to multiplication by a positive scalar.

Proposition 12.2. *Walras' law and homogeneity. Suppose each consumer is locally nonsatiated*

and demands are single-valued. Then

$$p \cdot z(p) = 0, \quad z(\lambda p) = z(p) \quad (\lambda > 0).$$

If $p \gg 0$ and markets clear in all but one commodity, Walras' law implies that the final market clears as well.

If preferences are continuous, strictly convex, and strongly monotone, the demand function is continuous on $\mathbb{R}_{++}^L \times \mathbb{R}_{++}$. Consequently z is continuous on strictly positive prices. Its domain should not be silently extended to a zero price: demand may become unbounded there. In an equilibrium with nonnegative prices, a good can have excess supply only at a zero price; under strong monotonicity and $p \gg 0$, every market clears.

For calculations it is convenient to normalize prices. We use either the simplex

$$\Delta = \{p \in \mathbb{R}_+^L : \mathbf{1}_L^\top p = 1\}$$

or the chart $p_L = 1$. These are equivalent for strictly positive prices.

Example 12.3. Cobb--Douglas exchange. With $u_i(x) = x_1^\alpha x_2^{1-\alpha}$ and $0 < \alpha < 1$,

$$x_i(p) = \left(\frac{\alpha p \cdot \omega_i}{p_1}, \frac{(1-\alpha) p \cdot \omega_i}{p_2} \right).$$

Substitution into $z(p) = 0$ reduces equilibrium to one relative-price equation, because the second equation follows from Walras' law.

12.2 Existence

Proposition 12.4. *Walrasian existence theorem. See [Arrow and Debreu, 1954, Debreu, 1959]. Consider a finite pure-exchange economy with consumption sets $\mathbb{R}_+^L$, individual endowments $\omega_i \gg 0$, and complete, transitive, continuous, convex, and strongly monotone preferences. Then a Walrasian equilibrium with $p^* \gg 0$ exists. Strict convexity is not required for existence; it makes each consumer's demand single-valued.*

The proof normalizes prices to Δ , truncates consumption at the compact aggregate feasible set, constructs a correspondence with nonempty compact convex values, and applies a fixed-point theorem. Strictly positive individual endowments provide positive wealth at every nonzero price, and strong monotonicity rules out a zero price for any commodity in equilibrium. These

hypotheses supply the boundary step that a fixed-point map defined only on $\mathbb{R}_{++}^L$ would otherwise omit.

12.3 Local uniqueness and regularity

Definition 12.5. Normalized local uniqueness. A normalized equilibrium $p^* \in \Delta$ is locally unique if some neighborhood of p^* contains no other normalized equilibrium.

Work in the chart $p_L = 1$ and write $\widehat{z}(p_1, \dots, p_{L-1}) = (z_1, \dots, z_{L-1})$. The last component is redundant by Walras' law.

Definition 12.6. Regular equilibrium. A differentiable equilibrium p^* is regular when the Jacobian $D\widehat{z}(p^*)$ is nonsingular.

Proposition 12.7. *Regularity implies local uniqueness. See[Mas-Colell et al., 1995]. If $D\widehat{z}(p^*)$ is nonsingular, the inverse function theorem implies that p^* is an isolated zero of $\widehat{z}$. This is a local statement; it does not rule out another equilibrium far away.*

Fix the chart $p_L = 1$ with its standard orientation. For a regular economy, define the signed index of an equilibrium as $\text{sign det}[-D\widehat{z}(p^*)]$. The index theorem[Mas-Colell et al., 1995] states that when the normalized equilibrium set is compact and excess demand points inward on the boundary of the chosen compact price region, the sum of these indices is one. Hence a regular economy has an odd number of equilibria. The conclusion is global and does not follow from the inverse function theorem alone.

12.4 What equilibrium theory does not identify

At a differentiability point, Walras' law and homogeneity imply the exact derivative restrictions

$$[D_p z(p)]^\top p = -z(p), \quad D_p z(p)p = 0.$$

At an equilibrium, the first identity becomes $[D_p z(p)]^\top p = 0$; these are restrictions, not a complete theory of comparative statics.

Proposition 12.8. *Sonnenschein--Mantel--Debreu theorem. See[Sonnenschein, 1972, Mantel, 1974, Debreu, 1974]. Subject to continuity, homogeneity of degree zero, Walras' law, and boundary divergence toward any zero-priced desirable good, aggregate excess demand can have essentially*

arbitrary shape on compact subsets of the strictly positive normalized price simplex. In particular, individual rationality does not generally imply uniqueness, stability, or monotone comparative statics for the aggregate equilibrium map.

The theorem is an impossibility result about restrictions on aggregate demand. It does not say that every arbitrary function is a demand system: the stated regularity, Walras, homogeneity, and boundary conditions remain necessary, and the result concerns a constructed finite exchange economy.

12.5 Uniqueness and gross substitutes

With a constant-returns production cone $Y \subseteq \mathbb{R}^L$ containing zero, a strictly positive price p supports the technology when $p \cdot y \leq 0$ for every $y \in Y$. If aggregate excess demand satisfies $z(p) \in Y$ and Walras' law, then choosing production $y = z(p)$ clears markets and maximizes profit. This observation separates the consumer condition from the technology-support condition.

Definition 12.9. Gross substitutes. An excess-demand function has the strict gross-substitutes property if, whenever only the price of good ℓ rises, excess demand for every other good $k \neq \ell$ strictly rises. Weak gross substitutes replaces strict inequalities with weak inequalities. The property is imposed on the relevant price domain and is stronger than Walras' law or homogeneity.

Proposition 12.10. *Gross-substitutes uniqueness theorem. See[Mas-Colell et al., 1995]. Under continuity, homogeneity, Walras' law, and a strict gross-substitutes condition, a pure-exchange equilibrium has at most one normalized strictly positive price vector. Existence still requires a separate existence theorem. Without strictness, gross substitutes can deliver a convex or lattice-structured equilibrium set rather than a unique price.*

12.6 The core

For clarity, define the core first in a pure-exchange economy. There are I consumers, endowments $\omega_i \in \mathbb{R}_+^L$, and a feasible allocation satisfies $\sum_i x_i = \sum_i \omega_i$. We use exact feasibility for blocking coalitions. Under free disposal and monotone preferences, one may replace equality by $\leq$ without changing the economic conclusion. A nonempty coalition $S \subseteq \{1, \dots, I\}$ can redistribute exactly its own aggregate endowment $\sum_{i \in S} \omega_i$.

Definition 12.11. Blocking, core, strong blocking, and strong core. Coalition S blocks a feasible allocation x if there are $\{x'_i\}_{i \in S}$ with $x'_i \succ_i x_i$ for every $i \in S$ and $\sum_{i \in S} x'_i = \sum_{i \in S} \omega_i$. The core

is the set of feasible allocations blocked by no coalition. Coalition S strongly blocks a feasible allocation x if there are $\{x'_i\}_{i \in S}$ satisfying $\sum_{i \in S} x'_i = \sum_{i \in S} \omega_i$, $x'_i \succeq_i x_i$ for every $i \in S$, and $x'_j \succ_j x_j$ for at least one $j \in S$. The strong core is the set of feasible allocations with no strong block. Every ordinary block is a strong block, so the strong core is contained in the core.

Proposition 12.12. *Walrasian allocations are in the strong core, and hence in the core. See [Debreu and Scarf, 1963, Mas-Colell et al., 1995]. Suppose preferences are locally non-satiated and (p^*, x^*) is a Walrasian equilibrium with $p^* \gg 0$. Suppose a coalition strongly blocks x^* . Consumer optimality and local nonsatiation imply $p^* \cdot x'_i \geq p^* \cdot \omega_i$ for every weakly improved member; the inequality is strict for the member who strictly improves. Summing over the coalition therefore gives $p^* \cdot \sum_{i \in S} x'_i > p^* \cdot \sum_{i \in S} \omega_i$, contradicting exact coalition feasibility. Thus no coalition strongly blocks x^* .*

The converse need not hold in a finite economy: the core can contain noncompetitive allocations. In the r -fold replica of a fixed finite economy, identify equal-treatment allocations with their per-type consumption vectors. Under continuous, convex, locally nonsatiated preferences and strictly positive aggregate resources, the Debreu--Scarf core-convergence theorem [Debreu and Scarf, 1963] says that every limit point of such normalized core allocations as $r \rightarrow \infty$ is a Walrasian allocation; conversely, every Walrasian allocation belongs to the core of every replica.

12.7 Problems

Part I: By hand

Problem 12.1 [Core]. Differential identities for excess demand.

Let aggregate excess demand $z : \mathbb{R}_{++}^L \rightarrow \mathbb{R}^L$ be continuously differentiable, homogeneous of degree zero, and satisfy Walras' law.

- (a) Differentiate homogeneity to show $D_p z(p)p = 0$.
- (b) Differentiate Walras' law to show $[D_p z(p)]^\top p = -z(p)$.
- (c) Specialize the second identity to an equilibrium and interpret the left null vector.

Part II: A coalition challenges a competitive allocation — chapter use case and verified scale-up

Problem 12.2 [Proof]. Regularity and local uniqueness.

Normalize prices by $p_L = 1$ and let $\hat{z}$ be the first $L - 1$ excess-demand equations.

- (a) Define a regular equilibrium.
- (b) Use the inverse function theorem to prove that a regular equilibrium is locally isolated.
- (c) Explain why local isolation implies finitely many equilibria only after compactness and boundary conditions are added.

Problem 12.3 [Proof]. Gross substitutes and uniqueness.

Assume aggregate excess demand satisfies strict gross substitutes on the relevant price domain: increasing one price strictly raises excess demand for every other good whenever the comparison is nondegenerate. Suppose two strictly positive normalized equilibrium price vectors exist.

- (a) Compare the largest relative price change across goods.
- (b) Use gross substitutes and Walras' law to derive a contradiction unless the two price vectors are proportional.
- (c) Explain why weak gross substitutes alone need not imply uniqueness and where strictness (or an irreducibility substitute) enters the proof.

Chapter use case

Seed: Problem 12.4. Fixed: all unlisted primitives and assumptions. Scaled: the blocking predicate from all-strict improvement to weak improvement for all members and strict improvement for one witness. Tool: Codex + Lean. Verification: Compile the Lean strong-coalition price-support declaration and match each hypothesis to the economic proof. Lean verifies the finite price-summing contradiction, not the derivation of price support from Walrasian optimization or the replica-economy convergence claim. This case closes the price-support and core thread; Problems 12.2 and 12.3 are separate regularity and uniqueness extensions.

Problem 12.4 [Core]. Competitive allocations and the strong core.

Use the strong-blocking convention in a finite pure-exchange economy with locally nonsatiated preferences. A nonempty coalition S strongly blocks x when it can choose $(y_i)_{i \in S}$ such that $\sum_{i \in S} y_i = \sum_{i \in S} \omega_i$, $y_i \succeq_i x_i$ for every $i \in S$, and $y_j \succ_j x_j$ for at least one $j \in S$. Define the associated strong core.

- (a) Prove that every Walrasian allocation belongs to the strong core.
- (b) Give an economic reason why the converse can fail with finitely many consumers.
- (c) State the content, but not a proof, of the core-convergence theorem for replica economies.

Part V

Advanced Equilibrium Applications

Chapter 13

Testable Restrictions on the Equilibrium Manifold

This chapter studies what a finite set of market observations can reveal about a pure-trade economy. The central tension is between individual utility maximization and market clearing: each consumer must rationalize their own choices, while the choices must add up to the observed aggregate endowment.

13.1 Observables and the equilibrium manifold

Fix a finite set of consumers $i = 1, \dots, I$, goods $\ell = 1, \dots, L$, and observations $t = 1, \dots, T$. At observation t , let $p^t \in \mathbb{R}_{++}^L$ denote the price vector, let $w_i^t > 0$ denote consumer i 's income, and let $\omega^t \in \mathbb{R}_+^L$ denote the aggregate endowment. The accounting restriction is $\sum_i w_i^t = p^t \cdot \omega^t$. Prices are normalized because only their positive scale is economically relevant.

Definition 13.1. (Pure-trade rationalization.) A finite dataset is rationalized by a pure-trade economy if there are continuous, locally nonsatiated, concave utility functions u_i and bundles $x_i^t \in \mathbb{R}_+^L$ such that, for every i, t , $x_i^t \in \arg \max\{u_i(x) : p^t \cdot x \leq w_i^t\}$, and, for every t , $\sum_i x_i^t = \omega^t$.

(Equilibrium manifold.) For a market excess-demand function $z(\hat{\omega}, p)$, where $\hat{\omega} = (\omega_1, \dots, \omega_I)$, the equilibrium manifold is $\mathcal{M} = \{(\hat{\omega}, p) : z(\hat{\omega}, p) = 0\}$. It records pairs of latent individual-endowment profiles and prices that can clear markets for fixed underlying preferences. The Brown--Matzkin observable set below is a projection of this theoretical manifold onto prices, individual incomes, and aggregate endowments.

13.2 Afriat inequalities and elimination of unobservables

For a finite collection of observed or latent bundles with strictly positive prices and expenditure exhaustion, Afriat's inequalities are

$$U_i^t \leq U_i^s + \lambda_i^s p^s \cdot (x_i^t - x_i^s) \quad \text{for all } s, t,$$

with $U_i^t \in \mathbb{R}$ denoting a utility-level certificate and $\lambda_i^s > 0$ denoting a marginal-utility-of-wealth certificate. They are equivalent to the existence of a monotone, concave utility representation for the finite choice data.

The equilibrium restrictions combine these inequalities with the budget equalities and market-clearing equalities

$$p^t \cdot x_i^t = w_i^t, \quad \sum_i x_i^t = \omega^t.$$

The unobserved variables are

$$(x_i^t, U_i^t, \lambda_i^t)_{i,t}.$$

Prices, individual incomes, and aggregate endowments are the data. Because λ_i^s multiplies latent x_i^t , the joint system is polynomial but not linear in all unknowns.

(Tarski--Seidenberg theorem.) The projection of a semialgebraic set is semialgebraic. Consequently, eliminating the unobserved variables from any finite polynomial system yields a quantifier-free semialgebraic condition in the observables. This theorem supplies an elimination principle; it does not say that the resulting restrictions are easy to compute or that every model has a nontrivial restriction.

13.3 Brown--Matzkin theorem

Brown and Matzkin (1996) apply this principle to pure-trade equilibrium. Under their regularity and invariance assumptions, finite observations on prices, individual incomes, and aggregate endowments are rationalizable if and only if they satisfy a finite family of semialgebraic restrictions obtained after eliminating the latent utility levels, marginal utilities of income, and consumption bundles. The restrictions are nontrivial: some finite datasets cannot lie on the equilibrium manifold of any economy in the maintained class. See[Brown and Matzkin, 1996]; its quantifier-elimination construction is not reproved in this text.

The theorem should not be confused with the Sonnenschein--Mantel--Debreu result. The latter

concerns the flexibility of aggregate excess demand when individual preferences are allowed to vary; Brown--Matzkin hold individual preferences fixed across observations and exploit variation in observed incomes and aggregate endowments.

13.4 Problems

Part I: By hand

Problem 13.1 [Core]. Equilibrium Afriat feasibility.

At each observation t , prices $p^t \gg 0$, individual incomes $(w_i^t)_i$, and aggregate endowment ω^t are observed, while individual allocations are latent.

- (a) Introduce latent consumptions x_i^t and write individual budget equalities $p^t \cdot x_i^t = w_i^t$ and aggregate market clearing $\sum_i x_i^t = \omega^t$.
- (b) Write Afriat inequalities for every consumer across observations.
- (c) Explain why joint feasibility is a finite semialgebraic revealed-equilibrium test, why the latent multipliers and bundles make the full system bilinear, and how quantifier elimination projects the restrictions onto the observables.

Part II: A counterfactual equilibrium identified set — chapter use case and verified scale-up

Problem 13.2 [Proof]. What equilibrium data identify.

Suppose only equilibrium prices and aggregate endowments are observed.

- (a) List the preference, allocation, and ownership objects that remain unobserved.
- (b) Explain why market clearing alone cannot identify individual demand.
- (c) Describe how additional observations, known individual endowments, or parametric restrictions can turn an underidentified model into a testable one.

Chapter use case

Seed: Problems 13.1, 13.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of equilibrium observations from two to T , with the maintained preference class and counterfactual fixed. Tool: Codex + Python. Verification: Verify feasibility and market clearing for every retained candidate, show how each added observation changes the bounds, and never replace set identification by a point estimate.

Problem 13.3 [Applied]. Counterfactual discipline.

A dataset passes a revealed-equilibrium feasibility test under a maintained class of preferences.

- (a) Explain why passing the test does not point-identify a counterfactual equilibrium.
- (b) Define an identified set for a counterfactual price or allocation.
- (c) State two economically meaningful restrictions that could tighten this set, and explain the cost of each restriction.

Chapter 14

Sequential Trade

14.1 Arrow--Debreu benchmark

There are $S \geq 1$ states, indexed by $\mathcal{S} = \{1, \dots, S\}$, and L goods in each state. Consumers are indexed by $i \in \{1, \dots, I\}$. Consumer i has state-contingent endowment $\omega_i = (\omega_{i1}, \dots, \omega_{iS})$ and preference $\succeq_i$ over contingent consumption plans $x_i = (x_{i1}, \dots, x_{iS})$.

Definition 14.1. (Arrow--Debreu equilibrium.) Prices $p = (p_1, \dots, p_S) \in (\mathbb{R}_{++}^L)^S$ and an allocation $x^* = (x_1^*, \dots, x_I^*)$ form an Arrow--Debreu equilibrium if, for each consumer, x_i^* maximizes $\succeq_i$ over $B_i^{AD}(p) = \{x_i : \sum_{s=1}^S p_s \cdot x_{is} \leq \sum_{s=1}^S p_s \cdot \omega_{is}\}$, and markets clear state by state: $\sum_i x_{is}^* = \sum_i \omega_{is}$ for every s .

14.2 Spot markets and sequential trade

Suppose contingent commodity markets are unavailable at date $t = 0$. There are K tradable assets with payoff matrix $R \in \mathbb{R}^{S \times K}$: one unit of asset k pays R_{sk} units of the first good in state s . Asset prices are $q \in \mathbb{R}^K$, and spot-good prices in state s are $p_s \in \mathbb{R}_{++}^L$.

Definition 14.2. (Radner budget set.) Given asset holdings $z_i \in \mathbb{R}^K$ and state-contingent consumption x_i , consumer i satisfies $q \cdot z_i \leq 0$ and $p_s \cdot x_{is} \leq p_s \cdot \omega_{is} + p_{s1}(Rz_i)_s$ for every s . The zero on the date-0 constraint is the normalization of zero initial asset endowment.

(Radner equilibrium.) A price system $(q, p_1, \dots, p_S)$ and plans $(z_i^*, x_i^*)_{i=1}^I$ form a Radner equilibrium when every consumer maximizes their preference over the Radner budget set, asset markets clear $\sum_i z_i^* = 0$, and goods markets clear in every state: $\sum_i x_{is}^* = \sum_i \omega_{is}$ for all s .

Proposition 14.3. (*Arrow--Radner equivalence.*) See [Radner, 1972, Mas-Colell et al., 1995]. Suppose consumption lies in $\mathbb{R}_+^{LS}$, preferences are continuous and strictly monotone, the payoff matrix has full row rank $\text{rank}(R) = S$, and asset prices satisfy no arbitrage. Full spanning and no arbitrage yield a strictly positive state-price vector ψ with $q = R^\top \psi$. For positive spot prices, define Arrow--Debreu commodity prices by $\tilde{p}_s = (\psi_s/p_{s1})p_s$. Then the consumer budget sets project to the same contingent-consumption budget, so the sets of consumption allocations supported by Arrow--Debreu and Radner equilibria coincide. If $\text{rank}(R) < S$, some contingent trades are unattainable and the equivalence need not hold.

The full-rank condition is the precise meaning of complete asset markets in this finite-state model. The proposition is an allocation-equivalence statement conditional on equilibrium prices; it does not by itself assert equilibrium existence.

14.3 Problems

Part I: By hand

Problem 14.1 [Core]. Spanning contingent consumption.

There are S states tomorrow and K traded assets with payoff matrix $R \in \mathbb{R}^{S \times K}$ and date-zero asset-price vector $q \in \mathbb{R}^K$. Assume no arbitrage.

- (a) State the rank condition under which every contingent net trade can be implemented by an asset portfolio.
- (b) If $S = 2$ and the assets have payoff vectors $(1, 1)$ and $(1, 0)$, construct the portfolio that delivers any target payoff (c_1, c_2) .
- (c) Let $\psi \in \mathbb{R}_{++}^S$ be state prices satisfying $q = R^\top \psi$. Explain how the date-zero portfolio budget maps into an Arrow-Debreu present-value budget.

Problem 14.2 [Core]. Incomplete markets.

Let $S = 3$ and suppose the only assets are a bond with payoff $(1, 1, 1)$ and an asset with payoff $(1, 0, 0)$.

- (a) Characterize the attainable payoff subspace.
- (b) Determine whether the claim $(0, 1, -1)$ is attainable.
- (c) Explain why competitive equilibrium with these assets need not be Pareto efficient relative to the full contingent-commodity economy.

Part II: Complete versus incomplete sequential markets — chapter use case and verified scale-up

Seed: Problems 14.1, 14.2. Fixed: all unlisted primitives and assumptions. Scaled: the asset span from rank $S - 1$ to rank S at a fixed number of states. Tool: Codex + Python; Lean target. Verification: Check portfolio replication, the state-price units map, and present-value budgets; include a rank-deficient claim that cannot be implemented. Arrow-Radner equivalence remains a future Lean target.

Problem 14.3 [Proof]. Arrow-Debreu and sequential equivalence.

There are S states, K assets with payoff matrix $R \in \mathbb{R}^{S \times K}$ of rank S , date-zero asset prices q , and state- s spot-price vectors p_s . Asset payoffs are denominated in the state numeraire. Consumer i chooses a portfolio z_i subject to $q \cdot z_i \leq 0$, then chooses x_{is} subject to $p_s \cdot x_{is} \leq p_s \cdot \omega_{is} + p_{s1}(Rz_i)_s$. Assume no arbitrage and strictly monotone continuous preferences.

(a) Map an Arrow-Debreu budget into date-zero asset purchases and state-by-state spot trades.

(b) Prove the reverse mapping from a sequential budget to a present-value budget.

(c) Identify which step fails under incomplete markets or binding portfolio constraints.

Chapter 15

Matching

15.1 One-to-one matching

Let M, W be finite disjoint sets of agents on the two sides of the market. For each $m \in M$, $\succ_m$ is a strict order on $W \cup \{m\}$; for each $w \in W$, $\succ_w$ is a strict order on $M \cup \{w\}$. Ranking oneself represents remaining unmatched. A matching is a map $\varphi : M \cup W \rightarrow M \cup W$ that assigns each agent either an acceptable partner on the other side or themselves, and satisfies $\varphi(\varphi(a)) = a$.

Definition 15.1. (Stability.) A matching is individually rational if no agent is matched with an unacceptable partner. A pair (m, w) not matched to one another blocks φ if $w \succ_m \varphi(m)$ and $m \succ_w \varphi(w)$. A matching is stable when it is individually rational and has no blocking pair.

15.2 Deferred acceptance

In the men-proposing deferred-acceptance algorithm, each unmatched man proposes to the most-preferred woman who has not rejected him. Each woman tentatively holds her most-preferred acceptable proposal received so far and rejects the rest. The process stops when no proposal is rejected.

Theorem 15.2. (Gale--Shapley theorem.) See[Gale and Shapley, 1962]. For every finite marriage problem, deferred acceptance terminates and returns a stable matching. Among all stable matchings, the outcome is weakly best for every man and weakly worst for every woman.

Termination is finite because a man proposes to each woman at most once. The stability and proposer-optimality statements are the Gale--Shapley theorem.

15.3 Housing markets and top trading cycles

In a housing market, each agent initially owns one house and has a strict preference over houses. The strong core consists of allocations for which no coalition can make every member weakly better off and at least one member strictly better off using only the houses owned by that coalition.

The top-trading-cycles algorithm directs each agent to point to their favorite remaining house and each house to point to its owner. Every directed cycle is executed, the participating houses and agents are removed, and the procedure is repeated.

Theorem 15.3. (*Top-trading-cycles theorem.*) See[Shapley and Scarf, 1974]. With strict preferences and one-house ownership, top trading cycles terminates, is Pareto efficient, and returns the unique allocation in the strong core; the same allocation is supported as the competitive allocation of the housing market.

15.4 Problems

Part I: By hand

Problem 15.1 [Core]. Deferred acceptance by hand.

Three men have rankings $m_1 : w_1 \succ w_2 \succ w_3$, $m_2 : w_1 \succ w_2 \succ w_3$, and $m_3 : w_2 \succ w_3 \succ w_1$. The women rank $w_1 : m_2 \succ m_1 \succ m_3$, $w_2 : m_1 \succ m_3 \succ m_2$, and $w_3 : m_3 \succ m_2 \succ m_1$.

- (a) Run men-proposing deferred acceptance round by round.
- (b) Verify stability of the resulting matching by checking potential blocking pairs.
- (c) Explain why termination is guaranteed in every finite market.

Part II: Housing allocation and coalitional stability — chapter use case and verified scale-up

Problem 15.2 [Proof]. Proposer optimality.

Consider a finite balanced one-to-one matching market in which every participant has a complete strict ranking of the other side and all partners are acceptable.

- (a) Prove that if a receiver rejects a proposer during deferred acceptance, that rejected proposer-receiver pair cannot be matched to each other in any stable matching.
- (b) Use this rejection lemma to prove proposer optimality among stable matchings.
- (c) Explain why the receiver side obtains its least-preferred stable matching.

Chapter use case

Seed: Problem 15.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of agents and houses from three to n . Tool: Codex + Python; Lean target. Verification: Recover the three-agent top-trading-cycles allocation; check permutation, individual rationality, Pareto comparisons, and coalition blocking by enumeration. Top trading cycles and housing-core claims remain future Lean targets.

Problem 15.3 [Core]. Top trading cycles.

Agents 1, 2, 3 initially own houses h_1, h_2, h_3 . Their rankings are 1 : $h_2 \succ h_1 \succ h_3$, 2 : $h_1 \succ h_2 \succ h_3$, and 3 : $h_1 \succ h_3 \succ h_2$.

- (a) Run top trading cycles.
- (b) Show that the outcome is individually rational and Pareto efficient.
- (c) Verify directly that no coalition blocks the outcome.

Chapter 16

Computable General Equilibrium

16.1 From accounts to counterfactuals

Computable general equilibrium (CGE) analysis uses a calibrated microeconomic model and a social accounting matrix to evaluate policy counterfactuals. Calibration chooses parameters so that the model reproduces a benchmark allocation and the associated accounting identities before the policy is changed.

16.2 A two-sector small open economy

Let sectors be indexed by $j \in \mathcal{J} := \{1, \dots, J\}$ and households by $h \in \mathcal{H} := \{1, \dots, H\}$; the benchmark application has $J = 2$ goods. Sector j uses labor $n_j \geq 0$ and sector-specific capital $K_j \geq 0$ to produce output Q_j . The corresponding aggregate factor endowments are $\bar{n}$ and $\bar{K}_j$. Let $P_j > 0$ denote the domestic price of good j , $\mathbf{P} := (P_1, \dots, P_J)$ the domestic price vector, W the wage, and r_j the rental rate on capital specific to sector j . A small open economy takes world prices as given. For sector j , a constant-elasticity-of-substitution production function is

$$Q_j = A_j \left[\alpha_j K_j^{-\rho_j} + (1 - \alpha_j) n_j^{-\rho_j} \right]^{-1/\rho_j},$$

with $A_j > 0$, $0 < \alpha_j < 1$, and $\rho_j > -1$, $\rho_j \neq 0$. The associated elasticity parameter is $\sigma_j = 1/(1 + \rho_j) > 0$.

Definition 16.1. (Cost and conditional factor demand.) Given output $\bar{Q}_j$ and factor prices

(r_j, W) , the unit-consistent cost function is

$$c_j(r_j, W, \bar{Q}_j) = \min_{K_j, n_j \geq 0} \{r_j K_j + W n_j : Q_j(K_j, n_j) \geq \bar{Q}_j\}.$$

When differentiable, conditional factor demands satisfy

$$K_j^D = \frac{\partial c_j}{\partial r_j}, \quad n_j^D = \frac{\partial c_j}{\partial W}.$$

This derivative formulation avoids confusing the price P_j with the CES parameter ρ_j .

16.3 Social accounting and calibration

The social accounting matrix records production, factor payments, household income, and expenditure flows that must balance at the benchmark. Let Y_h denote household h 's nominal income and let $\beta_{hj} \in [0, 1]$ be its expenditure share on good j , with $\sum_{j \in \mathcal{J}} \beta_{hj} = 1$. Under Cobb--Douglas preferences, household demand $x_{hj}(\mathbf{P}, Y_h)$ satisfies the expenditure equation $P_j x_{hj}(\mathbf{P}, Y_h) = \beta_{hj} Y_h$, so the observed benchmark share identifies $\beta_{hj} = P_j x_{hj}(\mathbf{P}, Y_h) / Y_h$. The calibrated model is checked by substituting the parameter values back into every accounting identity and market-clearing condition.

16.4 Equilibrium closure

A computable model is not defined by technology and accounting identities alone. Let $\bar{n}_h$ be household h 's labor endowment and $\bar{K}_{hj}$ its endowment of capital specific to sector j , where $\sum_{h \in \mathcal{H}} \bar{n}_h = \bar{n}$ and $\sum_{h \in \mathcal{H}} \bar{K}_{hj} = \bar{K}_j$. Household income must specify factor ownership, taxes, transfers, and the disposition of profits and tariff revenue. Writing T_h for transfers, Π_h for distributed profits, and Tax_h for direct taxes, a representative identity is

$$Y_h = W \bar{n}_h + \sum_{j \in \mathcal{J}} r_j \bar{K}_{hj} + T_h + \Pi_h - \text{Tax}_h.$$

Demand equations $x_{hj} = x_{hj}(\mathbf{P}, Y_h)$ then close the household side.

For each good $j \in \mathcal{J}$, let M_j and E_j denote imports and exports, Inv_j investment demand, and G_j government final demand. Let $a_{jk} \geq 0$ denote the amount of good j used as an intermediate

input per unit of sector k 's output. The resource balance is

$$Q_j + M_j = \sum_{h \in \mathcal{H}} x_{hj}(\mathbf{P}, Y_h) + G_j + \text{Inv}_j + E_j + \sum_{k \in \mathcal{J}} a_{jk} Q_k.$$

Factor markets satisfy

$$\sum_{j \in \mathcal{J}} n_j = \bar{n} = \sum_{h \in \mathcal{H}} \bar{n}_h, \quad K_j = \bar{K}_j = \sum_{h \in \mathcal{H}} \bar{K}_{hj} \quad \text{for sector-specific capital.}$$

Constant returns and competitive pricing impose zero profit, so the domestic producer price equals unit cost whenever the sector operates.

The government budget identifies the use of tax and tariff receipts, and the external account identifies the financing of net imports. Let $P_j^W > 0$ denote the world price of traded good j and let $\text{ER} > 0$ denote the exchange rate measured in units of domestic currency per unit of foreign currency. A balanced-trade closure, for example, imposes

$$\sum_{j \in \mathcal{J}} P_j^W M_j = \sum_{j \in \mathcal{J}} P_j^W E_j,$$

whereas a fixed foreign transfer replaces the right-hand side by exports plus that transfer. The model must choose one nominal normalization, such as $W = 1$ or $\text{ER} = 1$, and must not impose both as independent equilibrium equations.

A benchmark replication check solves this entire system at the calibrated parameters and reports the maximum absolute residual in every household, firm, market, government, and external-account equation. Counterfactual welfare must also be declared, for example equivalent variation relative to the benchmark utility and price system.

16.5 Counterfactual policies

For an imported good $j \in \mathcal{J}$ that is a perfect substitute for the domestic good, the law of one price with an ad valorem tariff $\tau_j \geq 0$ changes the domestic price according to

$$P_j = \text{ER} P_j^W (1 + \tau_j).$$

Free trade is the baseline $\tau_j = 0$; a tariff counterfactual changes τ_j and resolves the full equilibrium system. Other counterfactuals change tax, subsidy, transfer, or technology equations while leaving the accounting identities and calibrated preference and technology parameters in place.

16.6 Problems

Part I: By hand

Problem 16.1 [Core]. Calibrating a CES technology.

A constant-returns technology is $Q = A[\alpha K^{-\rho} + (1 - \alpha)n^{-\rho}]^{-1/\rho}$, with $\rho > -1$ and $\rho \neq 0$. A strictly positive interior cost-minimizing benchmark observation gives (Q, K, n, r, W) and an elasticity of substitution $\sigma > 0$, $\sigma \neq 1$. (The Cobb-Douglas limit $\sigma = 1$ is outside the displayed parameterization.)

- (a) Recover ρ from σ .
- (b) Use the cost-minimizing first-order condition to recover α .
- (c) Recover A and verify that the benchmark is exactly replicated.

Problem 16.2 [Core]. Social-accounting consistency.

A two-sector social accounting matrix records sectoral sales, intermediate purchases, labor and capital payments, household income, taxes, savings, and imports.

- (a) State the row-balance and column-balance conditions.
- (b) Explain why every market-clearing equation has a corresponding accounting identity but not every accounting identity is a behavioral equation.
- (c) Design a numerical residual check that would detect an incorrectly calibrated benchmark.

Part II: A tariff counterfactual from partial to general equilibrium — chapter use case and verified scale-up

Seed: Problems 16.1, 16.2, 16.3. Fixed: all unlisted primitives and assumptions. Scaled: model closure from fixed factor prices to one endogenous shared labor market linking import-competing and export sectors. Tool: Codex + Python. Verification: Replicate the partial-equilibrium benchmark, then report factor-market, goods-market, trade-balance, government-budget, and welfare residuals. Identify which change comes from the single added general-equilibrium channel.

Problem 16.3 [Applied]. A tariff counterfactual.

A small open economy takes the world price P^W and exchange rate ER as given. A tariff τ implies domestic price $P = \text{ER } P^W (1 + \tau)$. Let differentiable domestic demand and supply be $D(P)$ and $S(P)$, imports be $M(P) = D(P) - S(P) > 0$, and tariff revenue be returned lump-sum, $\text{TR}(\tau) = \tau \text{ER } P^W M(P(\tau))$. Interpret D as quasi-linear (or otherwise free of income effects) and

S as competitive supply generated by marginal cost, so consumer and producer surplus are valid partial-equilibrium welfare measures.

- (a) Derive the first-order change in the domestic price from a small tariff change.
- (b) Derive the symbolic first-order decomposition $d\mathcal{W} = d\text{CS} + d\text{PS} + d\text{TR}$ and simplify it at free trade. Do not assume numerical functional forms.
- (c) Explain which general-equilibrium channels are omitted by that approximation and must be restored in a CGE counterfactual.

Appendix A

Order Theory

This mathematical appendix collects the order-theoretic concepts used throughout the book. The orientation is economic: xRy means that x is ranked at least as high as y . Unless stated otherwise, X is a nonempty set.

A.1 Binary Relations

Definition A.1. A binary relation on X is a subset $R \subseteq X \times X$. We write xRy in place of $(x, y) \in R$. The inverse relation R^{-1} is defined by $xR^{-1}y \iff yRx$. The identity relation is $\Delta_X = \{(x, x) : x \in X\}$.

A.1.1 Properties

A relation R on X is:

Reflexive if xRx for every $x \in X$.

Complete (or total) if, for every $x, y \in X$, either xRy or yRx (or both). Completeness implies reflexivity by taking $x = y$.

Symmetric if xRy implies yRx .

Asymmetric if xRy implies $\neg(yRx)$. Every asymmetric relation is irreflexive.

Antisymmetric if xRy and yRx imply $x = y$. Antisymmetry does not mean the negation of symmetry.

Transitive if xRy and yRz imply xRz .

Acyclic if there do not exist $n \geq 1$ and $x_0, \dots, x_n \in X$ such that $x_0 = x_n$ and $x_{k-1}Rx_k$ for every $k = 1, \dots, n$. Acyclicity is normally imposed on a strict relation.

Example A.2. The equality relation on X is reflexive, symmetric, transitive, and antisymmetric, but it is incomplete whenever X contains two distinct elements.

A.1.2 Important Classes of Relations

Definition A.3. An equivalence relation is reflexive, symmetric, and transitive. A preorder is reflexive and transitive. A partial order is reflexive, transitive, and antisymmetric. A total preorder is a complete preorder. A linear order (or total order) is a complete partial order. A set equipped with a partial order is a partially ordered set, or poset.

In consumer theory, a rational weak preference relation is modeled as a total preorder. Antisymmetry is generally inappropriate for weak preference because distinct alternatives may be indifferent.

Example A.4. The usual relation $\geq$ is a linear order on $\mathbb{R}$. The componentwise relation on $\mathbb{R}^L$, defined by $xRy \iff x_\ell \geq y_\ell$ for every ℓ , is a partial order but is not complete when $L \geq 2$. For example, $(1, 2)$ and $(2, 1)$ are incomparable.

The lexicographic relation on $\mathbb{R}^2$ is defined by $xR_{\text{lex}}y$ if $x_1 > y_1$ or if $x_1 = y_1$ and $x_2 \geq y_2$. It is a linear order. Although complete and transitive, it has no continuous real-valued representation on $\mathbb{R}^2$ with its usual topology.

A.2 Symmetric and Asymmetric Parts

Definition A.5. The symmetric part I_R and asymmetric part P_R of a relation R are defined by

$$xI_Ry \iff xRy \text{ and } yRx, \quad xP_Ry \iff xRy \text{ and } \neg(yRx).$$

Proposition A.6. If R is a preorder, then I_R is an equivalence relation and P_R is asymmetric and transitive.

Proof. Reflexivity and symmetry of I_R follow immediately, and its transitivity follows from transitivity of R . Asymmetry of P_R is built into its definition. If xP_Ry and yP_Rz , then transitivity gives xRz . If zRx , then zRy , contradicting yP_Rz . Thus xP_Rz . $\square$

A.3 Equivalence Classes and Quotient Orders

Definition A.7. Let E be an equivalence relation. The equivalence class of x is $[x]_E = \{y \in X : yEx\}$. The quotient set is $X/E = \{[x]_E : x \in X\}$.

Example A.8. If $X = \{a, b, c\}$ and the only distinct equivalent elements are a and b , then $X/E = \{\{a, b\}, \{c\}\}$. For a preference preorder, the equivalence classes of I_R are indifference classes.

Proposition A.9. Let R be a preorder and let I_R be its symmetric part. On X/I_R , define

$$[x]_{I_R} \bar{R} [y]_{I_R} \iff xRy.$$

Then $\bar{R}$ is well defined and is a partial order.

Proof. Suppose $xI_R x', yI_R y'$, and xRy . Then $x'Rx, xRy$, and yRy' ; transitivity therefore gives $x'Ry'$, which proves that the definition is independent of representatives. Reflexivity and transitivity are inherited from R . If both $[x]\bar{R}[y]$ and $[y]\bar{R}[x]$, then $xI_R y$, so $[x] = [y]$. Thus $\bar{R}$ is antisymmetric. $\square$

A.4 Maximal and Greatest Elements

Definition A.10. For a nonempty subset $S \subseteq X$, the set of **maximal elements** is

$$\text{Max}(S, R) = \{x \in S : \nexists y \in S \text{ such that } yP_R x\}.$$

The set of **greatest elements** is

$$\text{Great}(S, R) = \{x \in S : xRy \text{ for every } y \in S\}.$$

A greatest element dominates every feasible alternative; a maximal element is merely undominated. Every greatest element is maximal. If R is complete on S , then the two notions coincide. Under a partial order there is at most one greatest element, while there may be many maximal elements. Under a preorder, several indifferent alternatives may all be greatest.

Example A.11. Let $S = \{a, b, c\}$ and define $R = \Delta_S \cup \{(a, b)\}$. Thus $aP_R b$ and every other pair of distinct alternatives is incomparable. Then a and c are maximal, but there is no greatest element because a and c are incomparable.

Proposition A.12. *Every nonempty finite subset of a set equipped with a total preorder has at least one greatest element and therefore at least one maximal element.*

Proof. Proceed by induction on the cardinality of S . The claim is immediate for a singleton. Suppose g is greatest in a finite set S , and add $a \notin S$. Completeness gives either aRg or gRa . In the first case, transitivity and gRy for every $y \in S$ imply that a is greatest in $S \cup \{a\}$. In the second case, g remains greatest. $\square$

Finiteness matters. For example, $(\mathbb{R}, \geq)$ is linearly ordered but has no greatest element.

A.5 Transitive Closure

Definition A.13. For relations $Q, R \subseteq X \times X$, composition is oriented by $x(R \circ Q)z \iff \exists y \in X : xQy \text{ and } yRz$. Let $R^1 = R$, and recursively define $R^{n+1} = R^n \circ R$. The transitive closure of R is

$$R^+ = \bigcup_{n \geq 1} R^n.$$

The reflexive-transitive closure is $R^* = \Delta_X \cup R^+$, where Δ_X is the identity relation defined above.

Proposition A.14. *The relation R^+ is the smallest transitive relation containing R .*

Proof. Concatenating an R -path of length m with one of length n yields a path of length $m + n$, so R^+ is transitive. Any transitive relation containing R must contain every R^n and hence their union. $\square$

A.6 Order Extensions

Definition A.15. Let R be a preorder on X . We call R^{ext} an extension of R if $R \subseteq R^{\text{ext}}$ and $P_R \subseteq P_{R^{\text{ext}}}$. Thus every original weak and strict comparison is preserved.

Theorem A.16. (*Szpilrajn extension theorem*) *Every partial order on a set has a linear-order extension. See [Szpilrajn, 1930].*

Proof. Proof sketch. The standard proof applies Zorn's lemma to the set of partial orders extending the given order, ordered by inclusion; this use of Zorn invokes the axiom of choice. The union of every chain is again a partial order, so a maximal extension exists. The key extension lemma is that if x and y are incomparable, adjoining xRy and taking the transitive closure

preserves antisymmetry: a new nontrivial cycle would contain the added edge and would imply an original path from y back to x , contradicting incomparability. Thus two incomparable elements could be consistently compared, contradicting maximality. Hence the maximal extension is complete and therefore linear. $\square$

Corollary A.17. *Every preorder has a total-preorder extension.*

Proof. Let I_R be the symmetric part of the preorder R . The induced relation $\bar{R}$ on X/I_R is a partial order by the earlier proposition. Next, let $\bar{R}^{\text{ext}}$ be a linear-order extension of $\bar{R}$, and define on X

$$xR^{\text{ext}}y \iff [x]_{I_R} \bar{R}^{\text{ext}} [y]_{I_R}.$$

Then R^{ext} is a total preorder. It contains R , and antisymmetry of $\bar{R}^{\text{ext}}$ ensures that every comparison in P_R remains strict. $\square$

A.7 Checklist for Economic Applications

- Use a total preorder for rational weak preference; use its asymmetric part for strict preference and its symmetric part for indifference.
- Do not call a strict relation a partial order: strict relations are irreflexive, whereas partial orders are reflexive.
- Distinguish maximal from greatest alternatives whenever the relation may be incomplete.
- Use quotient notation X/I_R , not set-difference notation, when collapsing indifference classes.
- State finiteness, compactness, continuity, or another attainment condition whenever existence of a maximal or greatest alternative is claimed.

A.8 Problems

Part I: By hand

Problem A.1 [Core]. Classifying relations.

Assume $|X| \geq 2$. For each relation below, determine whether it is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, transitive, and complete: equality on X ; strict set inclusion on 2^X ; weak set inclusion on 2^X ; and the componentwise order on $\mathbb{R}^2$.

Give a counterexample for every failed property.

Part II: Finite preference completion and multiple utilities — chapter use case and verified scale-up

Problem A.2 [Proof]. Quotienting a preorder.

Let R be a preorder on X and define xI_Ry if and only if xRy and yRx .

- (a) Prove that I_R is an equivalence relation.
- (b) Define $[x] \bar{R} [y]$ if and only if xRy . Prove that this definition is independent of representatives.
- (c) Prove that $\bar{R}$ is a partial order on X/I_R .

Problem A.3 [Proof]. Maximal, greatest, and finite existence.

Let R be a preorder on a nonempty finite set S .

- (a) Prove that S has an R -maximal element.
- (b) Give an example with several maximal elements and no greatest element.
- (c) Prove that if R is complete, maximal and greatest elements coincide.

Chapter use case

Seed: Problems A.1, A.2, A.3. Fixed: all unlisted primitives and assumptions. Scaled: the number $|X|$ of alternatives in a finite preorder, beginning with the hand example. Tool: Codex + Lean. Verification: Verify the preorder, quotient construction, and each proposed linear extension. Szpilrajn extension is currently verified; quotient pullback, intersection of all extensions, and the finite multiple-utility corollary remain explicit formal targets.

Problem A.4 [Proof]. Order extensions and multiple representation.

Let R be a preorder on a finite set X .

- (a) Pass to the quotient partial order on X/I_R and extend it to a linear order.
- (b) Pull the extension back to a complete preorder on X .
- (c) Show that R is the intersection of its complete preorder extensions and deduce a finite multiple-utility representation.

Appendix B

Technical Theorem Records

A starred theorem in the main text is a reader-level statement. This appendix records its exact environment, full equivalences, proof boundary, and formal-verification status. The star never means that omitted assumptions may be guessed and never means that a theorem is Lean verified.

B.1 WGARP and Coherent CMU

Result ID: CT-RP-WGARP-CMU. Let $L \geq 2$, $X = \mathbb{R}_+^L$, and $\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$ be finite, with $p^t \in \mathbb{R}_{++}^L$, $x^t \in X \setminus \{0\}$, and observed wealth $w^t = p^t \cdot x^t$. Equip $C(X)$ with the topology of uniform convergence on compact subsets. A compact coalition structure is a compact collection Ω of nonempty compact subsets of $C(X)$, under the induced Hausdorff topology.

Theorem B.1. *Full WGARP--CMU characterization. Under the preceding displayed environment, the following six statements are equivalent.*

1. The data are rationalized by an asymmetric, strictly increasing, continuous preference function $r : X \times X \rightarrow \mathbb{R}$.
2. The data are rationalized by a coherent, strictly increasing, continuous CMU function.
3. The data are rationalized by a skew-symmetric, coherent, continuous CMU function whose coalitions consist of strictly increasing concave utilities.
4. The data satisfy WGARP.
5. For every $s, t \in \mathbb{T}$ there exist $R^{s,t} \in \mathbb{R}$ and $\lambda_{s,t}^s, \lambda_{s,t}^t > 0$ such that

$$R^{s,t} = -R^{t,s}, \quad \lambda_{s,t}^s = \lambda_{t,s}^s, \quad R^{s,t} \geq \lambda_{s,t}^s p^s \cdot (x^s - x^t).$$

6. There are antisymmetric numbers $V^{s,t} = -V^{t,s}$ such that

$$\begin{aligned} p^s \cdot (x^s - x^t) \geq 0 &\implies V^{s,t} \geq 0, \\ p^s \cdot (x^s - x^t) > 0 &\implies V^{s,t} > 0 \end{aligned} \quad (s, t \in \mathbb{T}).$$

Proof record. The easy implications from a rationalizing preference function to WGARP and between the two numerical certificate systems are finite arguments. The reverse representation direction constructs symmetric pairwise Afriat utilities and assembles them into compact coherent coalitions. That research-level construction, including compactness and skew-symmetry, is proof-delegated to the exact finite-data theorem of [Aguiar et al., 2026]; this book does not relabel the delegation as an independent proof.

Boundary note. Coherence means pairwise intersection and need not imply a common utility in every coalition. A general coherent CMU may be incomplete or intransitive. The representation rationalizes the observed budgets but does not guarantee nonempty demand at all counterfactual budgets.

B.2 WARP and Strict Coherent CMU

Result ID: CT-RP-WARP-CMU. A preference function is strictly star-concave when, for every $x \neq y$ and $\alpha \in (0, 1)$,

$$r(\alpha x + (1 - \alpha)y, x) > \alpha r(x, x) + (1 - \alpha)r(y, x).$$

This condition concerns the first argument along line segments ending at the second argument; it is not global concavity of the two-argument function.

Theorem B.2. *Full WARP--CMU characterization. In the same finite budget environment, the following are equivalent: (i) WARP; (ii) rationalization by an asymmetric, strictly star-concave, continuous preference function; (iii) rationalization by a coherent, strictly star-concave, continuous CMU; and (iv) rationalization by a skew-symmetric, coherent, continuous CMU whose coalitions consist of strictly increasing strictly concave utilities.*

Proof record. The proof uses strict pairwise utility rationalizations and is separately delegated to the WARP theorem and online appendix of [Aguiar et al., 2026]. It is not obtained by merely replacing the acronym WGARP in the preceding theorem.

B.3 Formalization Record and Boundaries

CT-RP-CMU-SINGLETON. Textbook proof complete; local Lean target. The exact singleton reduction is elementary and introduces no compactness machinery beyond the singleton case.

CT-RP-CMU-CLD. Lean core verified through the pinned WGARP project and the compiled declaration `Econ803.Consumer.compensated_law_of_demand`. The certificate covers the asymmetric, strictly increasing preference-function result stated in the chapter.

CT-RP-WGARP-CMU and **CT-RP-WARP-CMU.** Cited theorems, not yet Lean complete. A complete formalization requires the pairwise Afriat construction, compact-open and Hausdorff topology, coherence, and the strict-rationalization dependencies.

The model does not by itself recover transitivity, global convexity, classical UMP--EMP duality, Slutsky symmetry, scalar-utility welfare comparisons, or global existence of counterfactual demand. Those conclusions remain owned by the benchmark chapters and require their displayed assumptions.

Appendix C

Glossary of Symbols

This glossary records the book's persistent notation. Each blue symbol links back to its first substantive use in the text. There, the blue first use links here. A symbol described as scoped may be reused only after the new local meaning is stated explicitly.

C.1 Sets and indices

L Number of commodities; $L \in \mathbb{N}_{>0}$. Scope: Book-wide.

ℓ, k Commodity-coordinate indices in $\{1, \dots, L\}$. Scope: Book-wide.

$\mathbb{R}_+^n$ Vectors in $\mathbb{R}^n$ whose coordinates are all nonnegative. Scope: Book-wide.

$\mathbb{R}_{++}^n$ Vectors in $\mathbb{R}^n$ whose coordinates are all strictly positive; equivalently $x \gg 0$. Scope: Book-wide.

X Ambient feasible set; in consumer chapters, the consumption set, normally $\mathbb{R}_+^L$. Scope: Scoped.

$i = 1, \dots, I$ Consumer index and number of consumers. Scope: Book-wide.

$j = 1, \dots, J$ Firm index and number of firms. Scope: Book-wide.

C.2 Operators

$a \cdot b$ Euclidean inner product. Scope: Book-wide.

$\|a\|_2$ Euclidean norm of a vector a . Scope: Book-wide.

$\mathbf{1}_n$ All-ones column vector in $\mathbb{R}^n$; the dimension is shown whenever it is not immediate. Scope: Book-wide.

I_n $n \times n$ identity matrix; the subscript records its dimension. Scope: Book-wide.

$a^\top$ Transpose of a vector or matrix; primes never denote transpose. Scope: Book-wide.

a' Alternative observation or object of the same type as a . Scope: Book-wide.

$A \setminus B$ Elements of A that do not belong to B . Scope: Book-wide.

$D_z f$ Jacobian of a vector-valued map f with respect to z . Scope: Book-wide.

$\nabla_z f$ Column gradient of a scalar-valued function f with respect to z . Scope: Book-wide.

$\leq, \geq, \gg$ Vector inequalities are componentwise; $x \gg 0$ means every coordinate is strictly positive. Scope: Book-wide.

2^X Power set of X . Scope: Book-wide.

$\arg \max, \arg \min$ Sets of maximizers and minimizers; a singleton selection is identified explicitly. Scope: Book-wide.

$\Pr, \mathbb{P}, \mathbb{E}$ Probability and expectation under the probability space declared in the surrounding model. Scope: Book-wide.

$\mathbb{I}\{\cdot\}$ Indicator of the event inside braces. Scope: Book-wide.

$\log$ Natural logarithm; information quantities are therefore measured in nats. Scope: Book-wide.

C.3 Optimization

$\mathcal{L}$ Lagrangian of the optimization problem currently under study. Scope: Scoped.

C.4 Choice and preference

$\succeq, \succ, \sim$ Weak preference and its asymmetric and symmetric parts. Scope: Book-wide.

$c: \mathcal{A} \rightrightarrows X$ Nonempty-valued choice correspondence on a family of feasible menus. Scope: Book-wide.

$\succeq^C$ Direct weak revealed choice: $x \succeq^C y$ when some menu contains both and selects x . Scope: Abstract-choice chapter.

$\succ^C$ Direct strict revealed choice: some menu selects x and rejects available y . Scope: Abstract-choice chapter.

$\succeq^{C*}$ Reflexive-transitive closure of direct weak revealed choice $\succeq^C$. Scope: Abstract-choice chapter.

C.5 Revealed preference

$\mathbb{T} = \{1, \dots, T\}$ Finite set of observation indices. Scope: Revealed-preference chapters.

$\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$ Dataset of price-choice observations indexed by $\mathbb{T}$. Scope: Book-wide.

$w^t = p^t \cdot x^t$ Observed expenditure, equal to wealth under budget exhaustion. Scope: Revealed-preference chapters.

$\succeq^{R,D}$ Direct weak revealed preference generated by affordability. Scope: Budget-demand chapters.

$\succ^{R,D}$ Strict direct revealed preference generated by strict affordability. Scope: Budget-demand chapters.

$\succeq^R$ Reflexive-transitive closure of budget-based $\succeq^{R,D}$. Scope: Budget-demand chapters.

(U^t, λ^t) Utility-level and positive marginal-utility-of-wealth certificates satisfying the Afriat inequalities. Scope: Afriat theorem.

$\succeq_\alpha^{R,D}, \succ_\alpha^{R,D}, \succeq_\alpha^R$ Efficiency-adjusted direct weak, direct strict, and revealed-closure relations at expenditure fraction α . Scope: CCEI section.

$\mathcal{S}_{\mathcal{O}}(p, w)$ Bundles consistent with GARP when appended to dataset $\mathcal{O}$ at a new budget. Scope: Consumer chapter.

$\text{CCEI}(\mathcal{O})$ Largest common expenditure fraction for which the adjusted dataset satisfies GARP. Scope: Consumer chapter.

$r : X \times X \rightarrow \mathbb{R}$ Two-argument numerical comparison rule used to define maximal demand. Scope: CMU section.

$x_r(p, w)$ Bundles in $B(p, w)$ that weakly defeat every feasible alternative under preference function r . Scope: CMU section.

Ω Compact collection of nonempty compact coalitions of continuous utility functions. Scope: CMU section.

U A nonempty compact set of continuous utility functions whose members evaluate a comparison unanimously. Scope: CMU section.

$r_\Omega(x, y)$ Coalitional multi-utility preference function $\max_{U \in \Omega} \min_{u \in U} [u(x) - u(y)]$. Scope: CMU section.

C.6 Consumer theory

x Consumption bundle; arguments such as $x(p, w)$ distinguish demand from a generic bundle. Scope: Book-wide default.

p Commodity-price column vector; normally $p \in \mathbb{R}_{++}^L$. Scope: Book-wide default.

w Scalar wealth or income available to a consumer. Scope: Book-wide default.

a_ℓ Strictly positive taste weights normalized to sum to one; equal expenditure shares only in the Cobb-Douglas case. Scope: Workhorse demand models.

$b_\ell(p, w), \mathbf{b}$ Expenditure share $p_\ell x_\ell(p, w)/w$ and its column vector; independent of wealth in the homothetic models considered here. Scope: Workhorse demand models.

ρ CES power parameter with $\rho < 1$ and Cobb-Douglas limit at zero; some production parameterizations use a different sign convention. Scope: CES consumer model.

σ Constant substitution elasticity $1/(1 - \rho) > 0$; distinguished from Marshallian own-price elasticity. Scope: CES consumer model.

γ Nonnegative subsistence bundle; discretionary wealth is $w - p \cdot \gamma$. Scope: Stone-Geary model.

$\mathcal{P}(p)$ Unit expenditure $e(p, 1)$ under the specified degree-one utility normalization; subscript TL denotes the translog index. Scope: Workhorse dual models.

Γ Symmetric matrix $(\gamma_{\ell k})$ of log-price quadratic coefficients, with zero row sums; distinct from the Stone-Geary vector γ . Scope: Translog expenditure model.

$B(p, w)$ Bundles affordable at prices p and wealth w : $\{x \in X : p \cdot x \leq w\}$. Scope: Book-wide.

$x(p, w)$ Marshallian demand correspondence; treated as a function only when uniqueness is assumed. Scope: Book-wide.

$h(p, \bar{u})$ Expenditure-minimizing demand correspondence at prices p and target utility $\bar{u}$. Scope: Book-wide.

u Direct utility; in uncertainty chapters it is the Bernoulli utility of a prize. Scope: Book-wide default.

$v(p, w)$ Maximum attainable utility at prices p and wealth w . Scope: Book-wide.

$e(p, \bar{u})$ Minimum expenditure required to attain utility $\bar{u}$ at prices p . Scope: Book-wide.

$\mu_p(x)$ Expenditure at reference prices p required to attain the utility of bundle x : $e(p, u(x))$. Scope: Welfare-analysis sections.

$S(p, w)$ $D_p x(p, w) + D_w x(p, w)x(p, w)^\top$. Scope: Book-wide.

$\mathbf{P}$ Strictly positive commodity-price domain $\mathbb{R}_{++}^L$. Scope: Consumer chapters.

$\mathbf{W}$ Strictly positive wealth domain $\mathbb{R}_{++}$. Scope: Consumer chapters.

C.7 Behavioral demand

$\hat{p}$ Price vector used in the consumer's perceived first-order condition. Scope: Sparse-max chapter.

x^{SM} Sparse-max demand satisfying perceived-price tangency and the true budget. Scope: Sparse-max chapter.

$a = (a_1, \dots, a_L)$ Coordinatewise attention weights in $[0, 1]^L$. Scope: Sparse-max chapter.

$\mathcal{C}$ Cost of attention or information acquisition. Scope: Attention chapter.

$\mathcal{A}$ Finite set of actions available to the rationally inattentive decision maker. Scope: Rational-inattention section.

μ_0 Full-support prior distribution over the finite state space $\mathcal{S}$. Scope: Rational-inattention section.

$q(a | s)$ Conditional probability of action a in state s ; equivalently, the direct information-and-decision channel. Scope: Rational-inattention section.

$\mathcal{I}(q)$ Shannon mutual information between the state and the action induced by channel q . Scope: Rational-inattention section.

C.8 Choice under uncertainty

Z Prize or consequence set; finite unless the infinite-outcome section explicitly broadens it. Scope: Uncertainty chapter.

μ, ν, η Probability mass functions or probability measures over prizes. Scope: Uncertainty chapter.

$\Delta(Z)$ Set of lotteries over Z ; for general spaces the measurable structure is stated explicitly. Scope: Uncertainty chapter.

δ_z Lottery assigning probability one to prize z . Scope: Uncertainty chapter.

$U(\mu)$ Expected Bernoulli utility of lottery μ . Scope: Uncertainty chapter.

F_μ Cumulative distribution function of lottery μ on an ordered prize space. Scope: Uncertainty chapter.

$\succeq^{\text{FOSD}}$ First-order stochastic-dominance relation, oriented toward the dominating distribution. Scope: Uncertainty chapter.

$\text{CE}(\mu; u)$ Sure prize with the same utility as lottery μ . Scope: Uncertainty chapter.

$\text{RP}(\mu; u)$ Expected prize minus certainty equivalent. Scope: Uncertainty chapter.

$\text{PP}(x, \varepsilon; u)$ Probability adjustment making the specified binary risk indifferent to a sure prize. Scope: Uncertainty chapter.

ϖ_s Strictly positive state probabilities summing to one. Scope: State-contingent chapter.

r Coefficient of relative risk aversion in the power-utility family. Scope: Risk-aversion section.

$\mathcal{S}$ Finite state space used in the surrounding model, including sequential-trade applications. Scope: Scoped.

C.9 Probability

Σ Borel sigma-algebra on the prize space. Scope: Infinite-outcome section.

$\Rightarrow$ Weak convergence of probability measures. Scope: Infinite-outcome section.

C.10 Aggregation

$x^A(p, \mathbf{w})$ Sum of individual Marshallian demands. Scope: Aggregate-demand chapter.

$\mathbf{w}$ Vector of individual wealth levels. Scope: Aggregate-demand chapter.

W Sum of individual wealth levels. Scope: Aggregate-demand chapter.

$a_i(p), b(p)$ Consumer-specific intercept and common wealth coefficient in Gorman indirect utility.
Scope: Aggregate-demand chapter.

$\mathcal{W}$ Social welfare function or explicitly stated welfare aggregate. Scope: Scoped.

C.11 Discrete choice

U_{ij}^* Latent utility of chooser i from alternative j . Scope: Discrete-choice chapter.

P_{ij} Probability that chooser i selects alternative j from the stated menu. Scope: Discrete-choice chapter.

τ Strictly positive Type-I extreme-value scale parameter. Scope: Discrete-choice chapter.

$(\Omega, \mathcal{F}, \mathbb{P})$ Probability space supporting random utility. Scope: Random-utility section.

U Measurable utility vector indexed by alternatives. Scope: Random-utility section.

$P = \{P_B\}_{B \in \mathcal{B}}$ Family of choice-probability distributions, one for each nonempty menu B . Scope: Stochastic-choice section.

$\mathcal{R}_Y$ Finite set of strict linear orders on the alternative universe Y . Scope: Random-utility section.

$\mathcal{V}(v)$ Expected maximum random utility as a function of the mean-utility vector $v \in \mathbb{R}^A$. Scope: Consumer-surplus section.

C.12 Firm theory

y Net-output vector, with positive coordinates denoting outputs and negative coordinates inputs.
Scope: Book-wide default.

Y Technologically feasible net-output plans. Scope: Firm chapter.

f Single-output production function. Scope: Firm chapter.

$\pi_Y(p)$ Support function of production set Y . Scope: Firm chapter.

$\mathcal{Y}_Y(p)$ Profit-maximizing production plans. Scope: Firm chapter.

z Nonnegative vector of physical inputs in a single-output technology. Scope: Firm chapter.

w Strictly positive vector of input prices. Scope: Firm chapter.

p_q Scalar price of output q . Scope: Single-output model.

$C(w, q)$ Minimum input expenditure required to produce at least q . Scope: Firm chapter.

$\mathcal{Z}(w, q)$ Cost-minimizing input correspondence. Scope: Firm chapter.

C.13 Equilibrium

ω Endowment vector; subscripts identify agents and superscripts identify observations or states when declared. Scope: Book-wide default.

θ_{ij} Consumer i 's ownership share of firm j . Scope: Competitive-equilibrium chapter.

$z(p)$ Aggregate demand minus aggregate resources at prices p . Scope: Equilibrium chapters.

C.14 Sequential trade

R Matrix of state-contingent security payoffs. Scope: Sequential-trade chapter.

q Date-0 security-price vector. Scope: Sequential-trade chapter.

ψ Strictly positive state-price vector satisfying $q = R^\top \psi$ when markets are complete. Scope: Sequential-trade chapter.

C.15 Matching

M, W Two finite sides of the one-to-one matching market. Scope: Matching chapter.

φ Involution matching map, with an unmatched agent mapped to themselves. Scope: Matching chapter.

C.16 Order theory

R Generic binary relation on a nonempty ground set X . Scope: Scoped.

R^{-1} Inverse relation: $xR^{-1}y$ iff yRx . Scope: Appendix A.

I_R Symmetric part $R \cap R^{-1}$. Scope: Book-wide.

P_R Asymmetric part of R . Scope: Appendix A.

$[x]_E$ Equivalence class of x under E . Scope: Appendix A.

X/E Set of equivalence classes of X under an equivalence relation E . Scope: Book-wide.

$\bar{R}$ Partial order induced by a preorder on its quotient by I_R . Scope: Appendix A.

$\text{Max}(S, R)$ Set of R -maximal elements of feasible subset S . Scope: Appendix A.

$\text{Great}(S, R)$ Set of R -greatest elements of feasible subset S . Scope: Appendix A.

Δ_X Diagonal or identity relation on X . Scope: Appendix A.

R^n n -fold relational composition, with composition orientation stated in the text. Scope: Appendix A.

R^+ Least transitive relation containing R . Scope: Appendix A.

R^* $\Delta_X \cup R^+$. Scope: Appendix A.

R^{ext} Complete-preorder extension obtained from a linear extension of the quotient order. Scope: Appendix A.

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